



Port Said Branch

Telecom Revenue & Churn Intelligence System

An End-to-End Data Warehouse and Predictive Analytics Platform

Mostafa Samir Ali

Nouran Mohsen ElBahaie

Nour Ehab Elshafey

Yomna Ali Gad

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Abstract

Telecommunications operators hold extensive records describing their subscribers, the services those subscribers consume, the contracts that bind them, and the revenue they generate. The presence of this data does not by itself produce insight. The records are distributed across operational tables optimised for transaction processing rather than analysis, and management questions — how much revenue is at risk, which customers are likely to leave, and what action would be most effective — cannot be answered directly from them.

This project designs and implements a Telecom Revenue and Churn Intelligence System: an end-to-end analytical platform that transforms a raw subscriber dataset into a governed data warehouse, a predictive model, and a suite of executive dashboards. The work spans conceptual data modelling, normalisation to Boyce-Codd Normal Form, physical implementation of an eleven-table operational schema in Microsoft SQL Server, a Python-based extract-transform-load pipeline, a star-schema data warehouse comprising five conformed dimensions and a customer-grain fact table with fifteen measures, exploratory and statistical analysis of churn behaviour, supervised classification with explainability analysis, and twenty-three interactive Power BI dashboard pages organised as a single analytical narrative.

The system is built on the IBM Telco Customer Churn dataset of 7,043 subscribers. The dimensional model was deliberately chosen as a star rather than a snowflake or galaxy schema, on the grounds that all analytical questions in the domain converge on a single grain — the customer — and that at this data volume the storage cost of denormalisation is negligible against the query and maintenance benefits. Machine learning treats the pronounced class imbalance through synthetic minority oversampling applied only to the training partition, and selects a decision threshold on business grounds rather than by maximising accuracy, recognising that a missed churner costs more than a false alarm.

The contribution of the work lies less in any single component than in their integration: an operational database, a dimensional warehouse, a predictive model, and a reporting layer that together take a management question from raw record to recommended action.

Keywords: customer churn, business intelligence, data warehouse, star schema, dimensional modelling, ETL, predictive analytics, telecommunications.

1 Introduction

1.1 Project Background

The telecommunications sector operates under a structural economic condition that shapes almost every commercial decision taken within it: the cost of acquiring a new subscriber substantially exceeds the cost of retaining an existing one. Markets in most territories are saturated, meaning that growth for one operator is largely growth taken from another, and the instruments of acquisition — handset subsidies, promotional pricing, commission to retail channels — are expensive and their effects short-lived. Retention, by contrast, operates on customers whose behaviour is already observable and whose service relationship is already established.

This asymmetry gives the measurement of subscriber attrition, conventionally termed *churn*, a significance out of proportion to its apparent simplicity. A churn rate is a single percentage, but the decisions that depend on it are numerous: which segments to defend, what a retention offer may cost before it destroys the value it protects, which service deficiencies are causing loss rather than merely correlating with it, and how much of next year's revenue is presently exposed.

Operators are not short of the data required to answer these questions. A modern billing and provisioning estate records contract type, payment instrument, service subscriptions, usage volumes, tenure, geography, referral activity and charge history for every subscriber. The difficulty is that this data is held in systems designed for a different purpose. Operational databases are normalised for transactional integrity and write performance; they answer *what is the current state of subscriber X* efficiently, and *what is the aggregate revenue at risk across all subscribers on month-to-month contracts who lack technical support* slowly, if at all.

1.2 Problem Statement

The problem this project addresses can be stated precisely:

A telecommunications operator possesses subscriber, service, contract and billing data sufficient in principle to identify which customers are likely to leave, why they leave, and what that departure costs. In practice this data is distributed across operational tables whose structure resists analytical querying, is not exposed to decision-makers in any usable form, and supports no predictive capability. The operator can therefore observe churn only after it has occurred, and cannot quantify its financial consequence or act on its causes.

Three distinct deficiencies follow from this. First, an *architectural* deficiency: no analytical data structure exists between the operational database and the decision-maker. Second, an *analytical* deficiency: no model

exists that estimates the probability of departure for an individual subscriber. Third, a *presentational* deficiency: no interface exists through which a manager, as opposed to an analyst, can interrogate the data.

Addressing any one of these in isolation produces limited value. A predictive model without a reporting layer produces scores that nobody reads. A dashboard built directly against a normalised operational schema is slow, fragile, and forces business logic into the presentation layer. The project therefore treats the three as a single problem.

1.3 Project Motivation

Four considerations motivated the selection of this problem.

The **economic weight of retention** makes churn among the few analytical problems in which a modest improvement in prediction translates directly and measurably into retained revenue. This gives the work a defensible business case rather than a purely technical one.

The **structural realism of the data** is unusual among teaching datasets. Subscriber data exhibits genuine class imbalance, genuine multicollinearity between service attributes, genuine categorical encoding difficulties, and genuine ambiguity between missing values and meaningful absences. These are the conditions under which real analytical work is performed.

The **breadth of disciplines required** was itself a motivation. A complete solution demands conceptual modelling, relational design and normalisation theory, ETL engineering, dimensional modelling, statistical analysis, supervised learning, model explainability, and dashboard design. Few academic projects exercise this range in a single coherent artefact.

Finally, the **separation between correlation and causation** is unusually consequential here. It is straightforward to observe that customers on month-to-month contracts churn more frequently. It is a different and harder claim that moving customers onto longer contracts would reduce churn, since contract choice is itself a signal of commitment. The project treats such findings as associations to be reported honestly rather than causal levers to be pulled.

1.4 Project Objectives

The work pursues one principal objective and nine subordinate ones.

Principal objective. To construct an integrated analytical system that enables a telecommunications operator to measure its financial position, understand the causes and cost of subscriber attrition, and convert both into decisions supported by evidence.

Subordinate objectives.

1. Design a conceptual entity-relationship model of the telecom subscriber domain.

2. Normalise that model and implement it as a physical operational schema in SQL Server.
3. Construct an ETL pipeline that loads source data into the operational schema with verifiable integrity.
4. Design and implement a dimensional data warehouse optimised for analytical query performance.
5. Characterise churn behaviour through exploratory and inferential statistical analysis.
6. Engineer domain-driven features that improve the separability of churned and retained subscribers.
7. Train, tune and evaluate supervised classifiers for churn prediction, treating class imbalance explicitly.
8. Explain model behaviour at both the population and the individual level.
9. Present the results as a coherent suite of interactive dashboards organised as an analytical narrative.

1.5 Project Scope

The scope of the work is bounded as follows.

Within scope. Data ingestion and quality assessment; conceptual, logical and physical database design; ETL implementation; dimensional warehouse construction; exploratory and statistical analysis; feature engineering; supervised classification and evaluation; model explainability; dashboard design and implementation; and documentation of all of the preceding.

Outside scope. Real-time or streaming ingestion; production deployment of the model as a service; automated retraining or model monitoring; integration with an operator's live billing estate; and any component requiring cost data, which the source dataset does not contain.

A formal statement of what was implemented against what was originally proposed is given in Section 3.3, and the reader is directed there before interpreting any claim of delivery in this document.

1.6 Contributions

The work makes four contributions.

It delivers an **end-to-end pipeline** rather than an isolated model, demonstrating the passage of a single dataset through operational storage, transformation, dimensional modelling, prediction and presentation.

It applies **normalisation theory to Boyce-Codd Normal Form** on data whose violations are genuine and subtle, arising from sentinel values embedded within data columns rather than from obvious redundancy.

It documents **dimensional design decisions with explicit rationale**, including the rejection of alternative schema types, the treatment of slowly changing dimensions, and the placement of degenerate dimensions within the fact table.

It organises the reporting layer as a **narrative rather than a collection**, in which each dashboard page answers one business question and passes a specific unanswered question to its successor.

1.7 Document Organisation

Chapters 2 to 4 establish the business and technical context. Chapters 5 to 11 document the design and construction of the data platform, proceeding from architecture through dataset, entity-relationship design, normalisation, operational implementation, ETL and data warehouse. Chapters 12 to 16 document the analytical work: exploratory analysis, feature engineering, modelling, explainability and business insight. Chapter 17 documents the business intelligence layer. Chapters 18 to 20 report testing, results, limitations and conclusions. Appendices contain the full data dictionary, DDL listings, DAX measure definitions and team contribution statements.

2 Business Case

2.1 The Telecommunications Retention Problem

A telecommunications operator sells a recurring service. Its revenue in any period is the product of its subscriber base and the average revenue that base generates. Growth is therefore available through two routes: increasing the number of subscribers, or increasing the revenue each produces. In a saturated market the first route is largely a matter of taking subscribers from competitors, which is expensive, and losing them to competitors, which is the phenomenon under study.

Churn is not a uniform loss. A subscriber who departs after two months on a promotional tariff, having consumed a subsidised handset, represents a different financial event from a subscriber who departs after four years on a premium bundle. The first may never have been profitable; the second was profitable and has ceased to be. An operator that measures churn only as a headcount percentage cannot distinguish between these, and will therefore misallocate its retention budget.

This observation motivates the central design decision of the analytical layer: churn is measured not only as a rate but as a **financial quantity**. Every analysis of departure is accompanied by an analysis of the revenue attached to it.

2.2 The Economics of Customer Churn

Four quantities govern the economics of retention.

Acquisition cost is what the operator spends to win a subscriber. **Customer lifetime value** is the total margin expected from that subscriber over the relationship. **Tenure** determines the point at which lifetime value exceeds acquisition cost. **Churn probability** determines whether that point will be reached.

The interaction of these produces a result that is not obvious: reducing churn among *low-value* subscribers may be worth less than reducing it among *high-value* subscribers by a much smaller margin. Retention effort should be directed by expected value, not by risk alone. A subscriber with an eighty per cent probability of leaving and a low monthly charge represents less exposure than a subscriber with a forty per cent probability and a high charge.

The system implements this reasoning directly. Risk scores produced by the predictive model are combined with revenue measures held in the fact table, so that management sees not a list of likely leavers but a ranking of revenue at risk.

2.3 Stakeholders and Their Decision Needs

Stakeholder	Principal question	System component that answers it
Executive management	What is our financial position and what is exposed?	Executive Financial Position; Executive Decision Center
Finance	Where does revenue originate and where is it lost?	Revenue Composition; Net Revenue and Refunds; Churn Financial Impact
Retention / CRM	Which subscribers should we contact, and in what order?	Risk scoring; Customer Revenue Value
Marketing	Which offers and contracts perform?	Contract Revenue Intelligence; Offer Performance
Customer support	Which service deficits associate with departure?	Support Interaction; Service Penetration
Network operations	Which access technologies underperform?	Network Type Performance
Data governance	Can these results be trusted?	Data Integrity Monitor

2.4 Business Questions

The system was specified by first enumerating the questions it must answer, and only then designing the structures that answer them.

Financial. What is the current revenue position? Of what components is revenue composed? What do ancillary services contribute? What is retained after refunds? What is lost to churn, in currency rather than headcount? Which contract type and which payment method perform best on a combined financial assessment?

Behavioural. Which subscribers are most likely to leave? How do tenure, demographics and service adoption relate to departure? Does the absence of support services associate with elevated churn?

Operational. Which access technology, service bundle, offer or geography carries the highest attrition? What is the effect of referral activity?

Governance. Are values missing, and where? Are absent values genuinely missing or meaningfully absent? Can the analytical results be reproduced from source?

2.5 Significance of the Business Case

The case is significant because it is *complete*. It begins at a measurable financial position, proceeds to a diagnosis of loss, locates that loss in specific operational conditions, and terminates in a recommended

action whose expected value can be estimated. Systems that stop at diagnosis leave the decision unmade; systems that begin at recommendation leave it unjustified. The four-part structure described in Chapter 17 is a direct expression of this progression.

3 Objectives and Scope

3.1 Functional Requirements

ID	Requirement
FR-1	Ingest subscriber, service, location, status and demographic source files
FR-2	Validate and cleanse source data prior to load
FR-3	Persist operational data in a normalised relational schema
FR-4	Transform operational data into a dimensional model
FR-5	Maintain referential integrity between fact and dimension tables
FR-6	Support exploratory and statistical analysis of the subscriber base
FR-7	Train a classifier that estimates churn probability per subscriber
FR-8	Explain model output at population and individual level
FR-9	Expose analytical results through interactive dashboards
FR-10	Permit filtering by contract, payment method, service and churn status
FR-11	Verify that reported values reconcile with the source database

3.2 Non-Functional Requirements

ID	Requirement	Basis
NFR-1	Analytical queries must resolve without perceptible delay in the reporting tool	Star schema; pre-computed bands; non-clustered indexing
NFR-2	Reported figures must reconcile exactly with the source database	Validation measures; row-count reconciliation
NFR-3	The pipeline must be reproducible from raw source	Scripted DDL and load; notebook-generated insert scripts
NFR-4	The dimensional model must accommodate additional fact tables	Conformed dimensions

ID	Requirement	Basis
NFR-5	Reports must be usable by non-technical management	One business question per page; consistent visual grammar
NFR-6	Data quality must be observable, not assumed	Dedicated integrity dashboard

3.3 Delivered Scope Against Proposed Scope

This section states plainly what was implemented and what was not. It should be read before any other claim of delivery in this document.

Component	Proposed	Actual status	Evidence
Entity-relationship design	Required	Implemented	churn_ERD_schema.md, ERD.png, draw.io source
Normalisation documentation	Required	Implemented	Normalisation to BCNF documented
Physical operational database	Required	Implemented	11 tables, telecom schema
Star-schema warehouse	Required	Implemented	5 dimensions, 1 fact, 15 measures
ETL pipeline	Required	Implemented	Notebook-driven script generation and load
Exploratory data analysis	Beyond proposal	Implemented	01_eda.ipynb
Feature engineering	Beyond proposal	Implemented	02_feature_engineering.ipynb
Predictive modelling	Beyond proposal	Implemented (results not persisted)	03_modeling.ipynb
Explainability (SHAP)	Beyond proposal	Implemented (results not persisted)	03_modeling.ipynb
Power BI dashboards	20 pages proposed	Implemented as 23 pages	Four-part dashboard suite
Database dictionary	Required	Partial	Schema documentation exists; formal dictionary compiled in Appendix B
SSRS paginated reports	6 reports proposed	Implemented — 6 reports	Captured in Section 17.10
Stored procedures	Required	Implemented — 4 procedures	Select_SP, Insert_SP, Update_SP, Delete_SP — Section 9.6

Component	Proposed	Actual status	Evidence
Churn logic automation	Required	Implemented — 3 functions	Churn Risk Level, CLTV Customer Classifier, Financial Loyalty — Section 9.7
Mobile tracking application	Bonus	Not implemented	—
SQL Server Integration Services	Not proposed	Deliberately excluded	Rationale in Section 10.6
SQL Server Analysis Services	Not proposed	Deliberately excluded	Rationale in Section 17.4
REST API service	Mentioned in repository README	Not implemented	No API source present
Interactive Python dashboard	Mentioned in repository README	Not implemented	No application source present
Retrieval-augmented assistant	Part of the stated project architecture	Declared, not evidenced	Appears as a layer in the team architecture; no artefact examined
Containerisation	Mentioned in repository README	Not implemented	No container definitions present

A note on evidence is warranted, because the boundary between what was built and what was described has moved during the preparation of this document and the distinction is maintained deliberately.

Three standards are applied. A component is recorded as **implemented** where an artefact was examined — a script, a report, a dashboard, a named database object. It is recorded as **declared** where it appears in the team's own architecture or documentation but no artefact was examined. It is recorded as **not implemented** where neither holds.

On that basis the database procedures and functions, the paginated reports and the dashboard suite are implemented; the retrieval-augmented assistant and agent layer are declared; and the REST service, interactive Python dashboard and containerisation described in the repository README are not implemented. Where this document states that something was built, evidence supports the statement.

4 Background and Technical Concepts

4.1 Customer Churn

Churn denotes the cessation of a service relationship by a subscriber. Two variants are distinguished in the literature. *Voluntary* churn occurs when the subscriber elects to leave, typically for a competitor, and is the variant susceptible to retention action. *Involuntary* churn occurs through non-payment, relocation or death, and is largely not. The dataset used here does not distinguish the two; the churn label records departure without recording its volition. This is a limitation of the source rather than of the analysis, and is noted in Chapter 19.

Churn analysis divides naturally along the standard analytics maturity progression. *Descriptive* analysis establishes what the churn rate is. *Diagnostic* analysis establishes which conditions associate with it. *Predictive* analysis estimates, for an individual subscriber, the probability of future departure. *Prescriptive* analysis recommends an action. This project performs all four, and the reporting layer is deliberately organised along the same progression.

4.2 Transaction Processing and Analytical Processing

The distinction between systems optimised for transactions and systems optimised for analysis is foundational to the architecture adopted here.

Property	OLTP	OLAP
Primary operation	Insert, update, delete	Read, aggregate
Typical query	Single row by key	Aggregate over many rows
Normalisation	High (3NF or beyond)	Deliberately denormalised
Optimised for	Write integrity, concurrency	Query throughput
Schema volatility	Stable	Extended as questions evolve
Historical data	Current state	Retained for trend analysis
Users	Applications and operators	Analysts and management

The reason a reporting tool is not connected directly to the operational schema follows from this table. Answering an aggregate question against a normalised schema requires traversing many joins, and the cost of that traversal is paid on every interaction with every dashboard. The warehouse pays the join cost once, at load time.

4.3 Dimensional Modelling

Dimensional modelling organises analytical data into *facts*, which are the measurements, and *dimensions*, which are the contexts by which measurements are sliced. Three schema topologies are conventionally available.

A **star schema** places one fact table at the centre, joined directly to denormalised dimension tables. Each dimension is one join from the fact.

A **snowflake schema** normalises the dimensions, so that a dimension may itself reference further tables. Storage falls; join depth rises.

A **galaxy** or **fact constellation** schema admits multiple fact tables sharing conformed dimensions, supporting analysis at more than one grain.

The selection made for this project, and the reasoning behind it, is documented in Section 11.1.

Three concepts recur in the design. A **surrogate key** is a system-generated integer primary key on a dimension, decoupling the warehouse from the natural keys of the source system. A **conformed dimension** is one whose definition is shared across fact tables, so that a measure sliced by it means the same thing everywhere. A **degenerate dimension** is a dimensional attribute of low cardinality retained in the fact table rather than extracted to a dimension of its own.

Slowly changing dimensions describe how a dimension responds to change in the source. Type 1 overwrites, retaining no history. Type 2 inserts a new row and closes the prior one, retaining full history at the cost of storage and query complexity. Type 3 retains a limited number of prior values in additional columns.

4.4 Class Imbalance

Churn datasets are imbalanced: retained subscribers substantially outnumber departed ones. A classifier optimising overall accuracy on such data can achieve a high score by predicting the majority class universally, while identifying no churner at all. Accuracy is therefore an inappropriate primary metric.

Three families of remedy exist. *Algorithmic* approaches weight the minority class more heavily in the loss function. *Undersampling* discards majority observations. *Oversampling* generates additional minority observations; the Synthetic Minority Over-sampling Technique does so by interpolating between existing minority observations and their nearest neighbours rather than duplicating them.

A methodological caution attends all resampling: it must be applied only to the training partition. Resampling before partitioning permits information from synthesised observations to appear in the test set, inflating the apparent performance of the model. The approach taken in this project is described in Section 14.1.

4.5 Model Evaluation Under Imbalance

Given imbalance, evaluation relies on the confusion matrix and the quantities derived from it. **Precision** is the proportion of predicted churners that genuinely churned. **Recall**, or sensitivity, is the proportion of genuine churners that the model identified. **F1** is their harmonic mean. The **receiver operating characteristic** curve plots true positive rate against false positive rate across all thresholds, and the area beneath it summarises discrimination independently of any single threshold.

The **precision-recall curve** is generally more informative than the ROC curve under strong imbalance, because it does not reward the model for correctly identifying the abundant negative class.

Calibration is distinct from discrimination. A model may rank subscribers correctly while producing probability estimates that are systematically too high or too low. Since this system uses probabilities to allocate retention budget, calibration is material and is assessed explicitly.

4.6 Model Explainability

A model that predicts without explaining is of limited use in a commercial setting, because a retention team must know not only that a subscriber is at risk but why. Tree ensembles expose an intrinsic feature importance, but it is computed from training-time split statistics and is known to be biased toward high-cardinality features.

SHapley Additive exPlanations derive from cooperative game theory. The contribution of each feature to a particular prediction is computed as its average marginal contribution across all possible orderings of the features. The resulting values are additive: they sum to the difference between the prediction and the base rate. This yields both a global ranking of drivers and a decomposition of any individual prediction — the latter being what a retention agent requires when preparing to contact a specific subscriber.

4.7 Business Intelligence and Analytical Narrative

A dashboard is not a report. A report presents figures; a dashboard supports interrogation. The design principle adopted throughout the reporting layer is that each page answers exactly one business question, and that the pages are ordered so that each raises the question its successor answers. The consequence is that a reader traverses the suite as an argument rather than browsing it as a collection. The specific sequence is set out in Chapter 17.

5 System Architecture

5.1 Architectural Overview

The system is organised as five layers, each consuming the output of its predecessor and each addressing a distinct concern. The separation is deliberate: it permits any layer to be modified without disturbing the others, and it locates each transformation at the point in the pipeline where it can be performed once rather than repeatedly.

The layers, their responsibilities and the transitions between them are shown in Figure 5.1. Each is described in turn in the sections that follow.

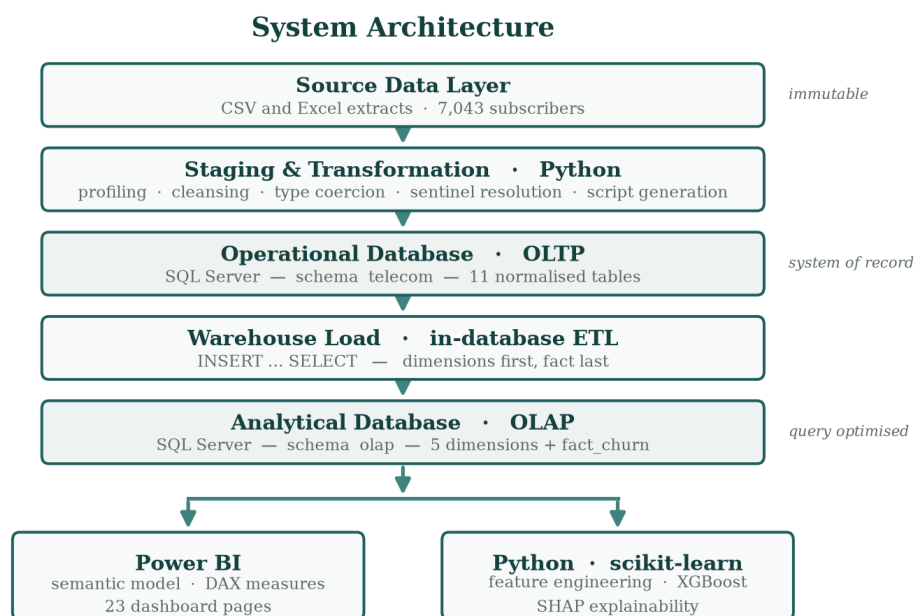


Figure 5.1 End-to-End System Architecture

5.2 Source Data Layer

The source consists of flat extracts of an IBM telecommunications subscriber dataset, supplied in both delimited and spreadsheet form. Two generations are present: an original twenty-one-column release covering 7,043 subscribers, and an extended release distributing the same subscriber population across five subject-oriented files covering demographics, location, services, status and population reference data. The extended release carries the financial and geographic attributes on which the revenue analysis depends and is therefore the basis of the operational schema.

The layer performs no transformation. Its function is to hold an immutable record of what was received, so that any downstream figure can be traced to its origin.

5.3 Staging and Transformation Layer

Transformation is performed in Python. The layer profiles each source file, reconciles type declarations against actual content, resolves the ambiguity between missing values and meaningful absences, standardises categorical encodings, and assembles the referential structure that the operational schema requires.

Its output is not data but **code**: a Transact-SQL script containing the insert statements that populate the operational schema. This choice is discussed in Section 10.3. Its practical consequence is that the load is reproducible and auditable — the script can be inspected, version-controlled and re-executed, which a direct database connection from a notebook cannot.

5.4 Operational Database Layer

The operational layer is a SQL Server database, `telecom_churn_db`, containing eleven tables in the `telecom` schema. The design is normalised: each table describes one subject, foreign keys enforce referential integrity, and junction tables resolve the many-to-many relationships between subscribers and services, and between subscribers and churn reports.

This layer represents the system of record. It is the answer to the question *what does the operator know about this subscriber*, and it is structured for correctness rather than for analytical speed.

5.5 Warehouse Load Layer

The transformation from operational to analytical structure occurs **inside the database**, through `INSERT ... SELECT` statements that read the normalised tables and write the dimensional ones. No external orchestration tool is involved.

The rationale is proportionality. At 7,043 subscribers the entire warehouse loads in seconds; introducing a dedicated integration service would add operational surface area without adding capability. The load order is significant and is enforced: dimensions are populated first so that their surrogate keys exist, and the fact table is populated last by a single joined statement that resolves those keys.

5.6 Analytical Database Layer

The analytical layer occupies the `olap` schema of the same database. It comprises five dimension tables and one fact table arranged as a star. Attributes that would otherwise require calculation at query time — age

bands, tenure bands, service counts, and the conversion of bit flags to readable labels — are materialised here at load time.

Placing the warehouse in the same database as the operational schema, rather than in a separate instance, is a decision of scale rather than principle. At this volume the isolation benefit of a separate instance does not justify the cross-database complexity it introduces.

5.7 Presentation and Analytics Layers

Two consumers read from the analytical layer.

Power BI imports the dimensional tables and builds a semantic model over them, with relationships defined on the surrogate keys and business logic expressed as DAX measures. Twenty-three dashboard pages are organised into four parts.

The Python analytics stack performs exploratory analysis, feature engineering, classification and explainability. It reads from the prepared dataset rather than from the warehouse, since model training requires the encoded and scaled feature matrix rather than the presentation-oriented dimensional structure.

5.8 Technology Stack

Layer	Technology	Function
Source	CSV, Excel	Raw subscriber extracts
Staging	Python 3, pandas, NumPy	Profiling, cleansing, script generation
Operational store	Microsoft SQL Server (T-SQL)	Normalised system of record
Warehouse load	T-SQL INSERT ... SELECT	Dimensional population
Analytical store	Microsoft SQL Server, <code>olap</code> schema	Star schema
Analysis	Jupyter, pandas, SciPy, statsmodels	EDA, hypothesis testing, VIF
Visualisation (analysis)	Matplotlib, seaborn	Statistical graphics
Modelling	scikit-learn, imbalanced-learn, XGBoost	Classification, resampling, tuning
Explainability	SHAP	Global and local attribution
Reporting	Power BI Desktop	Semantic model, DAX, dashboards
Modelling notation	draw.io	Entity-relationship diagrams
Version control	Git / GitHub	Source management

Data Flow from Source to Insight

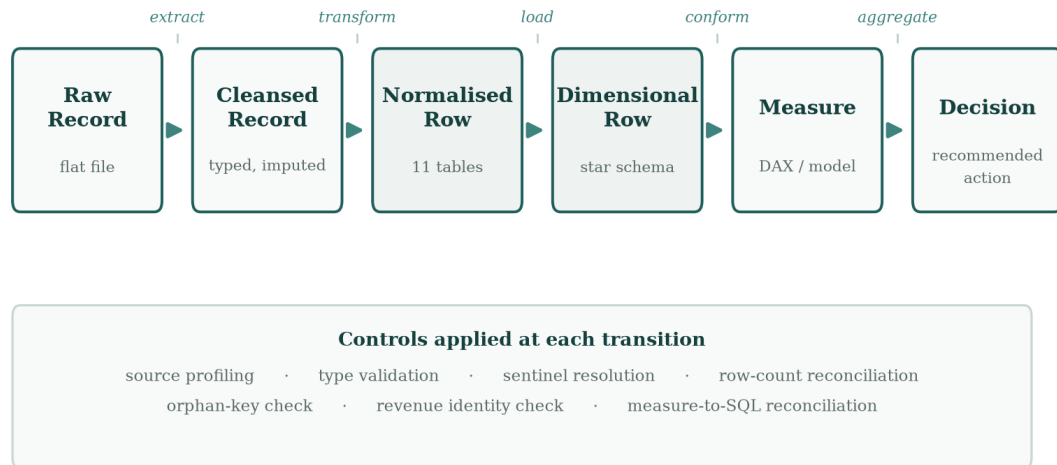


Figure 5.2 Data Flow from Source to Insight

6 Dataset Description

6.1 Provenance

The system is built on the IBM Telco Customer Churn dataset, a synthetic but structurally realistic representation of a telecommunications subscriber base, widely used as a reference dataset for churn analysis. It describes **7,043 subscribers**, each appearing exactly once. The population is fixed; no subscriber joins or leaves within the observation window.

This last property is consequential and is stated here because it constrains everything that follows. The dataset is a **snapshot**, not a time series. It records the state of each subscriber at one moment, including whether that subscriber had by then departed. It does not record when the departure occurred, nor the sequence of states preceding it.

6.2 Source File Inventory

File	Content
WA_Fn-UseC_-Telco-Customer-Churn.csv	Original release: 7,043 rows, 21 columns
metadata.csv	Column definitions for the original release
Telco_customer_churn.csv	Extended release, consolidated
Telco_customer_churn_demographics.csv	Age, gender, marital status, dependents
Telco_customer_churn_location.csv	City, state, country, coordinates, postal code
Telco_customer_churn_services.csv	Service subscriptions, usage, charges
Telco_customer_churn_status.csv	Churn label, value, score, reason, CLTV
Telco_customer_churn_population.csv	Population reference by postal area
CustomerChurn.csv	Consolidated working extract
telecom_churn_db.bak	SQL Server database backup of the loaded system

Each file is additionally present in Excel form. The two representations were verified to be equivalent; the delimited form was used for loading.

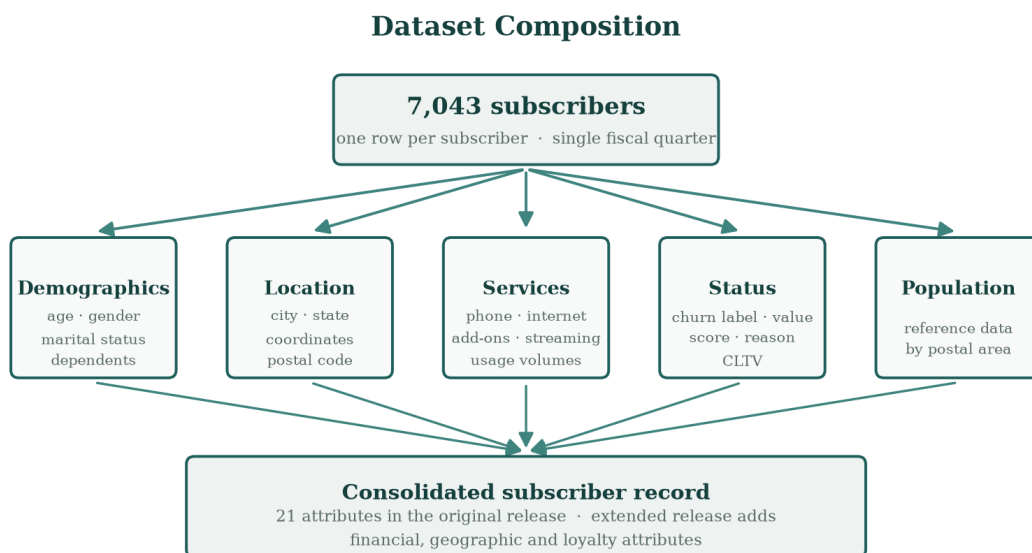


Figure 6.1 Dataset Composition Overview

6.3 Attribute Catalogue — Original Release

Attribute	Type	Domain
customerID	String	Natural key, format 7590-VHVEG
gender	String	Male, Female
SeniorCitizen	Integer	0, 1
Partner	String	Yes, No
Dependents	String	Yes, No
tenure	Integer	0–72 months
PhoneService	String	Yes, No
MultipleLines	String	Yes, No, <i>No phone service</i>
InternetService	String	DSL, Fiber optic, No
OnlineSecurity	String	Yes, No, <i>No internet service</i>
OnlineBackup	String	Yes, No, <i>No internet service</i>
DeviceProtection	String	Yes, No, <i>No internet service</i>
TechSupport	String	Yes, No, <i>No internet service</i>

Attribute	Type	Domain
StreamingTV	String	Yes, No, <i>No internet service</i>
StreamingMovies	String	Yes, No, <i>No internet service</i>
Contract	String	Month-to-month, One year, Two year
PaperlessBilling	String	Yes, No
PaymentMethod	String	Four values
MonthlyCharges	Float	Currency
TotalCharges	Float / String	Currency; 11 rows hold a blank string
Churn	String	Yes, No — the target

The values shown in italics are **sentinel values**: strings that encode a structural condition rather than a measurement. Their treatment is the central issue of Chapter 8 and is not a cosmetic matter.

6.4 Extended Release Attributes

The extended release adds the attributes on which the financial and geographic analysis depends: `total_revenue`, `total_refunds`, `total_extra_data_charges`, `total_long_distance_charges`, `avg_monthly_long_distance_charges`, `avg_monthly_gb_download`, `cltv`, `churn_score`, `churn_reason`, `number_of_referrals`, `offer`, `internet_type`, `unlimited_data`, `premium_tech_support`, together with `city`, `state`, `country`, `latitude`, `longitude` and `postal code`.

Two of these warrant particular note. `cltv` is a customer lifetime value estimate supplied with the dataset; it is an input, not a quantity this project computed. `churn_score` is likewise a supplied risk score, and must not be confused with the probability produced by the model trained in Chapter 14.

6.5 Data Quality Assessment

Observation	Extent	Resolution
TotalCharges typed as string	All rows	Coerced to numeric
TotalCharges blank	11 rows	These subscribers have <code>tenure = 0</code> ; imputed as one month of <code>MonthlyCharges</code>
SeniorCitizen encoded 0/1 while comparable flags use Yes/No	All rows	Standardised to a bit flag
'No phone service' sentinel in <code>MultipleLines</code>	682 rows	Resolved structurally; see Chapter 8

Observation	Extent	Resolution
'No internet service' sentinel in six add-on columns	1,526 rows	Resolved structurally; see Chapter 8
Duplicate rows	None found	—
Missing values after coercion and imputation	None	—

The imputation of `TotalCharges` deserves justification, since imputation is frequently performed carelessly. The eleven affected subscribers all have a tenure of zero months. Their total charges are not *unknown*; they are *not yet accrued*. Substituting a single month of their monthly charge is therefore not a statistical estimate of a missing quantity but a reconstruction of a known one.

6.6 Target Variable

The target appears in two forms. `Churn` (or `churn_label`) is the categorical form, taking values Yes and No. `churn_value` is its numeric encoding, taking 1 and 0 respectively. The two are equivalent; the numeric form is used in the fact table so that the churn rate is expressible as a simple ratio of sums, requiring no conditional logic in the reporting layer.

The observed distribution is **1,869 departed** against **5,174 retained**, a churn rate of **26.54 per cent**, read from the deployed reporting layer and reported in full in Section 18.2.

7 Entity-Relationship Design

7.1 Approach

The conceptual model was constructed before any physical structure, and deliberately in advance of examining the shape of the source files. The purpose of a conceptual model is to describe the domain as it exists commercially, not as it happens to have been extracted; deriving entities from file layouts inverts that relationship and produces a schema that encodes an accident of export rather than a structure of the business.

Nine entities were identified.

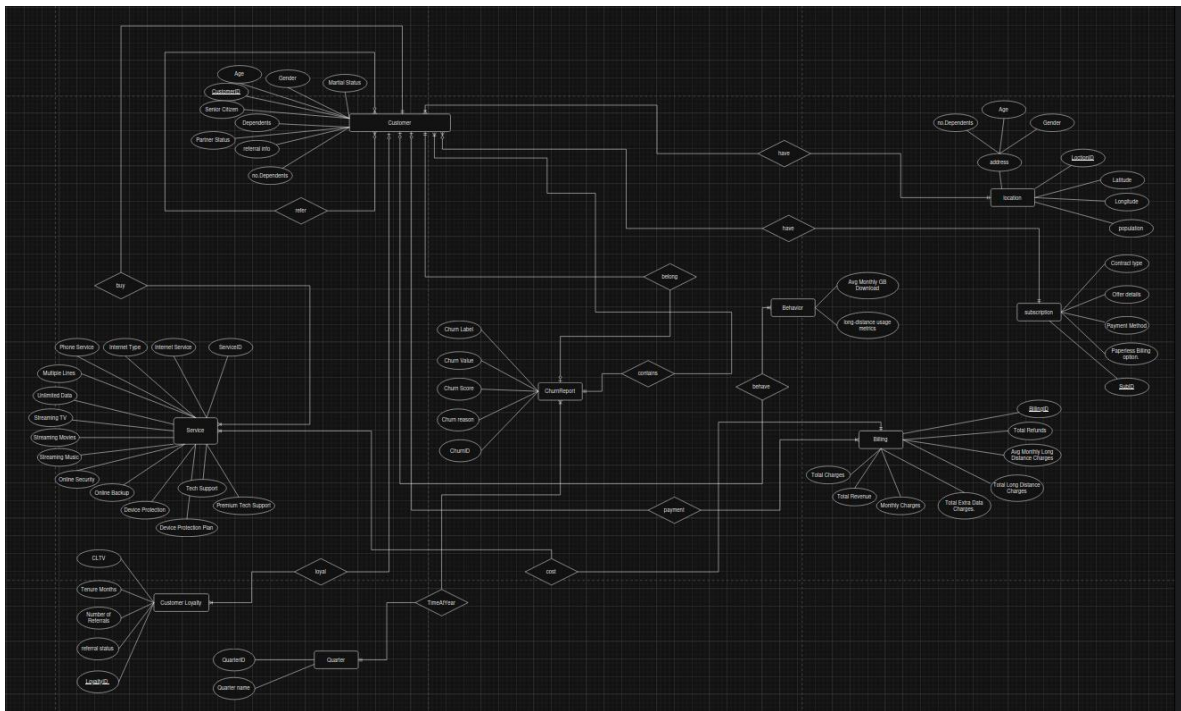


Figure 7.1 Conceptual Entity Relationship Diagram

7.2 Entity Catalogue

Entity	Represents
Customer	A telecommunications subscriber and their demographic attributes
Location	The geographic position associated with a subscriber
Subscription	The commercial agreement: contract type, payment instrument, offer

Entity	Represents
Service	The catalogue of services and add-ons available for subscription
Behavior	Usage measurements: data volume, long-distance activity
Billing	The financial record: charges, revenue, refunds
CustomerLoyalty	Tenure, lifetime value, referral activity, scoped to a fiscal period
ChurnReport	The analytical outcome: label, value, score, stated reason
Quarter	The fiscal period against which loyalty and financial data are scoped

Customer

The central entity. Attributes: `customer_id` (primary key), `age`, `gender`, `marital_status`, `senior_citizen`, `dependents`, `no_dependents`, `partner_status`, `referral_info`, `referred_by`. The `referred_by` attribute is a foreign key to `customer_id` within the same table, implementing the referral programme as a self-referencing relationship.

Location

Attributes: `location_id` (primary key), `address`, `latitude`, `longitude`, `population`, and area-level demographic aggregates. Related to Customer many-to-one.

Subscription

Attributes: `sub_id` (primary key), `customer_id` (foreign key), `contract_type`, `offer_details`, `payment_method`, `paperless_billing_option`. Holds the commercial terms of the relationship, separated from the services those terms cover.

Service

Attributes: `service_id` (primary key), `phone_service`, `internet_type`, `internet_service`, `multiple_lines`, `unlimited_data`, `streaming_tv`, `streaming_movies`, `streaming_music`, `online_security`, `online_backup`, `device_protection`, `device_protection_plan`, `tech_support`, `premium_tech_support`. Related to Customer many-to-many, resolved by a junction entity.

Behavior

Attributes: `behavior_id` (primary key), `customer_id` (foreign key), `avg_monthly_gb_download`, `long_distance_usage_metrics`. Isolated from Service because usage is measured continuously whereas subscription is a discrete state.

Billing

Attributes: `billing_id` (primary key), `customer_id` (foreign key), `total_charges`, `total_revenue`, `monthly_charges`, `total_extra_data_charges`, `total_long_distance_charges`, `avg_monthly_long_dist_charges`, `total_refunds`.

CustomerLoyalty

Attributes: `loyalty_id` (primary key), `customer_id`, `billing_id`, `quarter_id` (foreign keys), `cltv`, `tenure_months`, `number_of_referrals`, `referral_status`. Scoped to a fiscal quarter, permitting loyalty to be tracked over time should multi-period data become available.

ChurnReport

Attributes: `churn_id` (primary key), `churn_label`, `churn_value`, `churn_score`, `churn_reason`. Separated from Customer because the churn assessment is an analytical output that may be recomputed without altering the subscriber record — a separation that matters directly for machine-learning workflows, where labels are refreshed independently of features.

Quarter

Attributes: `quarter_id` (primary key), `quarter_name`, `start_date`, `end_date`, `fiscal_year`.

7.3 Relationship Catalogue

Relationship	From	Cardinality	To	Note
refer	Customer	M : M	Customer	Self-referencing referral programme
have	Customer	M : 1	Location	Subscriber's geographic position
have	Customer	M : 1	Behavior	Usage profile
subscription	Customer	M : 1	Subscription	Commercial terms
buy	Customer	M : M	Service	Resolved by <code>customer_service</code>
loyal	Customer	M : 1	CustomerLoyalty	Loyalty record
belong	Customer	M : M	ChurnReport	Resolved by <code>customer_churn_report</code>
contains	ChurnReport	1 : M	Behavior	Behaviours logged against a report
behave	ChurnReport	M : 1	Billing	Billing linked to churn assessment

Relationship	From	Cardinality	To	Note
payment	ChurnReport	M : 1	Billing	Payment context of the churn event
cost	CustomerLoyalty	M : 1	Billing	Financial context of the loyalty cycle
TimeAtYear	CustomerLoyalty	M : 1	Quarter	Fiscal scoping

7.4 Resolution of Many-to-Many Relationships

Two relationships are many-to-many and cannot be represented directly in a relational schema. Each is resolved by a junction entity carrying its own attributes.

`customer\_service` resolves Customer to Service, and carries `start_date` and `status`. These attributes belong to the *association* rather than to either participant: a subscription start date is a property of the subscriber's relationship to a service, not of the subscriber or the service alone.

`customer\_churn\_report` resolves Customer to ChurnReport, and carries `report_date`.

The self-referencing referral relationship on Customer is implemented not as a junction table but as a nullable foreign key `referred_by` on the entity itself, since the dataset records at most one referring subscriber per subscriber.

7.5 Foreign Key Summary

Foreign key	References
customer.referred_by	customer.customer_id
location.customer_id	customer.customer_id
subscription.customer_id	customer.customer_id
customer_service.customer_id	customer.customer_id
customer_service.service_id	service.service_id
behavior.customer_id	customer.customer_id
customer_churn_report.customer_id	customer.customer_id
customer_churn_report.churn_id	churn_report.churn_id
billing.customer_id	customer.customer_id
customer_loyalty.customer_id	customer.customer_id
customer_loyalty.billing_id	billing.billing_id

Foreign key

customer_loyalty.quarter_id

References

quarter.quarter_id

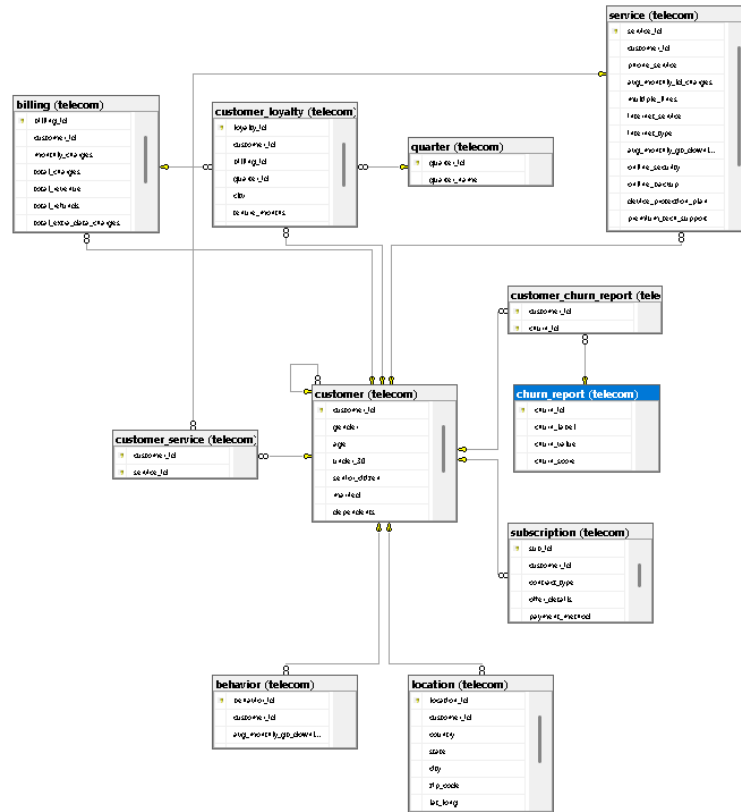


Figure 7.2 Logical Data Schema

8 Normalisation and Logical Design

8.1 Scope of This Chapter

Two normalisation exercises were performed, and this document distinguishes them because they operate on different inputs and reach different results.

The **first** takes the original twenty-one-column release and normalises it formally through First, Second, Third and Boyce-Codd Normal Forms, producing a nine-table schema. It is a rigorous demonstration on data whose dependency violations are genuine and instructive.

The **second** takes the extended release and produces the eleven-table `telecom` schema that was physically deployed and that the warehouse is built upon. It follows the same principles at a broader scope.

Sections 8.2 to 8.6 document the formal exercise. Section 8.7 relates its findings to the deployed schema.

8.2 First Normal Form

First Normal Form requires atomic values, a primary key, and the absence of repeating groups.

The source largely satisfies these conditions on inspection: every cell holds a single scalar, and `customerID` uniquely identifies every row. Two corrections were nonetheless applied. `SeniorCitizen` was stored as an integer while every comparable flag was stored as a Yes/No string; the encodings were unified to a bit type. The eleven blank values in `TotalCharges` were recognised as absent rather than zero.

Result: one table, one candidate key, all values atomic.

8.3 Second Normal Form

Second Normal Form requires First Normal Form and the absence of partial dependencies — every non-key attribute must depend on the whole primary key.

Because the primary key is a single column, partial dependency is impossible by construction: any dependency `customerID → X` is total. Formally, the relation is already in Second Normal Form.

Second Normal Form is nonetheless the appropriate stage at which to extract lookup relations for low-cardinality categorical attributes whose values repeat across thousands of rows. Storing the literal string 'Fiber optic' in over three thousand rows is redundant and renders any correction to that value an update against every affected row.

Lookup relation	Attributes	Distinct values
<code>dim_Gender</code>	<code>GenderID</code> , <code>GenderName</code>	2

Lookup relation	Attributes	Distinct values
dim_InternetServiceType	InternetServiceTypeID, ServiceTypeName	2
dim_ContractType	ContractTypeID, ContractTypeName	3
dim_PaymentMethod	PaymentMethodID, MethodName	4

Each lookup relation has two candidate keys — the surrogate identifier and the name — each of which determines the other. Both determinants are candidate keys, so each relation is already in Boyce-Codd Normal Form and requires no further attention.

8.4 Third Normal Form

Third Normal Form requires Second Normal Form and the absence of transitive dependencies: no non-key attribute may depend on the key only through another non-key attribute.

Two transitive dependencies were identified, and both arise from the sentinel values noted in Section 6.3.

Dependency 1. When `PhoneService = 'No'`, `MultipleLines` invariably holds `'No phone service'`. The value of `MultipleLines` is therefore determined by `PhoneService`, not independently by the subscriber:

```
customerID -> PhoneService -> MultipleLines
```

Dependency 2. When `InternetService = 'No'`, all six internet add-on attributes invariably hold `'No internet service'`:

```
customerID -> InternetService -> { OnlineSecurity, OnlineBackup,
                                   DeviceProtection, TechSupport,
                                   StreamingTV, StreamingMovies }
```

These are not clerical irregularities. A string sentinel occupying a data column is a dependency expressed as content rather than as structure, and it is invisible to any tool that inspects schema rather than values.

Resolution. Phone service and its dependent attribute were extracted to `CustomerPhoneService`; internet service and its six dependent attributes to `CustomerInternetService`. In both, the sentinel string was replaced by `NULL`, and a `CHECK` constraint was declared to enforce the semantic contract — that the dependent attributes may be null only when the determining attribute indicates absence of the service. The dependency thereby moves from data into structure, where the database can enforce it.

The remaining attributes were separated by subject: demographics to `Customer`, contract and charges to `CustomerBilling`, and the churn label to `CustomerChurn`. The last separation is more than tidiness — the churn label is an analytical outcome that is recomputed on a different cycle from the subscriber's service data, and coupling the two would force a rewrite of service records whenever labels were refreshed.

8.5 Boyce-Codd Normal Form

Boyce-Codd Normal Form requires that every determinant be a candidate key. It is stricter than Third Normal Form and detects anomalies in relations having overlapping candidate keys.

Relation	Candidate keys	Violation
<code>dim_Gender</code>	{GenderID}, {GenderName}	None — both determinants are candidate keys
<code>dim_InternetServiceType</code>	{ID}, {Name}	None
<code>dim_ContractType</code>	{ID}, {Name}	None
<code>dim_PaymentMethod</code>	{ID}, {Name}	None
<code>Customer</code>	{CustomerID}	None
<code>CustomerPhoneService</code>	{CustomerID}	None
<code>CustomerInternetService</code>	{CustomerID}	None
<code>CustomerBilling</code>	{CustomerID}	None
<code>CustomerChurn</code>	{CustomerID}	None

All relations satisfy Boyce-Codd Normal Form. Notably, it was the elimination of the sentinel-value dependencies that achieved this: a sentinel acts as an implicit determinant *within* a column, which is a Boyce-Codd violation presenting in the guise of a Third Normal Form problem.

Fourth and Fifth Normal Forms address multi-valued and join dependencies respectively. Neither was present — every attribute describes a single fact about one subscriber — and further decomposition would fragment the schema without benefit.

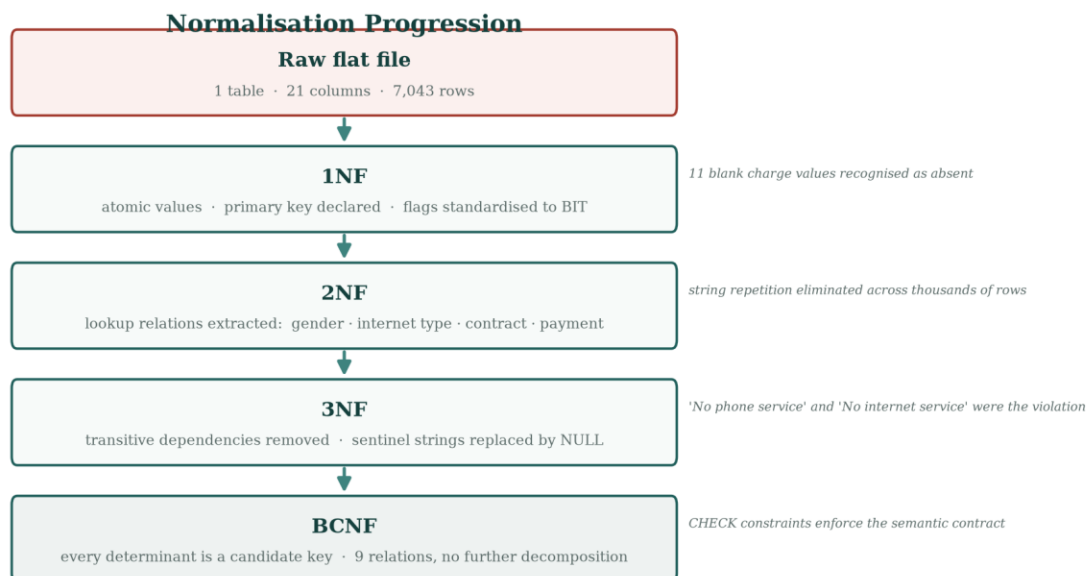


Figure 8.1 Normalisation Progression from 1NF to BCNF

8.6 Design Decisions

Natural key retained for Customer. `customerID` is already unique, opaque, and free of personally identifying content. A surrogate integer would add a mapping layer without adding normalisation benefit.

Null in preference to sentinel. Three reasons: it eliminates the transitive dependency; it is the standard relational representation of *not applicable*; and aggregate functions and modelling libraries treat it correctly without special-casing a magic string.

Satellite tables in preference to one wide table. Decomposition into five relations increases join count on retrieval. It was accepted because each satellite can be updated without locking the full subscriber record, the churn label is decoupled from service data, and nullable attributes are confined to the relations where they are meaningful rather than spread across the core relation. A view restores the wide representation for ad-hoc use.

Narrow integer types for lookup keys. Lookup relations hold at most four rows; a one-byte integer suffices and reduces foreign key storage across seven thousand subscriber rows.

8.7 Relation to the Deployed Schema

The deployed `telecom` schema comprises eleven tables rather than nine, because it was derived from the extended release, which carries geographic, financial, loyalty, usage and fiscal-period attributes absent from the original.

The principles are identical. Each table addresses one subject. Many-to-many relationships are resolved by junction tables. Foreign keys enforce referential integrity throughout. The correspondence is as follows.

Formal exercise (9 tables)	Deployed schema (11 tables)
Customer	telecom.customer
CustomerPhoneService, CustomerInternetService	telecom.service, telecom.customer_service
CustomerBilling	telecom.billing, telecom.subscription
CustomerChurn	telecom.churn_report, telecom.customer_churn_report
dim_* lookups	Attributes retained inline; equivalent lookups materialised in the olap layer
—	telecom.location (new subject)
—	telecom.behavior (new subject)
—	telecom.customer_loyalty (new subject)
—	telecom.quarter (new subject)

The deployed schema retains contract type and payment method as attributes of `subscription` rather than extracting them to lookup relations. This is a defensible relaxation: the values are stable, the storage saving at 7,043 rows is immaterial, and the lookup structures they would have formed are in any case materialised as dimension tables in the analytical layer, which is where they are actually queried.

9 Operational Database Implementation

9.1 Implementation Approach

The operational schema was implemented as a scripted deployment rather than through interactive design tools, so that the database can be recreated deterministically from source control. Three scripts constitute the deployment: schema creation, data insertion, and foreign key application.

Database: `telecom_churn_db`. Schema: `telecom`. Target platform: Microsoft SQL Server.

Foreign key constraints are applied **after** the data load rather than at table creation. This is deliberate: applying constraints during a bulk load forces per-row validation and imposes an insertion order that the generated script would otherwise not need to respect. Applying them afterwards validates the entire dataset in one pass and, if validation fails, fails loudly rather than silently accepting a partial load.

9.2 Table Inventory

#	Table	Subject	Grain
1	<code>telecom.customer</code>	Subscriber demographics	One row per subscriber
2	<code>telecom.location</code>	Geography	One row per location
3	<code>telecom.subscription</code>	Contract and payment terms	One row per subscription
4	<code>telecom.service</code>	Service and add-on catalogue	One row per service record
5	<code>telecom.customer_service</code>	Junction: subscriber to service	One row per association
6	<code>telecom.behavior</code>	Usage measurements	One row per subscriber
7	<code>telecom.billing</code>	Financial record	One row per subscriber
8	<code>telecom.customer_loyalty</code>	Tenure, lifetime value, referrals	One row per subscriber per quarter
9	<code>telecom.churn_report</code>	Churn assessment	One row per churn report
10	<code>telecom.customer_churn_report</code>	Junction: subscriber to churn report	One row per association
11	<code>telecom.quarter</code>	Fiscal calendar	One row per quarter

9.3 Representative Table Specification

`telecom.customer` is presented in full as representative of the implementation conventions applied throughout. The complete specification of all eleven tables appears in Appendix B.

Column	Type	Constraint	Description
<code>customer_id</code>	VARCHAR(20)	PRIMARY KEY, NOT NULL	Unique subscriber identifier
<code>age</code>	INT	NOT NULL	Subscriber age in years
<code>gender</code>	VARCHAR(10)	NOT NULL	Male, Female, Other
<code>marital_status</code>	VARCHAR(20)		Married, Single, Divorced
<code>senior_citizen</code>	BIT	DEFAULT 0	Senior citizen indicator
<code>dependents</code>	INT	DEFAULT 0	Number of dependents
<code>no_dependents</code>	BIT		Convenience flag
<code>partner_status</code>	VARCHAR(20)		Partner indicator
<code>referral_info</code>	VARCHAR(255)		Referral source detail
<code>referred_by</code>	VARCHAR(20)	FOREIGN KEY → <code>customer(customer_id)</code>	Self-referencing referral

9.4 Implementation Conventions

Naming. All identifiers use lower `snake_case`. Table names are singular where the row represents one instance of the entity. Junction tables are named by concatenating their participants.

Types. Identifiers are `VARCHAR(20)`; monetary quantities are `DECIMAL` with explicit precision and scale rather than a floating-point type, so that summation is exact and reporting figures reconcile without rounding drift; flags are `BIT`; counts are `INT`.

Constraints. Every table declares a primary key. Every logical relationship is declared as a foreign key rather than being maintained by convention. Junction tables use composite primary keys formed from their two foreign keys, which enforces uniqueness of the association without an additional surrogate.

9.5 Load Verification

The deployment is verified by a series of assertions executed after load: that each table's row count matches the count expected from source; that no orphan foreign key values exist; that no unintended nulls are present in non-nullable columns; and that `telecom.customer` contains exactly 7,043 rows.

9.6 Stored Procedures

Data management against the operational schema is performed through stored procedures rather than through ad-hoc statements issued by the application. Four procedures cover the full lifecycle of a record.

Procedure	Operation
Select_SP	Parameterised retrieval across customer accounts and service subscriptions
Insert_SP	Record creation with constraint validation
Update_SP	Field-level modification of existing records
Delete_SP	Removal with referential integrity enforced

Three properties follow from routing data access through procedures rather than through statements composed in the application.

The query plan is reused. A procedure is compiled once and its plan cached, where an ad-hoc statement may be recompiled on every execution.

The attack surface narrows. Parameters are bound rather than concatenated, which removes the injection vector that string-built SQL creates.

The contract is stable. The application depends on a procedure signature rather than on a table structure, so a column may be renamed or a table decomposed without the application changing — a property that matters directly here, since the schema passed through several normalisation stages before reaching its final form.

The procedures also supply the datasets behind the six paginated reports documented in Section 17.10, which is why those reports accept parameters at execution time.

9.7 User-Defined Functions

Three scalar functions place business logic inside the database, so that a definition is written once and every consumer — procedure, report, view or dimensional load — evaluates the same rule.

Function	Purpose
CLTV Customer Classifier	Segments the subscriber base by lifetime value, producing the high, medium and low value bands used in Part B of the reporting suite

Function	Purpose
Churn Risk Level	Derives a categorical risk level from tenure and contract type — the deterministic rule required by the project proposal, distinct from the probabilistic model of Chapter 15
Financial Loyalty	Combines accumulated revenue with relationship length into a single loyalty indicator

The placement decision is worth stating because the alternative was available and was rejected. Each of these rules could have been expressed as a DAX measure in the reporting layer or as a derived column in Python. Holding them in the database instead means that a subscriber classified as high value by a report, by a stored procedure and by the warehouse load is classified identically in all three, because all three call the same function. Logic duplicated across layers diverges; logic held in one layer cannot.

`Churn Risk Level` must not be confused with the model output. It is a rule-based classification derived from two attributes. The probability produced by the classifier in Chapter 15 is a separate quantity computed from the full feature set, and the two are reported separately throughout.

10 ETL Pipeline

10.1 Pipeline Overview

The pipeline moves data from flat source files to the analytical star schema in two distinct stages, implemented in two different technologies for two different reasons.

Stage one — source to operational. Implemented in Python. The source files require profiling, type reconciliation, sentinel resolution and referential assembly, all of which are more naturally expressed in a data-manipulation library than in SQL.

Stage two — operational to analytical. Implemented in Transact-SQL. Both source and target reside in the same database, and the transformation is fundamentally a set of joins and projections, which is what SQL exists to express.

The two stages, the artefact that passes between them and the order in which the target tables are populated are shown in Figure 10.1.

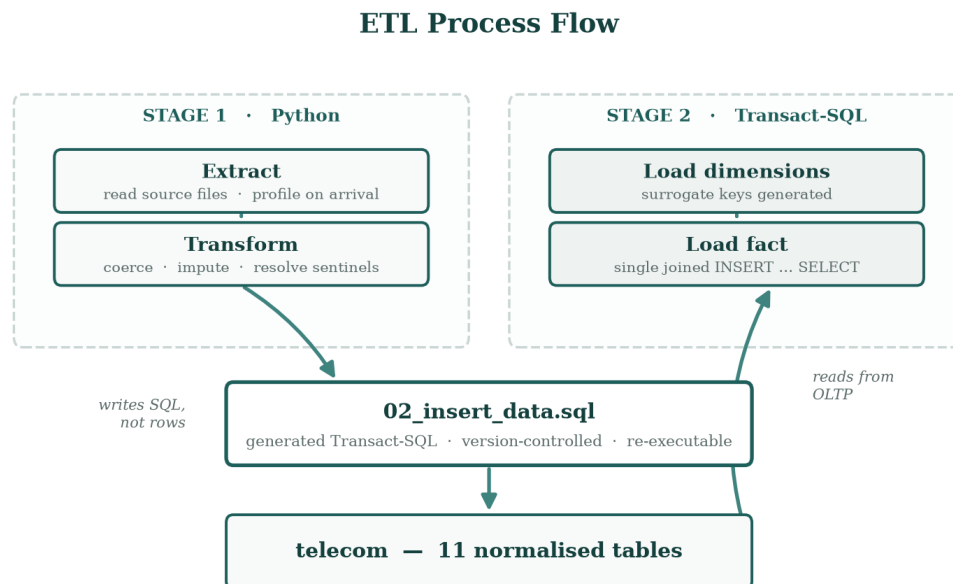


Figure 10.1 ETL Process Flow

10.2 Extraction

Extraction reads the delimited source files into memory. Each file is profiled on arrival: row count, column count, declared type against inferred type, distinct value count per column, and null count per column. The

profile is compared against expectation, and a discrepancy halts the pipeline rather than propagating silently.

The extended release is used in preference to the original, because it carries the financial and geographic attributes on which the revenue analysis depends. The five subject files are joined on the subscriber identifier to reconstruct the full subscriber record.

10.3 Transformation and Script Generation

The transformation stage performs the operations documented in Section 6.5 — type coercion, imputation of the eleven zero-tenure charge values, standardisation of flag encodings, and resolution of the sentinel values into structural nulls — and then performs an operation that distinguishes this pipeline from a conventional one.

Rather than writing to the database, it writes SQL. The notebook emits `02_insert_data.sql`, a Transact-SQL script of approximately five megabytes containing the insert statements that populate all eleven operational tables.

This design merits explanation, since a direct database connection would be the more obvious choice.

The load becomes an artefact. A script is a file: it can be committed, reviewed, diffed against a previous version, and executed by someone who does not have the Python environment installed. A connection from a notebook leaves no trace of what was inserted.

The load becomes separable from its generation. The database administrator who executes the load need not run, trust or understand the notebook.

The load becomes reproducible. Executing the same script twice produces the same database state. Re-running a notebook against a live connection may not, if the notebook's state has changed between runs.

Failure becomes legible. A constraint violation reports the statement that caused it, which can be located directly in the script.

The cost of the approach is script size and the absence of incremental loading. At the volume concerned, neither is material.

10.4 Loading the Operational Schema

The generated script is executed against `telecom_churn_db`. Insertion order respects the dependency graph: parent tables before child tables, and both before the junction tables that reference them. Foreign key constraints are applied after the load completes, for the reasons given in Section 9.1.

10.5 Loading the Analytical Schema

The warehouse load is executed entirely within the database by `new_OLAP.sql`. It performs, in order:

1. Creation of the `olap` schema if absent.
2. Drop and recreation of all dimension and fact tables, making the load idempotent — the script may be re-executed without manual cleanup.
3. Population of each dimension by `INSERT ... SELECT` from the corresponding operational tables, with derived attributes computed inline.
4. Population of `fact_churn` by a single joined `INSERT ... SELECT` that resolves every dimension surrogate key.
5. Creation of non-clustered indexes on the foreign key columns of the fact table.

Order is not incidental. Dimensions must exist and be populated before the fact table can resolve their surrogate keys; the fact load is therefore necessarily last.

The derived attributes computed during dimension load — `age_band`, `tenure_band`, `tenure_band_sort`, `service_count`, and the conversion of bit flags to 'Yes'/'No' labels — are computed here rather than in the reporting tool. Computing them once at load time in preference to on every query is the central economy of the dimensional approach.

10.6 Why Not a Dedicated Integration Service

The Microsoft stack provides SQL Server Integration Services for exactly the task this chapter describes, and its absence here is a deliberate choice rather than an omission. The reasoning is recorded because the alternative is the conventional one and its rejection requires justification.

Proportionality. Integration Services earns its cost through scheduling, restartable checkpoints, incremental extraction, parallel data flows and connectivity to heterogeneous sources. This pipeline loads 7,043 rows from flat files into a single database instance, and completes in seconds. None of those capabilities is exercised. Introducing the service would add a deployment, a runtime dependency and a package format to maintain, in exchange for facilities the workload does not use.

Reproducibility. The chosen approach produces a plain Transact-SQL script as its intermediate artefact. That script is text: it can be committed to version control, reviewed line by line, diffed against a previous version, and executed by anyone with a database client. An Integration Services package is a binary-adjacent XML file whose meaning is legible only inside its designer, and whose diffs are not reviewable in any practical sense. For a project whose central claim is that every figure is traceable to its source, an auditable load script is worth more than a graphical one.

Portability. The pipeline as built requires Python and a SQL Server connection. It does not require the Integration Services runtime to be installed, licensed or configured, which matters for a system that must be reproducible on a marking machine rather than only on the machine that built it.

Where the transformation actually belongs. The operational-to-analytical step is a set of joins and projections between two schemas of the same database. Expressing that in `INSERT ... SELECT` places the work in the engine that holds the data. Routing it through an external service would extract rows from SQL Server only to write them back, which is slower and adds a failure mode.

The decision is scale-dependent and would reverse. At multi-source, multi-million-row, scheduled-incremental volumes, the case for a managed integration service becomes decisive. This is recorded in Section 20.4 as the trigger condition for revisiting the choice.

10.7 Data Quality Controls

Control	Stage	Purpose
Source profiling	Extraction	Detect structural change in source files
Type coercion validation	Transformation	Ensure numeric columns contain numbers
Sentinel resolution	Transformation	Convert encoded absence to structural null
Referential assembly	Transformation	Ensure every foreign key has a parent
Row count reconciliation	Load	Source count equals target count
Orphan key check	Post-load	No fact row references a missing dimension
Revenue identity check	Post-load	See Section 11.9

11 Data Warehouse Design and Implementation

11.1 Schema Type Decision

Three dimensional topologies were considered. The star schema was selected.

Star against snowflake

Criterion	Star	Snowflake
Query complexity	One join per dimension	Multi-join chains through normalised dimensions
Reporting engine performance	Excellent — columnar engines favour flat dimensions	Slower — more relationships to traverse
Measure expression complexity	Low — written against flat tables	Higher — requires relationship navigation
Storage overhead	Slight denormalisation, of the order of megabytes at this volume	Marginal saving at this volume
Maintenance	Simple	Dimension hierarchies to maintain
Data volume	7,043 rows	Benefit appears at hundreds of millions of rows

At 7,043 subscribers, the storage cost of denormalisation is well under ten megabytes. The query simplicity, engine compatibility and measure readability obtained in exchange decisively outweigh it. A snowflake schema would add join depth for no measurable return at this scale.

Star against galaxy

A galaxy schema — multiple fact tables sharing conformed dimensions — was considered and rejected. Every analytical question in the domain converges on a single grain: the customer and the churn event attached to them. A second fact table would be justified if individual invoice lines or support interactions were being tracked, but the source aggregates financial data to the subscriber, making it a natural measure group within one fact table rather than a separate fact.

The decision is not permanent. The dimensions were designed as conformed precisely so that a second fact table can be added later without redesign.

11.2 Grain

The grain of `fact_churn` is **one row per subscriber**, representing the complete observable state of that subscriber across their relationship with the operator.

This is a **snapshot** fact table, not a transaction fact table and not an accumulating snapshot. The distinction matters. A transaction fact records an event at a point in time; an accumulating snapshot records a process with a defined lifecycle and updates as the process advances. A periodic snapshot records state as at a moment. The source is a point-in-time assessment of a subscriber population, which is precisely a snapshot.

Grain is declared before any other design decision because every subsequent decision depends on it. A measure may be placed in the fact table only if it is additive at the declared grain, and a dimension may be joined only if it is single-valued at that grain.

11.3 Slowly Changing Dimensions

All dimensions are implemented as **Type 1** — an incoming change overwrites the existing value and no history is retained.

The justification is that the source contains no history to retain. It is a static snapshot; there is no second observation of any subscriber against which a change could be detected. Implementing Type 2 handling would add versioning columns, effective-date ranges and current-row flags to serve a requirement that the data cannot express.

Were the system extended to receive periodic feeds, `dim_customer` and `dim_subscription` would be the natural Type 2 candidates, since contract type and demographic attributes are the attributes whose change carries analytical meaning.

11.4 Key Strategy

Surrogate keys. Every dimension takes an integer identity column as its primary key. Three benefits follow. Integer joins outperform string joins, and the natural key here is a twenty-character string. Type 2 handling, should it later be required, needs a key that can repeat for the same business entity, which a natural key cannot. And reporting engines index and compress integer relationships more efficiently.

Natural keys retained. Business keys such as `customer_id` are carried on the dimension as alternate keys. They are not used for joining but preserve traceability from any warehouse row back to its operational source — indispensable when a reported figure must be reconciled.

Conformed dimensions. `dim_customer`, `dim_location` and `dim_date` are designed to be reusable without modification by any future fact table.

Degenerate dimensions. `churn_reason` and `churn_label` are retained in the fact table rather than extracted. Both have low cardinality and are used as filter attributes; a separate dimension would add a join to retrieve an attribute already present.

11.5 Schema Structure

The completed structure is shown in Figure 11.1. Each dimension joins the fact table on a single surrogate key, and no dimension joins another — the defining property of the star.

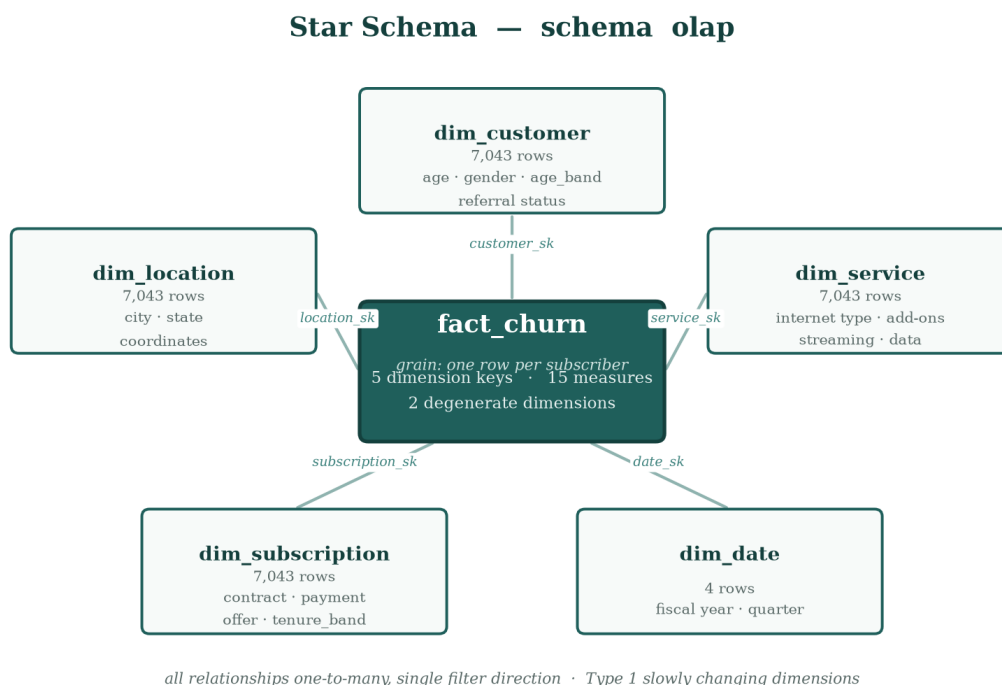


Figure 11.1 Star Schema — Five Dimensions and `fact_churn`

11.6 Dimension Tables

Dimension	Grain	Rows	Operational source
<code>olap.dim_customer</code>	One per subscriber	7,043	<code>telecom.customer</code>
<code>olap.dim_location</code>	One per location	7,043	<code>telecom.location</code>
<code>olap.dim_service</code>	One per service record	7,043	<code>telecom.service</code>
<code>olap.dim_subscription</code>	One per subscription	7,043	<code>telecom.subscription</code>
<code>olap.dim_date</code>	One per fiscal quarter	4	<code>telecom.quarter</code>

`dim\_customer` carries customer_sk, customer_id, age, gender, marital status, dependents, senior citizen indicator, age_band and referral status.

`dim\_subscription` carries subscription_sk, contract type, payment method, paperless billing indicator, offer, tenure_band and tenure_band_sort.

`dim\_service` carries service_sk, phone service, internet service, internet_type, multiple lines, online security, online backup, device protection, technical support, premium_tech_support, streaming television, streaming films and unlimited_data.

`dim\_location` carries location_sk, city, state, country, latitude, longitude and postal code.

`dim\_date` carries date_sk, fiscal_year, quarter identifier, quarter name and quarter number.

The tenure_band_sort column deserves comment. A band expressed as text sorts alphabetically, which places '13-24' before '1-12'. An accompanying integer sort column resolves this without requiring the reporting tool to encode the ordering.

11.7 Fact Table

olap.fact_churn holds five dimension foreign keys — customer_sk, location_sk, service_sk, subscription_sk, date_sk — two degenerate dimensions, and fifteen measures.

11.8 Measure Catalogue

Measure	Type	Description	Analytical use
churn_value	INT	1 = departed, 0 = retained	Churn rate numerator
churn_score	INT	Supplied risk score, 0–100	Risk distribution
cltv	INT	Customer lifetime value	Value segmentation
tenure_months	INT	Months as subscriber	Cohort analysis
number_of_referrals	INT	Referrals made	Loyalty indicator
monthly_charges	DECIMAL	Recurring monthly charge	ARPU, revenue at risk
total_charges	DECIMAL	Cumulative charges	Lifetime value proxy
total_revenue	DECIMAL	Total revenue recognised	Revenue reporting
total_refunds	DECIMAL	Refunds issued	Service quality indicator
total_extra_data_charges	DECIMAL	Data overage charges	Usage revenue
total_long_distance_charges	DECIMAL	Long-distance charges	Usage revenue

Measure	Type	Description	Analytical use
avg_monthly_gb_download	INT	Mean monthly data volume	Usage segmentation
avg_monthly_ld_charges	DECIMAL	Mean monthly long-distance charge	Usage indicator
service_count	INT	Count of active services	Bundle and upsell analysis
customer_count	INT	Constant 1	Additive subscriber count

The `customer_count` column is a conventional device. Storing a literal 1 permits subscriber counts to be expressed as a sum, which is additive across every dimension, rather than as a distinct count, which is not and which performs poorly.

Measures are stored where they are additive and derived where they are not. Ratios — churn rate, average revenue per user, refund rate — are computed in the reporting layer, because a ratio of sums is not the sum of ratios and storing a pre-computed average would produce incorrect results whenever it was aggregated.

11.9 Revenue Identity and Its Validation

Inspection of the fact table established that total revenue is not an independent measure but a composition of the others:

```
total_revenue = total_charges
               + total_long_distance_charges
               + total_extra_data_charges
               - total_refunds
```

Where a subscriber has no long-distance activity, no data overage and no refund, `total_revenue` equals `total_charges` exactly; where any of these is non-zero, the difference is accounted for by the identity.

This is a materially useful finding, and it was converted into a standing validation control. Two measures are defined in the semantic layer:

```
Gross Revenue Before Refunds =
  [Total Charges]
  + [Total Long Distance Charges]
  + [Total Extra Data Charges]

Revenue Formula Check =
  [Gross Revenue Before Refunds]
  - [Total Refunds]
  - [Total Revenue]
```

Revenue Formula Check must evaluate to zero, or to a value within decimal rounding tolerance of zero, across every combination of filters. Any departure indicates a load defect, a transformation error or a source inconsistency. It is a continuous assertion on the correctness of the warehouse rather than a one-off test.

Verification against the deployed model

The control was evaluated in the reporting layer and the identity holds **exactly**, both in aggregate and within every contract type.

Component	Value	Share of gross
Base charges	16,060,725.24	75.10 %
Long-distance revenue	5,275,906.10	24.67 %
Extra-data revenue	48,320.00	0.23 %
Gross revenue before refunds	21,384,951.34	100.00 %
Less refunds	(13,819.65)	0.06 %
Total revenue	21,371,131.69	

Disaggregated by contract, the residual between gross and recognised revenue reproduces the refund figure reported independently on the leakage page:

Contract	Gross	Recognised revenue	Implied refunds	Reported leakage
Month-to-month	6,168,249.81	6,162,488.22	5,761.59	5.8 K
One year	6,175,084.64	6,171,794.31	3,290.33	3.3 K
Two year	9,041,616.89	9,036,849.16	4,767.73	4.8 K
Total	21,384,951.34	21,371,131.69	13,819.65	13.82 K

Two measures computed by different routes — one as a residual of the composition, the other as a direct sum of the refund column — agree to the cent. Revenue Formula Check returns 0.00. The warehouse load is arithmetically sound.

11.10 Analytical Optimisations

Optimisation	Rationale
Bit flags converted to 'Yes'/'No' text	Renders directly as slicer values without a calculated column
age_band materialised alongside raw age	Demographic segmentation without runtime binning

Optimisation	Rationale
tenure_band with an integer sort companion	Cohort analysis with correct ordering
service_count pre-computed	Common bundle metric, avoids repeated conditional summation
Descriptive label column on every dimension	Suitable for direct use on chart axes and legends
Non-clustered indexes on fact foreign keys	Supports the join pattern the reporting tool issues

11.11 Relationship Configuration

From	To	Cardinality	Filter direction
dim_customer[customer_sk]	fact_churn[customer_sk]	One to many	Single, dimension to fact
dim_location[location_sk]	fact_churn[location_sk]	One to many	Single
dim_service[service_sk]	fact_churn[service_sk]	One to many	Single
dim_subscription[subscription_sk]	fact_churn[subscription_sk]	One to many	Single
dim_date[date_sk]	fact_churn[date_sk]	One to many	Single

Filtering is single-directional in every case. Bidirectional filtering was avoided because it introduces ambiguity in the presence of multiple relationship paths and degrades query performance without being required by any analysis performed here.

11.12 A Structural Constraint

dim_date contains four rows, and the loaded subscriber data occupies **one fiscal quarter**.

This is stated plainly because it constrains the analysis materially. No genuine time-series analysis is possible: there is no second period against which to compute a trend, no seasonality to detect, and no basis for temporal forecasting. Any figure presented as a projection in this system is an extrapolation from internal ratios under stated assumptions, not a forecast derived from historical behaviour, and is labelled as such wherever it appears.

The dimension is retained despite carrying no analytical variation, because it is structurally correct and because the addition of further periods would require no redesign.

12 Testing and Validation

12.1 Approach

Validation was applied at each interface of the pipeline rather than only at its end, on the principle that a defect is cheapest to locate at the stage that introduced it.

12.2 Source Data Testing

Test	Expectation
Row count of source extract	7,043
Uniqueness of customerID	No duplicates
Column count and names	Match documented schema
Numeric columns parse as numeric	After coercion, no failures
Null audit	Only the eleven known blank charge values

12.3 Operational Database Testing

Test	Expectation
Row count per table	Matches source-derived expectation
Primary key uniqueness	Enforced by constraint
Foreign key integrity	No orphan references after constraint application
Not-null enforcement	No unintended nulls
Junction table cardinality	No duplicate associations

12.4 Warehouse Testing

Test	Expectation
fact_churn row count	7,043
Dimension row counts	7,043, 7,043, 7,043, 7,043 and 4 respectively
Surrogate key uniqueness	One row per key in each dimension
Orphan fact rows	None — every dimension key resolves

Test	Expectation
Derived band coverage	Every subscriber assigned a band; no nulls
Revenue identity	Revenue Formula Check evaluates to zero

12.5 Semantic Layer Testing

Reported figures were reconciled against direct SQL queries on the same data. A measure that does not reproduce the result of its equivalent query indicates an error in relationship configuration, filter context or measure definition. This reconciliation was performed for the principal financial measures and for the subscriber count.

12.6 Model Testing

Model evaluation was performed on a held-out partition that took no part in training and to which no synthetic observation was added. The evaluation procedure is documented in Section 14.6.

13 Exploratory Data Analysis

13.1 Objectives

Exploratory analysis was undertaken to characterise the subscriber population, to establish which attributes separate departed from retained subscribers, to test whether apparent separations are statistically supportable, and to identify redundancy among attributes before modelling.

13.2 Environment and Data Preparation

The analysis was conducted in Jupyter with pandas 2.3.3 and NumPy 2.4.2, using Matplotlib and seaborn for graphics, SciPy for hypothesis testing and statsmodels for variance inflation factors.

Preparation confirmed the dataset shape as **7,043 rows by 21 columns**, coerced `TotalCharges` from text to numeric, standardised flag encodings, verified the absence of duplicate rows, and confirmed that no nulls remained after imputation of the eleven zero-tenure charge values.

13.3 Target Variable Analysis

The distribution of the churn label was examined as both a count and a proportion. The dataset is imbalanced, with retained subscribers outnumbering departed ones — the condition that governs the modelling strategy in Chapter 14 and that makes accuracy an unsuitable primary metric.

The reporting layer gives the observed split as **5,174 retained (73.46 per cent)** against **1,869 departed (26.54 per cent)** — a ratio of roughly 2.8 to 1. The imbalance is pronounced but not extreme; it is severe enough to make accuracy misleading and to warrant resampling, and mild enough that the minority class is well populated at 1,869 observations.

13.4 Univariate Analysis — Numerical Attributes

Three numerical attributes were examined: `tenure`, `MonthlyCharges` and `TotalCharges`.

For each, the analysis produced a histogram with a kernel density estimate to characterise the distribution's shape; a box plot split by churn status to expose differences in location and spread; a quantile-quantile plot to assess departure from normality, which determines whether parametric or non-parametric tests are appropriate; and descriptive statistics computed separately for departed and retained subscribers.

Outliers were identified by the interquartile range rule. Values outside the interval bounded by the first quartile less one and a half times the interquartile range, and the third quartile plus the same quantity, were flagged. Flagged values were **retained** rather than removed: a subscriber with an unusually high monthly

charge is a genuine high-value subscriber, not a recording error, and removing such observations would discard exactly the population the retention analysis most needs to consider.

13.5 Univariate Analysis — Categorical Attributes

The categorical attributes were plotted as a grid of frequency distributions, establishing the composition of the subscriber base across gender, partner and dependent status, service subscriptions, contract type, billing preference and payment method. The purpose is to establish base rates: a category comprising two per cent of subscribers cannot account for a large share of churn regardless of its internal churn rate, and interpreting a rate without its base is a common source of error.

13.6 Bivariate Analysis Against Churn

Each categorical attribute was assessed by computing the churn rate within each of its levels and comparing that rate against the overall baseline, which was drawn on each chart as a reference line. This presentation answers the operative question directly: does membership of this category raise or lower the probability of departure relative to the population as a whole?

Numerical attributes were assessed by overlaying kernel density estimates for departed and retained subscribers on common axes, which exposes differences in distribution rather than merely in mean.

Two focused analyses were performed. **Tenure cohort analysis** grouped subscribers by length of relationship and computed the churn rate within each cohort, addressing whether attrition risk is concentrated early in the relationship. **Contract type deep-dive** examined the composition of the subscriber base by contract type alongside the distribution of monthly charges within each type, addressing whether contract type is confounded with price.

13.7 Statistical Significance Testing

Visual separation is not evidence of a real difference. Each numerical attribute was therefore subjected to a **Mann-Whitney U test** comparing its distribution among departed subscribers against its distribution among retained subscribers.

The Mann-Whitney U test was selected in preference to Student's t-test because the quantile-quantile plots established that the attributes are not normally distributed. The Mann-Whitney U test is non-parametric: it makes no distributional assumption and tests whether one distribution is stochastically greater than the other.

13.8 Correlation and Multicollinearity

Categorical attributes were encoded numerically and a full correlation matrix computed and rendered as a heatmap. Attributes were then ranked by their correlation with the churn indicator to produce a preliminary ordering of candidate drivers.

Correlation with the target identifies candidate predictors; correlation *among* predictors identifies redundancy. **Variance inflation factors** were therefore computed for each attribute. The variance inflation factor measures how far the variance of a regression coefficient is inflated by linear dependence on the other predictors; a high value indicates that an attribute carries little information not already present elsewhere. This matters for the interpretability of linear models, where multicollinearity destabilises coefficients and makes their signs unreliable, and it informs the feature selection performed in Chapter 14.

13.9 Findings

14 Feature Engineering

14.1 Objectives

Feature engineering converts the cleaned dataset into a matrix suitable for supervised learning, and introduces derived attributes that express domain knowledge the raw columns do not.

14.2 Cleaning and Standardisation

Three operations preceded feature construction.

`TotalCharges` was imputed for the eleven zero-tenure subscribers as one month of `MonthlyCharges`, for the reason given in Section 6.5.

The sentinel values 'No phone service' and 'No internet service' were collapsed to 'No'. This differs from the treatment applied in the operational schema, where they became structural nulls, and the difference is deliberate: a relational schema benefits from distinguishing *not applicable* from *declined*, whereas a feature matrix requires a value, and for modelling purposes a subscriber without internet and a subscriber who declined an internet add-on are equivalent — neither has the service.

`customerID` was dropped. An identifier carries no predictive information and, being unique, would permit a sufficiently flexible model to memorise the training set.

14.3 Engineered Features

Eight features were constructed, each expressing a hypothesis about subscriber behaviour.

Feature	Definition	Hypothesis
<code>tenure_group</code>	Tenure banded into intervals	Behaviour differs qualitatively between early and mature relationships; the effect of tenure is not linear
Charge per month ratio	$\text{TotalCharges} / (\text{tenure} + 1)$	Mean spend across the relationship; the increment avoids division by zero at zero tenure
Charge gap	<code>MonthlyCharges</code> less the historical mean	A positive gap indicates the subscriber currently pays above their own historical average — a recent increase, which may precipitate departure

Feature	Definition	Hypothesis
service_count	Count of active add-on services	Engagement proxy; a subscriber with many services faces higher switching cost
Paperless and electronic cheque flag	Conjunction of both conditions	The combination is associated in the literature with elevated attrition; the interaction may carry information neither component carries alone
Month-to-month flag	Contract type indicator	Isolates the highest-risk contract as a binary term
Fibre optic flag	Internet type indicator	Higher charges and, in this dataset, higher attrition
No internet flag	Internet absence indicator	Minimal service, minimal charge, distinctive behaviour

Each was validated by computing the churn rate across its levels before it was admitted to the feature set. A derived feature that does not separate the classes adds dimensionality without information.

14.4 Categorical Encoding

Encoding was selected per attribute according to its measurement scale, rather than applying one scheme uniformly.

Attribute class	Scheme	Justification
Binary (Yes/No, Male/Female)	Direct mapping to 0/1	Two levels require one column; no information is added by more
Ordinal (Contract)	Ordinal encoding	Month-to-month, one year and two year stand in genuine order; the encoding preserves it
Nominal (InternetService, PaymentMethod)	One-hot with first level dropped	No natural order exists; integer coding would impose a false one. Dropping one level avoids the dummy variable trap

Encoding was verified by confirming that no object-typed columns and no nulls remained in the resulting matrix.

14.5 Scaling

`StandardScaler` was applied to the continuous attributes only. Binary indicators and ordinal integers were left unscaled: they are already on comparable scales, and standardising a zero-one indicator distorts its interpretation without benefit.

The fitted scaler was persisted. This is a requirement rather than a convenience — an observation presented to the trained model at inference time must be transformed by the *same* scaler, fitted on the *same* training statistics. Re-fitting a scaler at inference produces a silent and undetectable inconsistency between training and serving.

14.6 Feature Selection

Two independent selection criteria were applied and their results combined.

Mutual information measures the reduction in uncertainty about the target obtained by observing an attribute. Unlike correlation it detects non-linear association and is applicable to mixed attribute types.

Random forest importance measures the mean decrease in node impurity attributable to each attribute across the ensemble. It captures interaction effects that a univariate measure cannot.

The two rankings were combined into a single ordering. Attributes ranked low by both were dropped: agreement between methods founded on different principles is stronger evidence of irrelevance than either alone.

14.7 Persistence

Four artefacts were written: the full processed dataset, the feature matrix, the target vector, and the ordered list of feature names in JSON.

The last is essential. A trained model consumes columns by position. If the feature matrix is reconstructed at inference with columns in a different order, every prediction is silently wrong, with no error raised. Persisting the column order and asserting against it at inference converts a silent failure into a loud one.

15 Predictive Modelling

15.1 Objective

To estimate, for each subscriber, the probability of departure, and to select and calibrate a model whose operating point reflects the commercial asymmetry between the two kinds of error.

15.2 Partitioning and Resampling

The data was partitioned into training and test sets by **stratified** split, preserving the class ratio in both partitions. An unstratified split can, under imbalance, produce a test set whose class balance differs materially from the population, rendering the evaluation unrepresentative.

SMOTE was then applied to the training partition only. This ordering is the critical methodological point of the chapter. Resampling before partitioning allows synthetic observations — generated by interpolating between real minority observations — to be distributed across both partitions. The test set then contains observations derived from training observations, and the resulting performance estimate is optimistic by an amount that cannot be determined after the fact. The test set here contains only real observations in their original proportion.

As executed. An 80/20 split produced a training partition of **5,634 observations** and a test partition of **1,409**, each carrying **30 features**. The churn rate is **26.5 per cent in both partitions**, confirming that stratification held. SMOTE was then applied to the training partition alone.

```
# — Stratified split – preserve class ratio —
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.20,
    stratify=y,
    random_state=RANDOM_STATE
)

print(f'Train: {X_train.shape} | Test: {X_test.shape}')
print(f'Train churn rate: {y_train.mean()*100:.1f}%')
print(f'Test churn rate: {y_test.mean()*100:.1f}%')

Train: (5634, 30) | Test: (1409, 30)
Train churn rate: 26.5%
Test churn rate: 26.5%
```

Figure 15.1 Stratified partition with class-ratio verification

15.3 Candidate Models

Five algorithms spanning distinct inductive biases were evaluated.

Model	Family	Rationale for inclusion
Logistic Regression	Linear	Interpretable baseline; coefficients are directly readable as log-odds
Random Forest	Bagged trees	Captures interactions; robust to unscaled inputs and outliers
Gradient Boosting	Boosted trees	Sequential error correction; typically strong on tabular data
XGBoost	Regularised boosting	Regularisation and efficient handling of sparse encodings
K-Nearest Neighbours (k=11)	Instance-based	Non-parametric contrast; makes no assumption about decision boundary shape

Including a linear baseline is deliberate. If a heavily tuned ensemble does not materially exceed logistic regression, the additional complexity is not justified, and the interpretable model should be preferred.

As configured. Logistic regression and random forest were instantiated with `class_weight='balanced'`; XGBoost with `scale_pos_weight` set to the ratio of negative to positive labels; k-nearest neighbours with `k = 11`. All five were fitted on the SMOTE-augmented training partition and evaluated on the untouched test partition through a common harness returning accuracy, precision, recall, F1, ROC-AUC, PR-AUC and training time.

```
# — Define candidate models —
MODELS = {
    'Logistic Regression': LogisticRegression(
        max_iter=1000, class_weight='balanced', random_state=RANDOM_STATE
    ),
    'Random Forest': RandomForestClassifier(
        n_estimators=200, class_weight='balanced',
        random_state=RANDOM_STATE, n_jobs=-1
    ),
    'Gradient Boosting': GradientBoostingClassifier(
        n_estimators=200, learning_rate=0.1, random_state=RANDOM_STATE
    ),
    'XGBoost': XGBClassifier(
        n_estimators=200, learning_rate=0.1,
        scale_pos_weight=(y_train == 0).sum() / (y_train == 1).sum(),
        eval_metric='logloss', random_state=RANDOM_STATE, n_jobs=-1
    ),
    'KNN': KNeighborsClassifier(n_neighbors=11, n_jobs=-1)
}

print(f'Comparing {len(MODELS)} models on SMOTE-augmented training data...')
Comparing 5 models on SMOTE-augmented training data...
```

Figure 15.2 Candidate model definitions

```
def evaluate_model(model, X_tr, y_tr, X_te, y_te):
    """Fit a model and return a dict of evaluation metrics."""
    t0 = time.time()
    model.fit(X_tr, y_tr)
    train_time = time.time() - t0

    y_pred = model.predict(X_te)
    y_prob = model.predict_proba(X_te)[:, 1] if hasattr(model, 'predict_proba') else None

    return {
        'Accuracy' : accuracy_score(y_te, y_pred),
        'Precision' : precision_score(y_te, y_pred, zero_division=0),
        'Recall' : recall_score(y_te, y_pred, zero_division=0),
        'F1' : f1_score(y_te, y_pred, zero_division=0),
        'ROC-AUC' : roc_auc_score(y_te, y_prob) if y_prob is not None else np.nan,
        'PR-AUC' : average_precision_score(y_te, y_prob) if y_prob is not None else np.nan,
        'Train time': round(train_time, 2)
    }

results = {}
for name, model in MODELS.items():
    print(f'Training {name}...', end=' ')
    results[name] = evaluate_model(model, X_train_sm, y_train_sm, X_test, y_test)
    print(f"F1={results[name]['F1']:.3f} ROC-AUC={results[name]['ROC-AUC']:.3f} ({results[name]['Train time']}s)")

results_df = pd.DataFrame(results).T.sort_values('ROC-AUC', ascending=False)
print('\n— Baseline Comparison —')
print(results_df.round(4).to_string())
```

Figure 15.3 Evaluation harness and baseline comparison loop

A note on the imbalance correction

Two imbalance corrections are applied in sequence, and the interaction between them is worth recording.

For logistic regression and random forest the effect is nil. `class_weight='balanced'` derives its weights from the label distribution it is fitted on; that distribution is already balanced after SMOTE, so both classes receive a weight of approximately 1.0. The parameter is redundant but harmless.

For XGBoost the effect is not nil. `scale_pos_weight` is computed from `y_train` — the **original, imbalanced** training labels, giving a ratio of roughly 2.8 — and the model is then fitted on `y_train_sm`, which SMOTE has already balanced. The positive class is therefore weighted approximately 2.8 times on data where it no longer needs weighting at all.

The consequence is predictable in direction: recall is inflated, precision depressed, and — more importantly for this system — the predicted probabilities are pushed upward and are no longer calibrated. Since Chapter 19 uses those probabilities to assign risk tiers and to compute revenue at risk, the distortion propagates into the business layer rather than staying inside the model.

The calibration curve in Section 15.7 is the diagnostic for this, and it is examined there.

15.4 Cross-Validation

The three strongest candidates were subjected to stratified k-fold cross-validation. A single train-test split yields a point estimate whose variance is unknown; cross-validation yields a distribution of scores, from which the stability of a model can be assessed as well as its central tendency. A model with a slightly lower mean and substantially lower variance may be preferable in deployment.

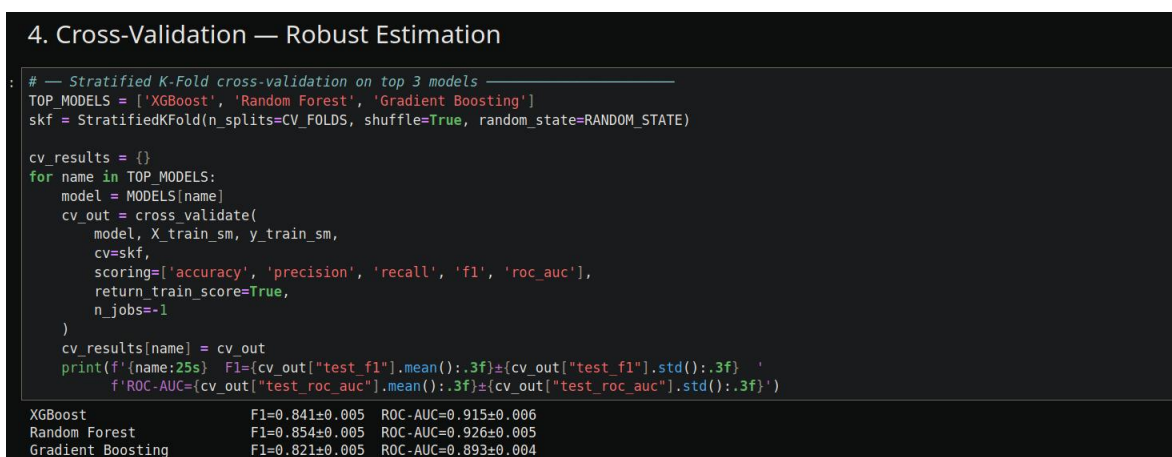


Figure 15.4 Cross-Validation Score Distribution

15.5 Hyperparameter Optimisation

Selection criterion

The criterion by which XGBoost was carried forward is stated here precisely, because it is easy to describe incorrectly.

Model selection was driven by ROC-AUC, not by recall. The baseline comparison ranks the five candidates with `sort_values('ROC-AUC', ascending=False)`. The randomised search is configured with `scoring='roc_auc'` and `refit=True`, so both the search and the final refit optimise ROC-AUC. Recall is computed and reported for every candidate, but it is not the quantity on which any selection decision turns.

Recall enters at one point only: threshold calibration. Section 16.4 selects the operating point under the constraint $recall \geq 0.75$, *maximise precision*. That is a decision about where to cut a probability, taken after the model is already chosen. Conflating the two would misdescribe the method — and it is the kind of claim an examiner can check against the code in seconds.

The accurate summary is therefore: **XGBoost was selected on discrimination and then operated at a recall-constrained threshold.**

One further detail is worth disclosing rather than leaving to be discovered. The three models carried into cross-validation are assigned as a literal list — `TOP_MODELS = ['XGBoost', 'Random Forest', 'Gradient Boosting']` — rather than read from the ranked baseline table. The choice is consistent with the expected ordering for tabular data of this kind, but it is a hard-coded assumption rather than a result. If the executed baseline ranks a different model into the top three, the code will not follow it.

Search configuration

XGBoost was tuned by **randomised search** over its hyperparameter space: 40 sampled configurations, five-fold stratified cross-validation, scored on ROC-AUC. The grid spans `n_estimators`, `max_depth`, `learning_rate`, `subsample`, `colsample_bytree`, `min_child_weight`, `gamma`, `reg_alpha` and `reg_lambda`.

Randomised search was preferred to exhaustive grid search on the standard argument that, when only a subset of hyperparameters materially affects performance, random sampling explores the influential dimensions more thoroughly for a given computational budget than a grid that spends equal effort on every dimension.

5. Hyperparameter Tuning (XGBoost)

```

In [10]: # — Randomized search over XGBoost hyperparameter space —————
xgb_param_grid = {
    'n_estimators' : [100, 200, 300, 400],
    'max_depth'    : [3, 4, 5, 6],
    'learning_rate' : [0.01, 0.05, 0.1, 0.2],
    'subsample'     : [0.7, 0.8, 0.9, 1.0],
    'colsample_bytree': [0.7, 0.8, 0.9, 1.0],
    'min_child_weight': [1, 3, 5],
    'gamma'         : [0, 0.1, 0.2, 0.5],
    'reg_alpha'      : [0, 0.01, 0.1],
    'reg_lambda'     : [1, 1.5, 2],
}

xgb_base = XGBClassifier(
    scale_pos_weight=(y_train == 0).sum() / (y_train == 1).sum(),
    eval_metric='logloss',
    random_state=RANDOM_STATE,
    n_jobs=-1
)

random_search = RandomizedSearchCV(
    xgb_base,
    param_distributions=xgb_param_grid,
    n_iter=40,
    scoring='roc_auc',
    cv=StratifiedKFold(n_splits=5, shuffle=True, random_state=RANDOM_STATE),
    verbose=1,
    random_state=RANDOM_STATE,
    n_jobs=-1,
    refit=True
)

print('Running RandomizedSearchCV (40 iterations x 5-fold)...')
t0 = time.time()
random_search.fit(X_train_sm, y_train_sm)
print(f'Done in {time.time()-t0:.1f}s')

print(f'\nBest ROC-AUC (CV): {random_search.best_score_:.4f}')
print(f'Best params:')
for k, v in random_search.best_params_.items():
    print(f'  {k}<22) = {v}')

Running RandomizedSearchCV (40 iterations x 5-fold)

```

Figure 15.5 Randomised Hyperparameter Search Results

15.6 Final Evaluation

The tuned model was evaluated once on the held-out test set. Evaluation comprised the confusion matrix in both raw and row-normalised form; the receiver operating characteristic curve with its area; the precision-recall curve, treated as the more informative summary under imbalance; and the full classification report giving precision, recall and F1 per class.

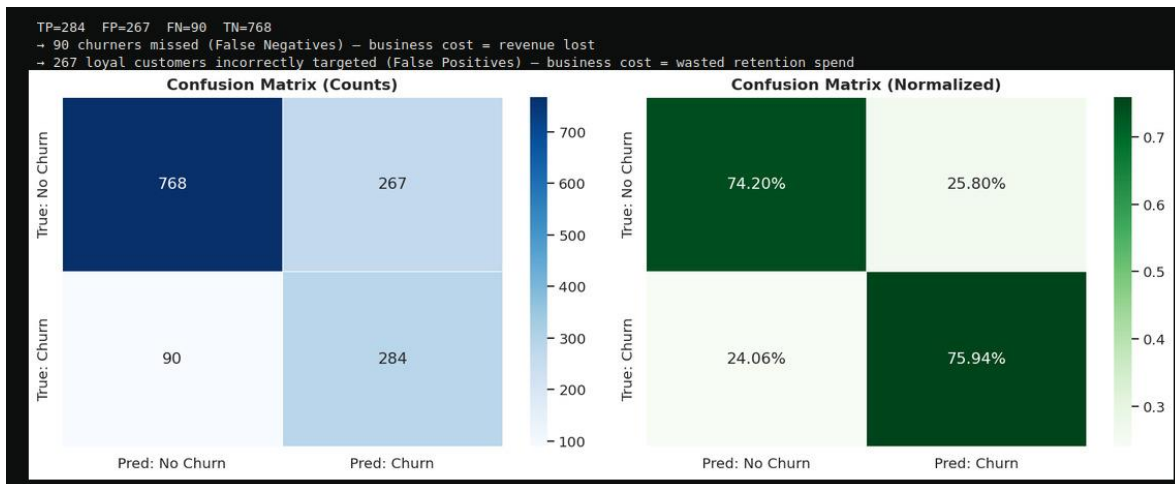


Figure 15.6 Confusion Matrix — Raw and Normalised

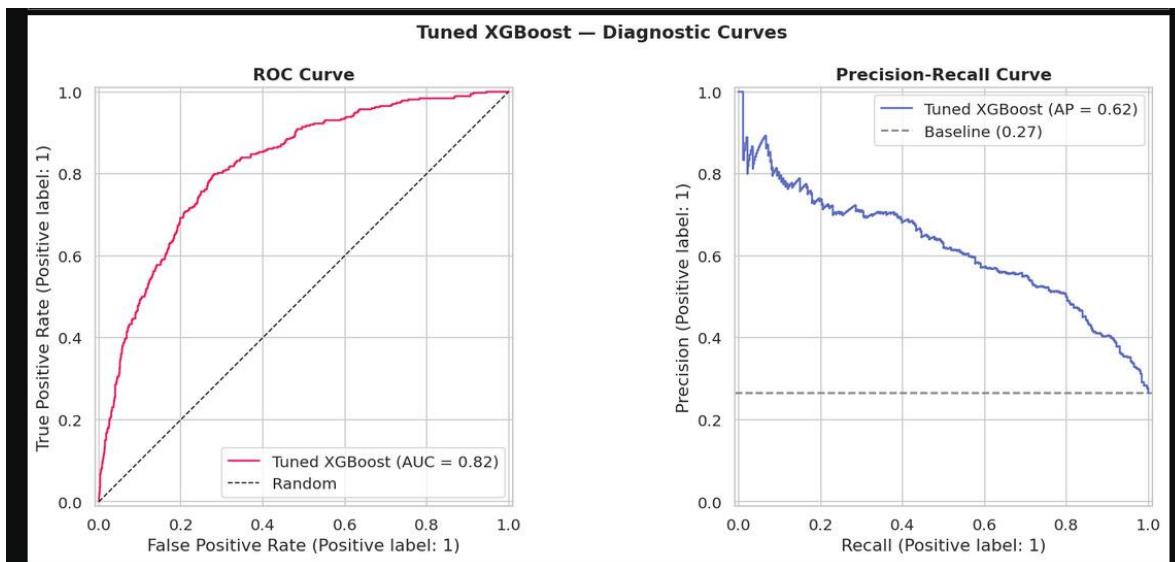


Figure 15.7 Receiver Operating Characteristic and Precision-Recall Curves

Baseline comparison

All five candidates were evaluated on the held-out test partition. The table is reproduced as printed by the notebook.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	Train (s)
Logistic Regression	0.9766	0.8470	0.8005	0.9143	0.8389	0.8587	0.30
Gradient Boosting	0.9756	0.8303	0.8246	0.9124	0.8332	0.8475	4.49
Random Forest	0.9771	0.8599	0.8497	0.9015	0.8283	0.8196	1.09

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	Train (s)
XGBoost	0.9740	0.8071	0.8647	0.9098	0.8221	0.8267	0.71
KNN	0.9728	0.8935	0.8128	0.9141	0.8190	0.8625	0.00

XGBoost leads on **recall alone**. On accuracy it is fourth, on precision fifth, on F1 fourth, on ROC-AUC fourth and on PR-AUC fourth. This matters for how the selection is described, and Section 15.5 returns to it.

Cross-validation

Five-fold stratified cross-validation on the three carried-forward models:

Model	F1	ROC-AUC
Random Forest	0.854 ± 0.005	0.926 ± 0.005
XGBoost	0.841 ± 0.005	0.915 ± 0.006
Gradient Boosting	0.821 ± 0.005	0.893 ± 0.004

Random Forest leads on both. Variance is low and near-identical across all three, so no stability argument distinguishes them.

Tuning

Forty sampled configurations over five folds, 200 fits, completed in 12.2 seconds. Best cross-validated ROC-AUC **0.9195**. Selected configuration: `n_estimators 400`, `max_depth 6`, `learning_rate 0.1`, `subsample 0.8`, `colsample_bytree 0.9`, `min_child_weight 3`, `gamma 0.1`, `reg_alpha 0`, `reg_lambda 2`.

Final test-set performance

The tuned model produced the following confusion matrix on the test partition:

	Predicted: No Churn	Predicted: Churn
Actual: No Churn	768	267
Actual: Churn	90	284

1,409 observations, 374 of them churners (26.5 per cent), consistent with the partition described in Section 15.2.

Derived directly from those four counts:

Metric	Value
Accuracy	$(284 + 768) / 1409 = \mathbf{0.7466}$
Precision	$284 / 551 = \mathbf{0.5154}$
Recall	$284 / 374 = \mathbf{0.7594}$
F1	$\mathbf{0.6141}$
ROC-AUC	$\mathbf{0.82}$ (from the plotted curve)
Average precision (PR-AUC)	$\mathbf{0.62}$ (from the plotted curve)

The normalised confusion matrix reports 75.94 per cent on the churn row, which is the recall figure above, computed independently by the plotting code.

In business terms: the model catches 284 of 374 departures and misses 90. It also flags 267 retained subscribers who would be contacted unnecessarily. Against a base rate of 26.5 per cent, an average precision of 0.62 means the ranking is materially better than random — a subscriber in the top decile of predicted risk is far more likely to depart than one drawn at random — even though the precision at the default threshold is only 0.52.

15.7 Calibration

Discrimination and calibration are distinct properties. A model may order subscribers by risk correctly while systematically overstating or understating the absolute probability. Because this system uses probability to allocate retention effort and to compute revenue at risk, absolute calibration is material and was assessed by comparing predicted probabilities against observed frequencies across probability bins.

The result confirms the concern raised in Section 15.3. The calibration curve sits below the diagonal across the entire upper range: subscribers assigned a predicted probability of 0.95 churn at an observed rate of approximately 0.70; those assigned 0.75 churn at approximately 0.40; those assigned 0.55 at approximately 0.30. The model is **systematically over-confident**, which is precisely the behaviour predicted when `scale_pos_weight` is applied on top of SMOTE-balanced data.

The consequence is concrete rather than academic. The risk tiers defined in Section 19.1 use raw predicted probability as their boundary, so a subscriber placed in the Critical tier at 0.75 or above has a true departure probability closer to 0.40. The tier populations are therefore overstated and the revenue-at-risk figures derived from them are inflated.

Two remedies exist and either is straightforward: remove the redundant `scale_pos_weight` and refit, or wrap the fitted model in `CalibratedClassifierCV` — which the notebook already imports but does not apply.

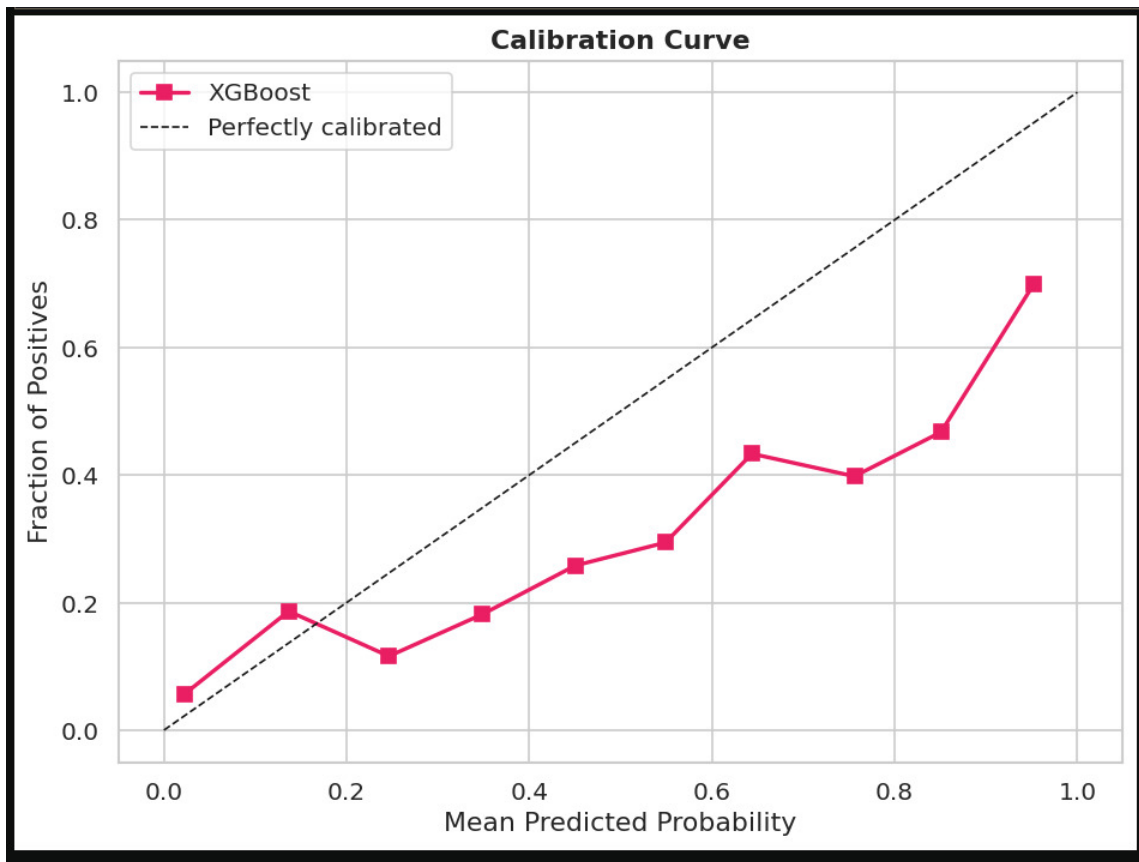


Figure 15.8 Model Calibration Curve

15.8 Persistence

The selected model and its metadata — training timestamp, hyperparameters, feature list and selected threshold — were persisted together. Persisting the model alone is insufficient: a model without its feature order and threshold cannot be reproduced or served correctly.

16 Explainability and Threshold Calibration

16.1 Intrinsic Feature Importance

XGBoost exposes an intrinsic importance derived from the frequency and quality of splits on each attribute. It is inexpensive and directionally useful, but it is computed from training-time statistics, is known to favour high-cardinality attributes, and says nothing about the direction of an effect — only that an attribute was useful for splitting.

Result. Gain-based importance is dominated by a single attribute. `Contract` scores approximately 0.32 — more than three times the next attribute and roughly a quarter of all gain in the ensemble. It is followed by `has_fiber` (≈ 0.095), `has_no_internet` (≈ 0.085), `is_month_to_month` (≈ 0.053) and `InternetService_No` (≈ 0.045). Everything below that sits under 0.03 and forms a long, flat tail.

Two observations follow.

The model and the dashboards agree. Contract type is the strongest separator in the reporting layer (45.84 per cent against 2.55 per cent, Section 18.4) and it is also what the classifier relies on most. Two independent methods converging on the same driver is stronger evidence than either alone.

Four of the top five are engineered or encoded contract and internet attributes. `has_fiber`, `has_no_internet` and `is_month_to_month` are all features constructed in Chapter 14; they were not present in the source data. That is a direct vindication of the feature engineering, and it is worth stating, because engineered features are often added and never shown to have earned their place.

`gender` sits near the bottom of the ranking, consistent with the null result reported in Section 18.10.

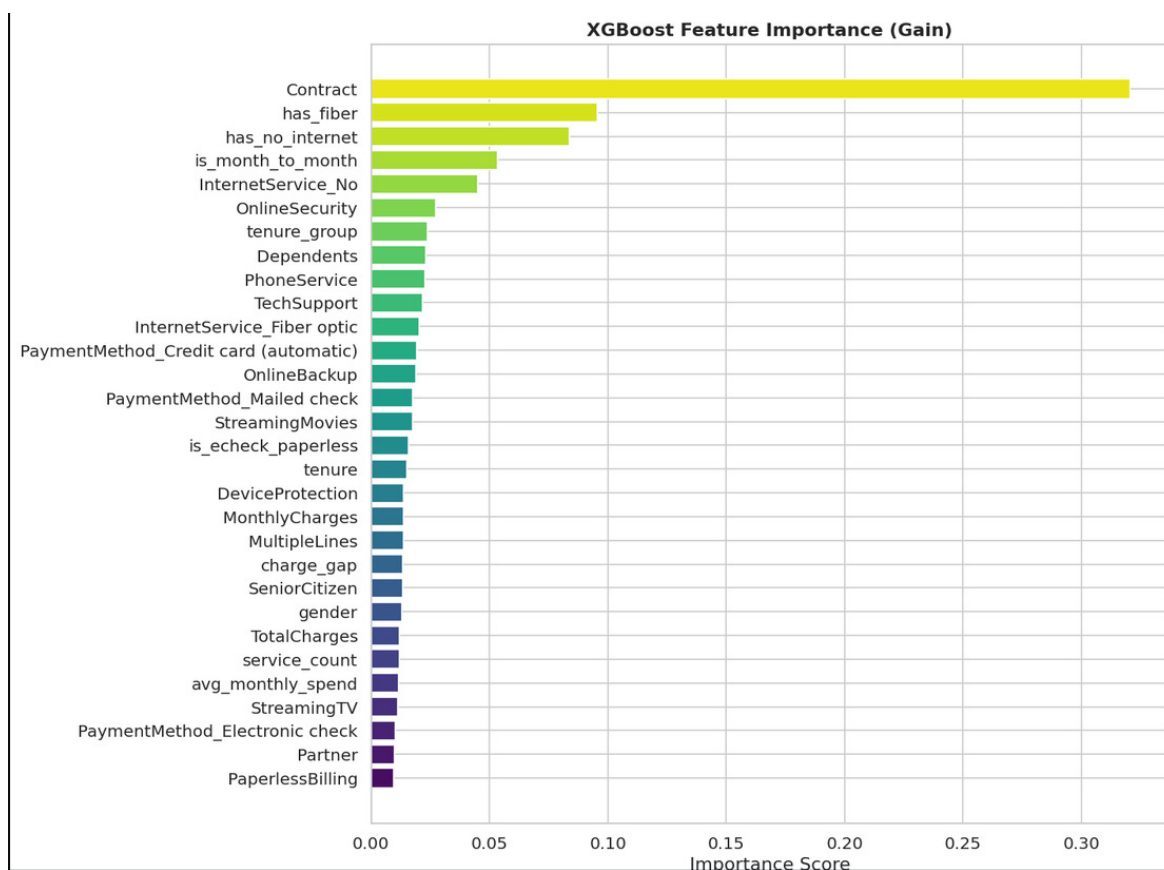


Figure 16.1 XGBoost Built-in Feature Importance

16.2 SHAP Global Analysis

SHAP values were computed to obtain attributions that are theoretically grounded and additive. Two global views were produced.

The **beeswarm plot** displays, for every attribute, the distribution of SHAP values across all subscribers, with point colour encoding the attribute's value. It conveys three things simultaneously: which attributes matter most, in which direction each pushes the prediction, and whether the effect is consistent across the population or concentrated in a subgroup.

The **bar summary** ranks attributes by mean absolute SHAP value, giving a single ordering of global influence.

16.3 SHAP Local Analysis

A **waterfall plot** was produced for one high-risk subscriber, decomposing that individual prediction from the population base rate to the final probability, attribute by attribute.

This is the output with the most direct operational value in the entire modelling chapter. A retention agent preparing to contact a subscriber requires not a score but a reason: this subscriber's risk is elevated principally by short tenure and a month-to-month contract, and mitigated slightly by the number of services held. That statement is actionable in a way a probability is not.

16.4 Business-Weighted Threshold Selection

A classifier emits a probability; converting it to a decision requires a threshold. The conventional default of 0.5 is appropriate only when the two error types carry equal cost, which here they do not.

A **false negative** is a subscriber who will depart and whom the model does not flag. The cost is the loss of that subscriber's future revenue, plus the acquisition cost of a replacement.

A **false positive** is a retained subscriber who is flagged and contacted unnecessarily. The cost is the marginal expense of the retention contact and any incentive extended.

The former materially exceeds the latter. The threshold was therefore lowered below 0.5, trading precision for recall, and selected under an explicit operating constraint: **maximise precision subject to recall of at least 0.75**. The constraint expresses a commercial position — that the operator is prepared to accept a defined volume of unnecessary contacts in order to identify three quarters of genuine departures.

The threshold that maximises F1 was computed alongside for comparison, since F1 weights the two errors equally and therefore represents a different, and here inappropriate, commercial stance.

17 Business Intelligence and Reporting Layer

17.1 Design Principles

The reporting layer was governed by four rules, applied without exception.

One business question per page. A page exists to answer a question that can be stated in a sentence. A page that answers three questions answers none of them clearly.

Sequence as argument. Pages are ordered so that each raises the question its successor answers. The reader traverses an argument rather than browsing a collection.

Explicit non-duplication. Because the suite was built by four contributors, each page carries an explicit exclusion list — analyses that belong elsewhere and must not be repeated. An attribute may appear as a tooltip or a filter in more than one place, but the same business question is asked exactly once in the entire suite.

Consistent visual grammar. A given quantity is rendered by the same mark type, the same colour and the same axis convention wherever it appears.

17.2 Suite Structure

Part	Title	Question	Pages
A	Executive Revenue Suite	How is revenue earned, where is it lost, and what should be decided?	10
B	Customer Behaviour and Churn Analysis	Why do subscribers leave?	6
C	Operational and Specialised Analytics	Where does the effect appear operationally?	6
D	Data Integrity Monitor	Can these results be trusted?	1

The order is the analytical progression established in Chapter 2: measure the position, diagnose the cause, locate it operationally, then verify that the data supporting all three is sound.

17.3 Semantic Model

The dimensional tables are imported into the reporting tool and related on their surrogate keys as specified in Section 11.11. Business logic resides entirely in DAX measures held in a dedicated measure table; surrogate key columns are hidden from the report view, since they carry no meaning to a reader.

17.4 Why Not a Dedicated Analysis Services Model

The semantic layer — relationships, hierarchies, measures and aggregation logic — is held inside the Power BI dataset rather than in a separate SQL Server Analysis Services tabular model. As with the integration decision in Section 10.6, the alternative is the conventional enterprise pattern and its rejection is recorded rather than left implicit.

The engine is the same. A Power BI dataset and an Analysis Services tabular model are both hosted by the VertiPaq columnar engine and both consume DAX. Moving the model to Analysis Services would not change how a measure is written, how it is evaluated, or how fast it returns. The storage format and the query language are identical.

What Analysis Services adds is organisational, not analytical. Its advantages appear when a single semantic model must be shared by many reports built by many teams, governed centrally, secured by role at row level, refreshed on an independent schedule, or queried by tools other than Power BI. This project has one model, one reporting tool and one audience. None of those conditions holds.

What it would cost. A server instance to deploy and maintain, a separate development project, a deployment step between model change and report change, and a second place where a measure definition can drift out of step with the report that uses it.

The single-model constraint is met differently. The concern Analysis Services addresses — four contributors defining the same measure four different ways — is real here, since the suite was built by four people. It is addressed in this project by holding all measures in one shared dataset with a single measure table, and by the validation control in Section 11.9 that asserts arithmetic consistency across the model. Section 18.8 records three places where that discipline was not fully maintained, which is a fair illustration of the risk; it is a governance failure, not a failure of the platform choice.

The trigger for revisiting the decision is stated in Section 20.4: a second reporting tool, an external consumer of the model, or a row-level security requirement.

17.5 Measure Definitions

Total Customers	= COUNTROWS(fact_churn)
Total Revenue	= SUM(fact_churn[total_revenue])
Total Charges	= SUM(fact_churn[total_charges])

```

Total Monthly Charges = SUM(fact_churn[monthly_charges])
Total Refunds          = SUM(fact_churn[total_refunds])
Total Long Distance Charges = SUM(fact_churn[total_long_distance_charges])
Total Extra Data Charges   = SUM(fact_churn[total_extra_data_charges])

Total Ancillary Revenue =
    [Total Long Distance Charges] + [Total Extra Data Charges]

Net Revenue              = [Total Revenue] - [Total Refunds]
Refund Rate %            = DIVIDE([Total Refunds], [Total Revenue])
Churn Rate %             = DIVIDE([Churned Customers], [Total Customers])
Revenue Lost %           = DIVIDE([Lost Revenue], [Total Revenue])

Revenue per Customer =
    DIVIDE([Total Revenue], DISTINCTCOUNT(fact_churn[customer_id]))

Customers with Refunds =
    CALCULATE([Total Customers], fact_churn[total_refunds] > 0)

Avg Revenue Lost per Churned Customer =
    DIVIDE([Lost Revenue], [Churned Customers])

Gross Revenue Before Refunds =
    [Total Charges] + [Total Long Distance Charges] + [Total Extra Data Charges]

Revenue Formula Check =
    [Gross Revenue Before Refunds] - [Total Refunds] - [Total Revenue]

```

Two definitions require comment. **Revenue per Customer** and average revenue per user are **not** the same quantity: the first is mean lifetime revenue, the second mean monthly revenue. They are maintained as distinct measures because conflating them produces figures that are wrong by a factor equal to mean tenure. **Revenue Formula Check** is the validation control described in Section 11.9 and is expected to evaluate to zero under every filter combination.

17.6 Part A — Executive Revenue Suite

Each page below is specified by its question, its indicators and its visuals. The transition note states the question the page leaves unanswered, which its successor addresses.

A.1 Executive Financial Position

What is the operator's current financial position?

Indicators: total revenue; monthly recurring revenue; total subscribers; retained revenue; revenue per subscriber; churn rate.

Visuals: retained against lost revenue; revenue by contract type; revenue share by payment method; executive financial summary matrix.

Slicers: contract type, payment method, churn label.

Excluded: internet type, tenure band and offer, each of which is the subject of a page elsewhere in the suite.

As delivered. Total revenue \$21.37M · MRR \$456.12K · 7,043 subscribers · retained revenue \$17.69M · revenue per subscriber \$3,034.38 · churn rate 26.54%. Revenue splits \$12.7M bank withdrawal, \$8.2M credit card, \$0.5M mailed cheque. The executive summary matrix carries the finding that governs the rest of the suite:

Contract	Revenue	Subscribers	ARPU	Churn rate
Month-to-month	\$6,162,488.22	3,610	1,707.06	45.84 %
One year	\$6,171,794.31	1,550	3,981.80	10.71 %
Two year	\$9,036,849.16	1,883	4,799.18	2.55 %
Total	\$21,371,131.69	7,043	3,034.38	26.54 %

Transition: the position is known; its composition is not.

A.2 Revenue Composition Analysis

Of what is total revenue composed?

Indicators: total revenue; gross revenue before refunds; base charges; total ancillary revenue; total refunds.

Visuals: revenue build-up waterfall; revenue mix by contract; gross against total revenue by contract; component summary matrix.

As delivered. Base charges \$16,060,725.24 (75.10 %) · long-distance \$5,275,906.10 (24.67 %) · extra-data \$48,320.00 (0.23 %) · refunds \$13,819.65 (0.06 %). Long-distance is not a marginal line item — it is a quarter of all revenue, and it is the component least visible in the raw dataset.

Transition: the ancillary component is identified but not analysed.

A.3 Ancillary Revenue Performance

What do long-distance and data overage charges contribute, and from where?

Indicators: total ancillary revenue; long-distance revenue; extra-data revenue; ancillary share; mean monthly long-distance charge.

Visuals: long-distance against extra-data; ancillary revenue by contract; by payment method; per subscriber.

As delivered. Total ancillary revenue \$5.32M, of which long-distance \$5.28M and extra-data \$48.32K — 24.9 % of revenue, \$755.96 per subscriber. By contract: two year \$2,310,915.34 · one year \$1,543,092.65

· month-to-month \$1,470,218.11. Long-distance outweighs extra-data by a factor of more than a hundred; the two are not comparable revenue streams and should not be presented as a pair without that caveat.

Transition: gross revenue is understood; the retained amount is not.

A.4 Net Revenue and Revenue Leakage Analysis

After refunds, what has the operator actually retained?

Indicators: gross revenue; revenue leakage; net revenue; revenue retention percentage.

Visuals: gross-to-net waterfall; leakage by contract; leakage by payment method.

As delivered. Gross revenue \$21.38M · leakage \$13.82K · net revenue \$21.37M · **revenue retention 99.94 %**. Leakage by contract: month-to-month \$5.8K, two year \$4.8K, one year \$3.3K. By payment method: bank withdrawal \$8.5K, credit card \$4.8K, mailed cheque \$0.5K.

The finding here is a negative one and is worth stating as such: **refunds are not a material source of loss**. At six hundredths of one per cent of revenue, leakage is an order of magnitude too small to warrant management attention. The page earns its place by ruling out a hypothesis, and by confirming — through a second, independent route — the revenue identity validated in Section 11.9.

Transition: the largest single source of loss lies elsewhere. It is not refunds.

A.5 Churn Financial Impact

What has subscriber departure cost?

Indicators: lost revenue; lost monthly revenue; departed subscribers; revenue lost percentage; mean revenue lost per departed subscriber; lifetime value lost.

Visuals: retained against lost revenue; lost revenue by contract; by payment method; revenue loss with churn rate by contract.

Excluded: loss by internet type and by tenure band, which belong to Part C and Part B respectively.

As delivered. 1,869 departed subscribers · monthly revenue lost **\$139.1K** · revenue attached to departed subscribers **\$3.68M** · lifetime value lost **\$8M**.

Monthly loss concentrates almost entirely in one contract: month-to-month **\$121K**, one year \$14K, two year \$4K. Month-to-month accounts for **87 per cent of all monthly revenue lost**. By payment method: bank withdrawal \$104K, credit card \$27K, mailed cheque \$8K.

The page also carries the only direct evidence in the entire system of *why* subscribers left, ranked by the monthly revenue attached to each stated reason:

Stated reason	Monthly revenue lost
Competitor had better devices	\$24K
Competitor made a better offer	\$23K
Attitude of support person	\$15K
Don't know	\$10K
Competitor offered more data	\$9K
Competitor offered higher download speeds	\$8K
Attitude of service provider	\$7K
Product dissatisfaction	\$6K
Price too high	\$6K
Network reliability	\$5K

Transition: the magnitude of loss is known; the value of the individual subscriber is not.

A.6 ARPU and Revenue Value

What is a subscriber worth, and is the operator losing its more valuable ones?

Indicators: average revenue per user overall, retained and departed; the gap between them; mean lifetime value by churn status.

Visuals: ARPU by churn status; average CLTV by churn status.

Excluded: value-tier segmentation, which is Part B.

As delivered. ARPU **\$3.03K** overall — **\$3.42K** among retained subscribers against **\$1.97K** among departed, a gap of **–\$1.45K**. Mean lifetime value, by contrast, is **\$4.5K retained against \$4.1K departed**.

That pair of figures is the most instructive on the page and is easily misread. Realised revenue per departed subscriber is **42 per cent lower** than per retained subscriber, but their *estimated lifetime value* is only 9 per cent lower. The subscribers being lost are not intrinsically less valuable — they are of comparable potential and simply left before that potential was realised.

The implication reverses the comfortable reading of Section 18.5. The operator is not shedding low-value subscribers; it is failing to retain ordinary ones long enough to earn from them. The gap is tenure, not quality.

A.7 Billing Channel Performance

Does paperless billing associate with better financial performance or greater risk?

Indicators: paperless subscribers; paperless share; average revenue per user for each channel; paperless revenue share; churn rate by channel.

Visuals: revenue by channel; average revenue per user by channel; churn rate by channel; subscriber distribution.

The page is titled Billing Channel Performance rather than Billing Efficiency, as originally proposed. The dataset records neither collection time nor invoice issuing cost; efficiency cannot be measured from it, and a title claiming otherwise would misrepresent the analysis.

A.8 Payment Method Intelligence

Which payment method performs best on a combined financial assessment?

Indicators: best revenue method; best ARPU; lowest churn; best retained revenue.

Visuals: performance overview combining revenue and ARPU; churn rate by method; retained revenue by method.

As delivered.

Payment method	Revenue	ARPU	Churn rate	Retained revenue
Bank withdrawal	\$12,672K	\$3.24K	34.00 %	\$9.91M
Credit card	\$8,157K	\$2.97K	14.48 %	\$7.32M
Mailed cheque	\$542K	\$1.41K	36.88 %	\$0.45M

No single method wins outright, which is exactly why the page exists. Bank withdrawal leads on revenue, ARPU and retained revenue, but churns at more than twice the rate of credit card. Credit card generates a third less revenue while retaining far better.

The spread is wide enough to matter — 14.48 per cent against 34.00 per cent — but the caution of Section 18.7 applies unchanged: payment instrument is chosen by the subscriber, not assigned by the operator, so the difference may reflect who selects each method rather than any effect of the method itself.

A.9 Contract Revenue Intelligence

Which contract type performs best on a combined financial assessment?

Indicators: best contract revenue; best contract ARPU; lowest contract churn; best retained revenue.

Visuals: performance overview combining revenue and ARPU; churn rate by contract; retained revenue by contract.

As delivered.

Contract	Revenue	ARPU	Churn rate	Retained revenue
Two year	\$9,037K	\$4.80K	2.55 %	\$8.70M
One year	\$6,172K	\$3.98K	10.71 %	\$5.31M
Month-to-month	\$6,162K	\$1.71K	45.84 %	\$3.67M

Unlike payment method, this page has an unambiguous winner. Two-year contracts lead on **all four** indicators simultaneously — highest revenue, highest ARPU, lowest churn and highest retained revenue.

The month-to-month column is the mirror image. It generates almost the same gross revenue as one year (\$6.16M against \$6.17M) from **more than twice the subscribers**, at a third of the ARPU, and retains only \$3.67M of it.

The contract generating the highest gross revenue is not treated as best by default. Assessment is multi-indicator, since a contract may lead on revenue while leading also on loss.

A.10 Executive Decision Center

Given the preceding analysis, what should management do?

Indicators: top revenue contract; best payment method; largest leakage source; highest financial risk; net revenue; total leakage percentage.

Visuals: opportunity against risk matrix; revenue leakage waterfall; key findings panel; recommended actions panel.

The page does not restate the preceding visuals. It converts the strongest findings into three or four recommended actions.



Figure 17.1 Executive Financial Position



Figure 17.2 Revenue Composition Analysis

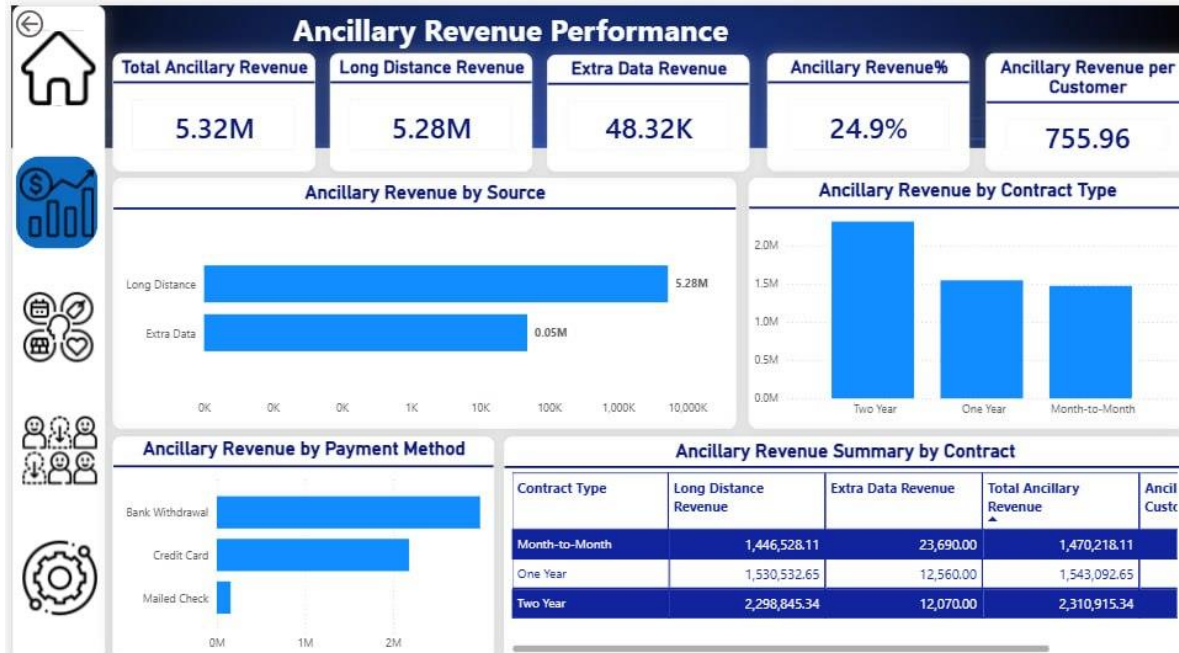


Figure 17.3 Ancillary Revenue Performance



Figure 17.4 Net Revenue and Revenue Leakage Analysis

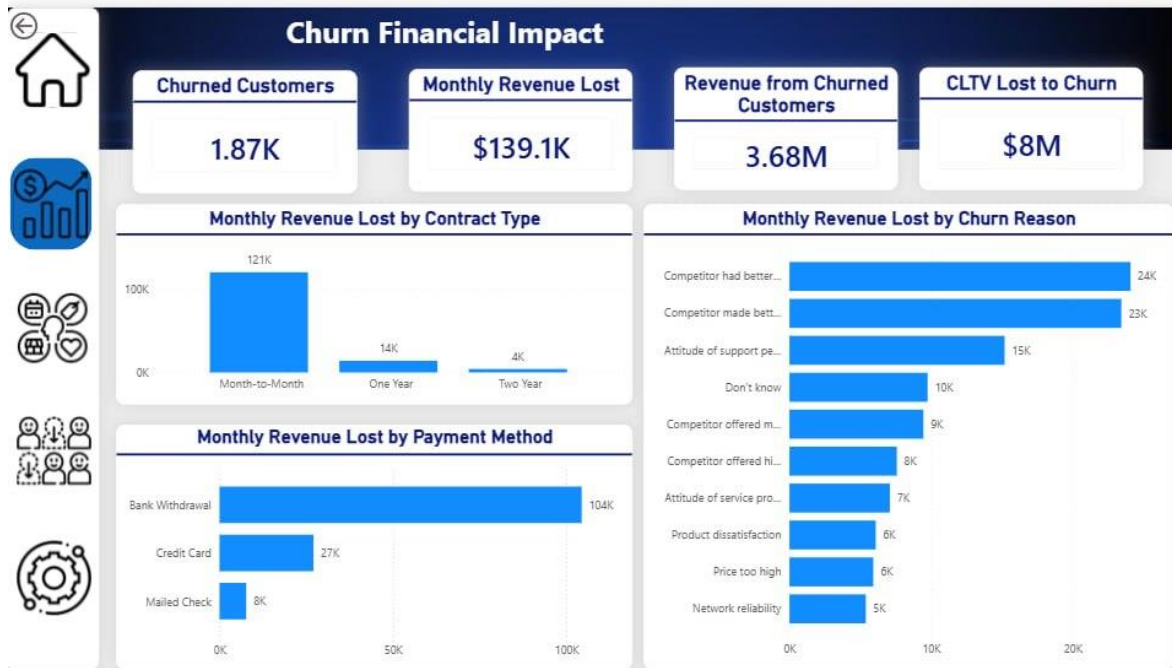


Figure 17.5 Churn Financial Impact

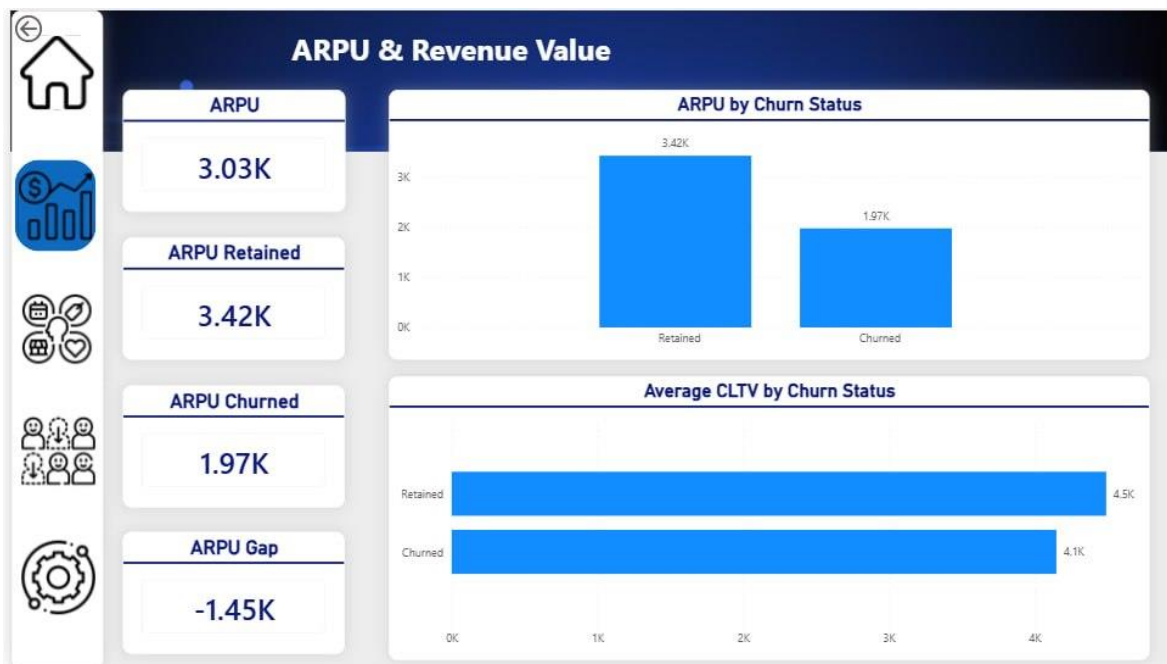


Figure 17.6 Customer Revenue Value

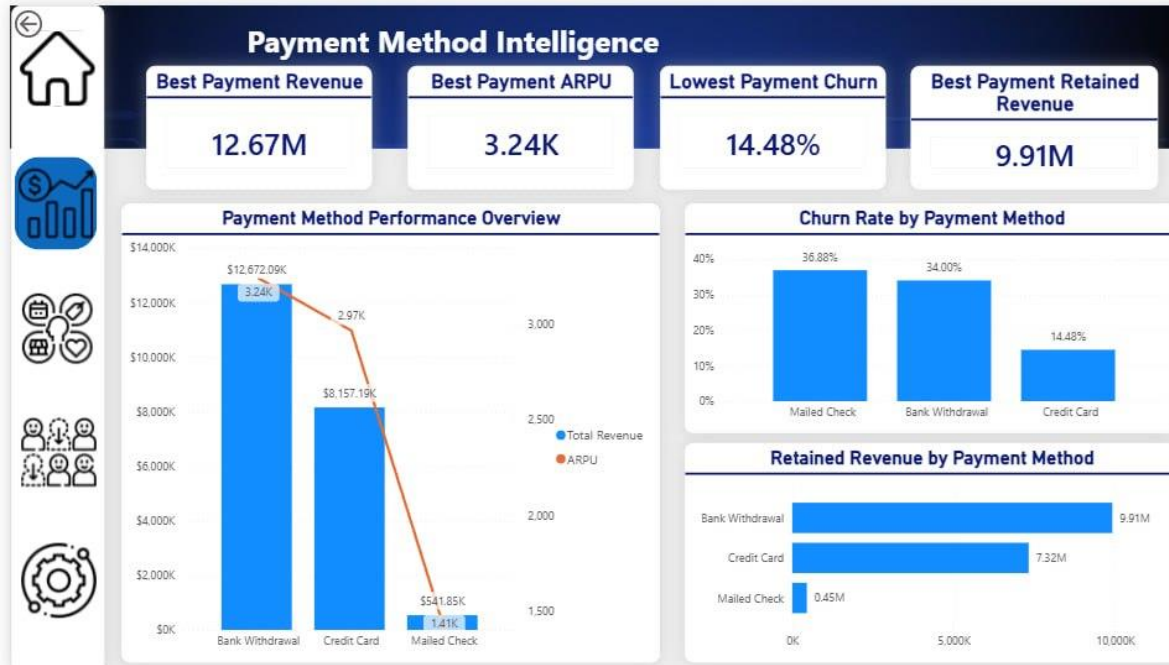


Figure 17.7 Payment Method Intelligence



Figure 17.8 Contract Revenue Intelligence

17.7 Part B — Customer Behaviour and Churn Analysis

B.1 Tenure and Loyalty

Does risk vary with the length of the relationship?

Mean tenure is **37.6 months among retained subscribers and 18.0 among departed** — retained subscribers have been with the operator more than twice as long. Attrition falls monotonically and steeply across tenure bands.

Tenure band	Churn rate	Mean monthly charge
0–12 months	47.44 %	\$56.10
13–24 months	28.71 %	\$61.36
25–48 months	20.39 %	\$65.93
48+ months	9.51 %	\$73.95

The two columns move in opposite directions, and that is the finding. Long-tenure subscribers pay **more** per month, not less, yet churn at a fifth of the rate of new ones. Price is therefore not the dominant driver of early attrition; the first year of the relationship is. Referral activity concentrates in the same place: 7.0K of the 13.7K referrals come from the 48-month-plus band.



Figure 17.9 Tenure and Loyalty — cohort analysis

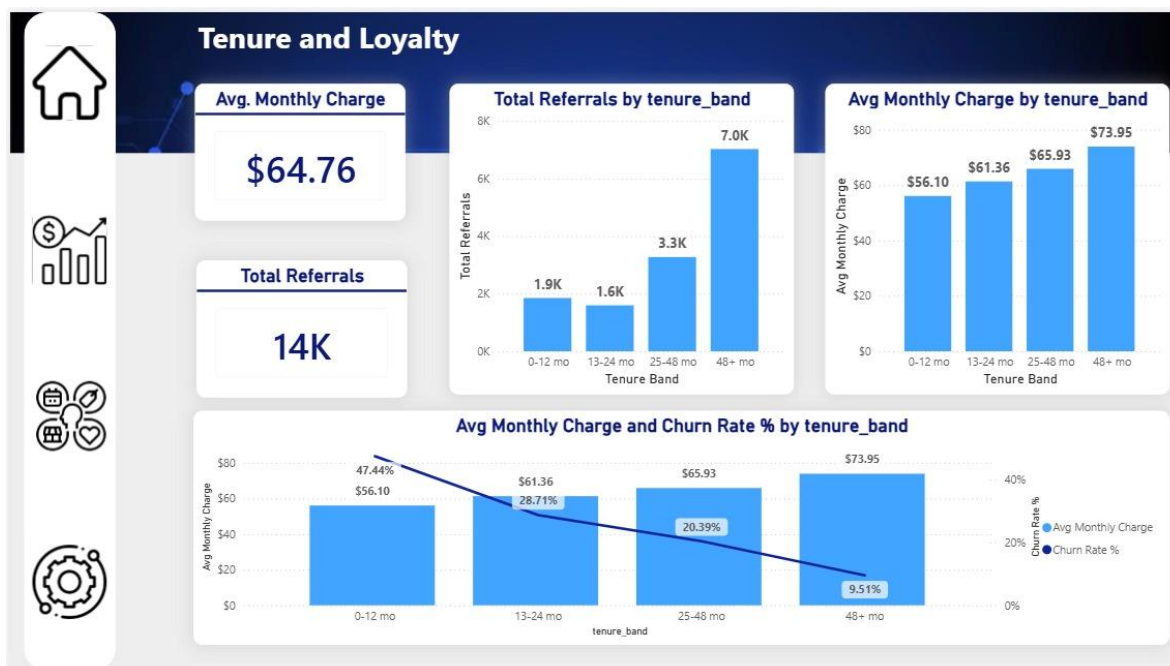


Figure 17.10 Tenure and Loyalty — referrals and charges by band

B.2 Service Penetration

Does breadth of service adoption relate to retention?

Mean services held: **5.1 among retained subscribers, 5.3 among departed.** Departed subscribers hold marginally *more* services, not fewer — which contradicts the intuition that a broader bundle raises switching cost.

The relationship with service count is not monotonic but **inverted-U**: churn rises from 8.82 % at one service to a peak of **46.90 %** at around three, then falls away to 4.85 % at eleven. Subscribers with very few services have little to leave; subscribers with very many are deeply embedded. The exposed population sits in the middle.

This page also gives the authoritative churn rates by access technology: fibre optic **40.72 %**, cable 25.66 %, DSL 18.58 %, none 7.40 % — and these are the figures used throughout this document.



Figure 17.11 Service Penetration — adoption and service count

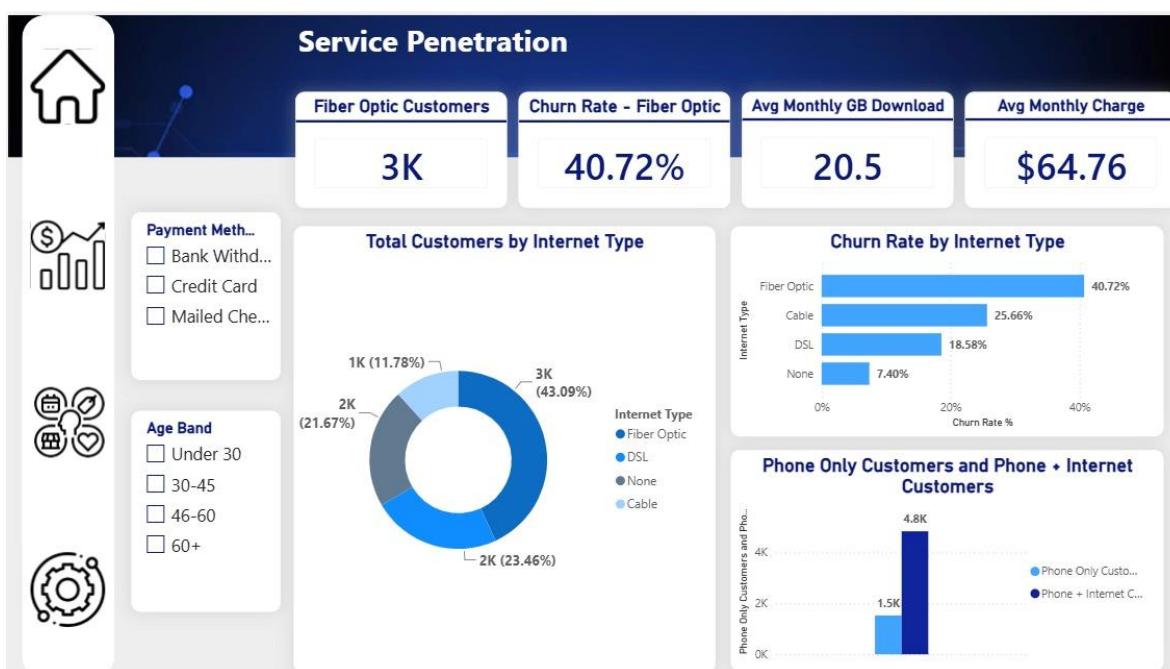


Figure 17.12 Service Penetration — access technology

B.3 Support Interaction

Does the absence of support services associate with departure?

Subscribers without technical support churn at **31.19 %**; those with it, at **15.17 %** — a difference of sixteen percentage points, better than a twofold ratio.

Adoption of premium technical support rises sharply with tenure: 12.95 % in the first year, 21.09 %, 30.87 %, and 47.03 % beyond four years. Support adoption and tenure are therefore confounded, and the support effect cannot be cleanly separated from the tenure effect without controlling for one of them.

Adoption across the four support services is even: online backup 2,429 · device protection 2,422 · premium technical support 2,044 · online security 2,019.

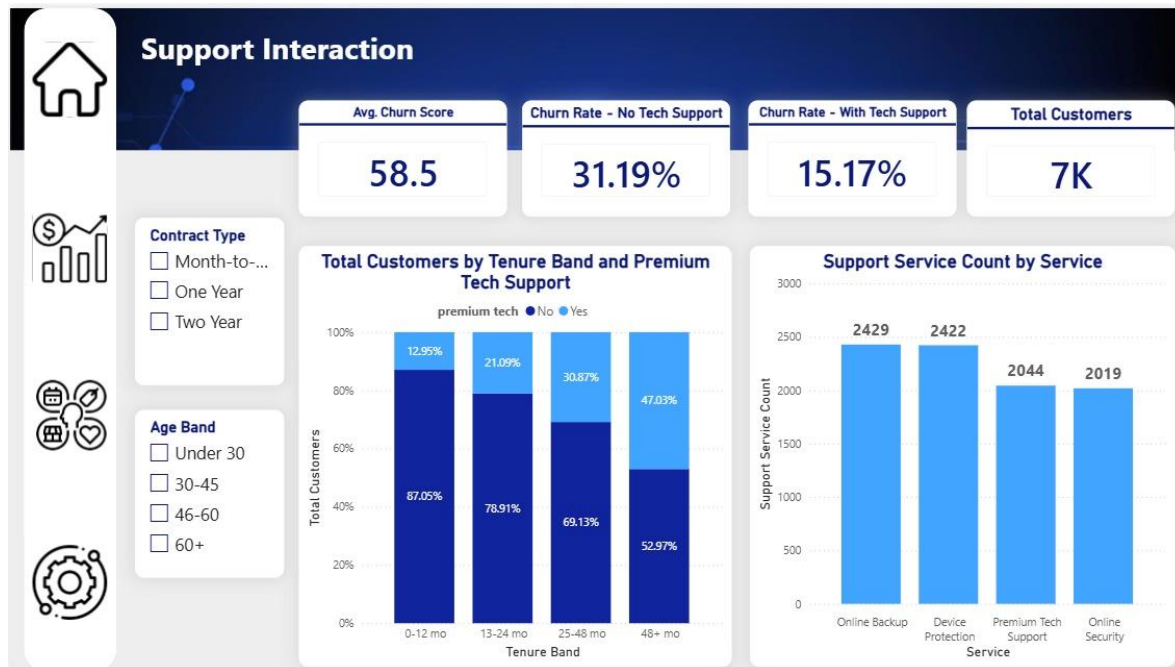


Figure 17.13 Support Interaction

B.4 Streaming Habits

Does media consumption relate to retention?

Segment	Churn rate
No streaming	22.80 %
Dual streamers	29.43 %
Streaming television	30.07 %
Streaming films	29.94 %

Streaming subscribers churn some seven points above non-streamers, and the three streaming variants are indistinguishable from one another. The stronger separation is data allowance: **unlimited data 31.65 % against 15.97 % without** — very nearly double.

Both findings point the same way as B.2 and Section 18.6: high-consumption subscribers on high-tier plans are the exposed group. Mean monthly download is 26.4 GB for streamers against 15 GB for non-streamers.



Figure 17.14 Streaming Habits

B.5 Demographic Profiles

Which demographic groups carry elevated risk?

Attribute	Churn rate
Senior citizens (≈1,000 subscribers)	41.68 %
Non-senior	23.61 %
Age 60+	36.46 % · 46–60: 24.31 % · 30–45: 23.90 % · under 30: 21.70 %
Without dependents	32.55 %
With dependents	6.52 %
Single	32.96 %
Married	19.66 %

Two results and one non-result.

Household composition is the strongest demographic separator in the dataset. Subscribers with dependents churn at 6.52 % against 32.55 % without — a fivefold difference, larger than any other demographic effect measured. Marital status points the same way.

Age matters at the top end only. The three bands below sixty sit within three points of one another; the 60+ band jumps to 36.46 %, and senior citizens to 41.68 %.

Gender does not matter. Female and male rates are within a percentage point in every age band. This is worth reporting explicitly: a null result correctly identified is a finding, and it prevents a segmentation that would waste effort.

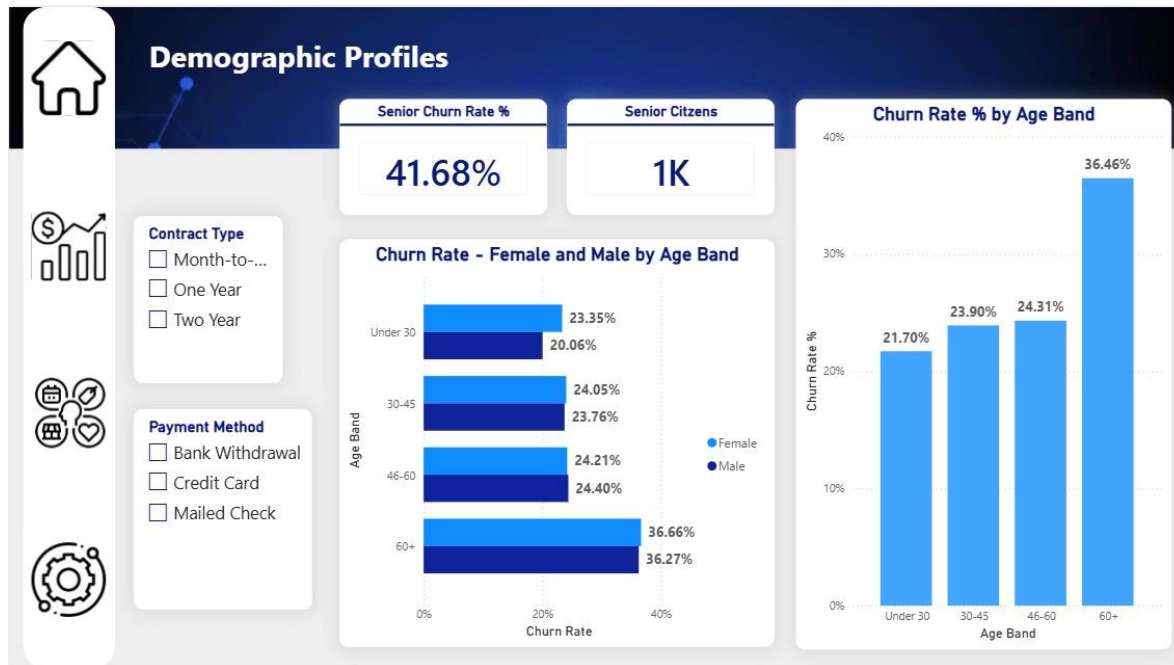


Figure 17.15 Demographic Profiles — age, gender and senior status

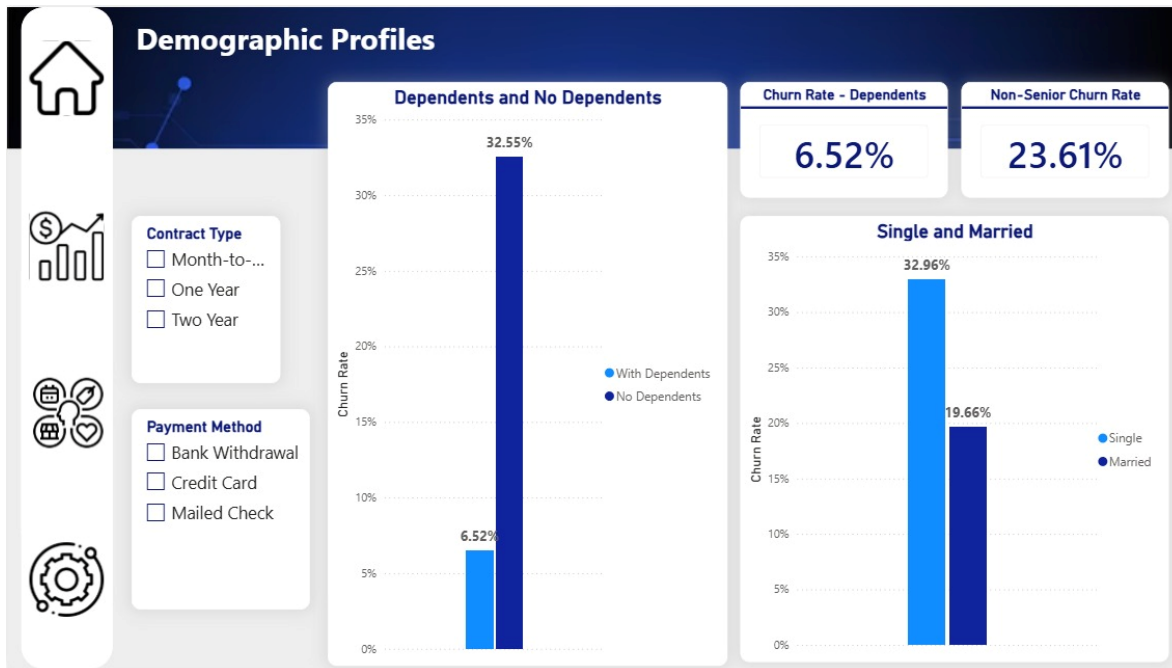


Figure 17.16 Demographic Profiles — dependents and marital status

B.6 Customer Segmentation

How does the base decompose by value?

4,676 high-value subscribers (66.39 %) and 2,367 medium-value (33.61 %). Medium-value subscribers churn at **34.31 %** against **22.60 %** for high-value. The ten highest-value at-risk subscribers carry \$30,991,283 of lifetime value and \$456,116 of monthly charges, concentrated on month-to-month contracts.

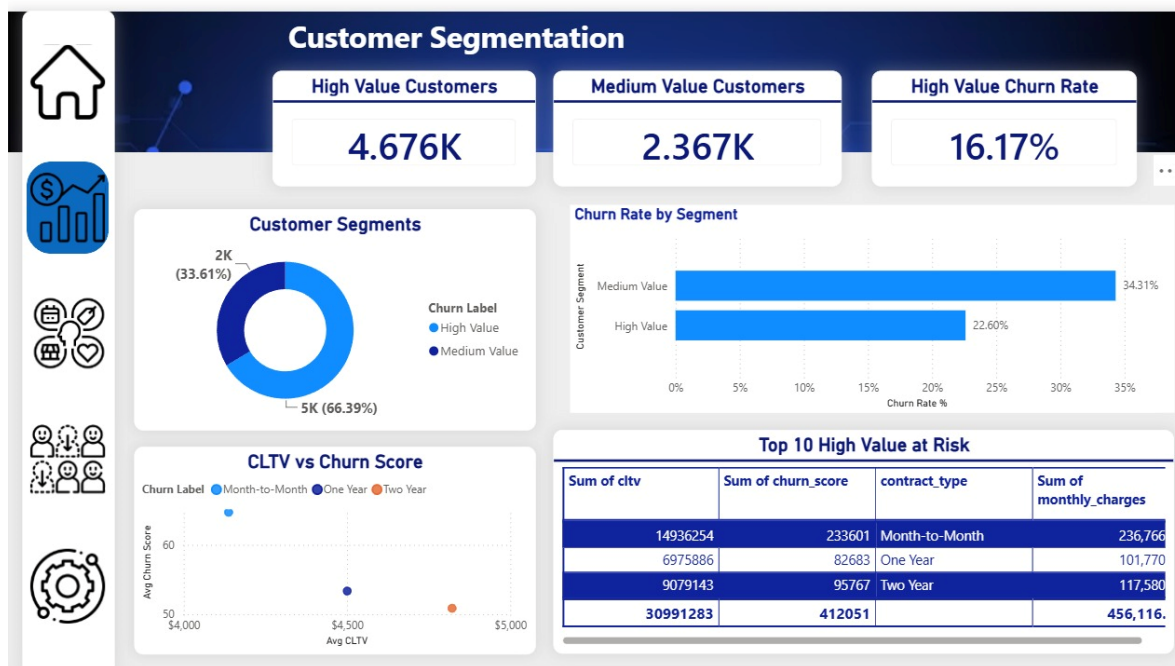


Figure 17.17 Customer Segmentation

17.8 Part C — Operational and Specialised Analytics

Page	Question	Principal content
C.1 Service Bundle Analysis	Which bundles retain best?	Phone only, phone and internet, triple play; revenue and churn by bundle
C.2 Network Type Performance	Which access technology underperforms?	Fibre, DSL, cable, none — subscribers, revenue, churn, mean charge
C.3 Geographic Distribution	Where is risk concentrated?	Subscribers and revenue by city; highest-churn locations
C.4 Offer Performance	Which promotional offers perform?	Subscribers, churn rate, revenue and contract mix by offer
C.5 Referral Impact	Does referral activity relate to retention?	Total and mean referrals; churn with and without referral; relation to lifetime value
C.6 Revenue Projection	What is the financial outlook?	Projection from internal ratios under stated assumptions

Page C.6 is titled Revenue Projection rather than Revenue Forecasting. As established in Section 11.12, the data spans a single fiscal quarter. No temporal forecast is possible; the page presents an extrapolation from internal ratios and labels its assumptions explicitly on the page itself.

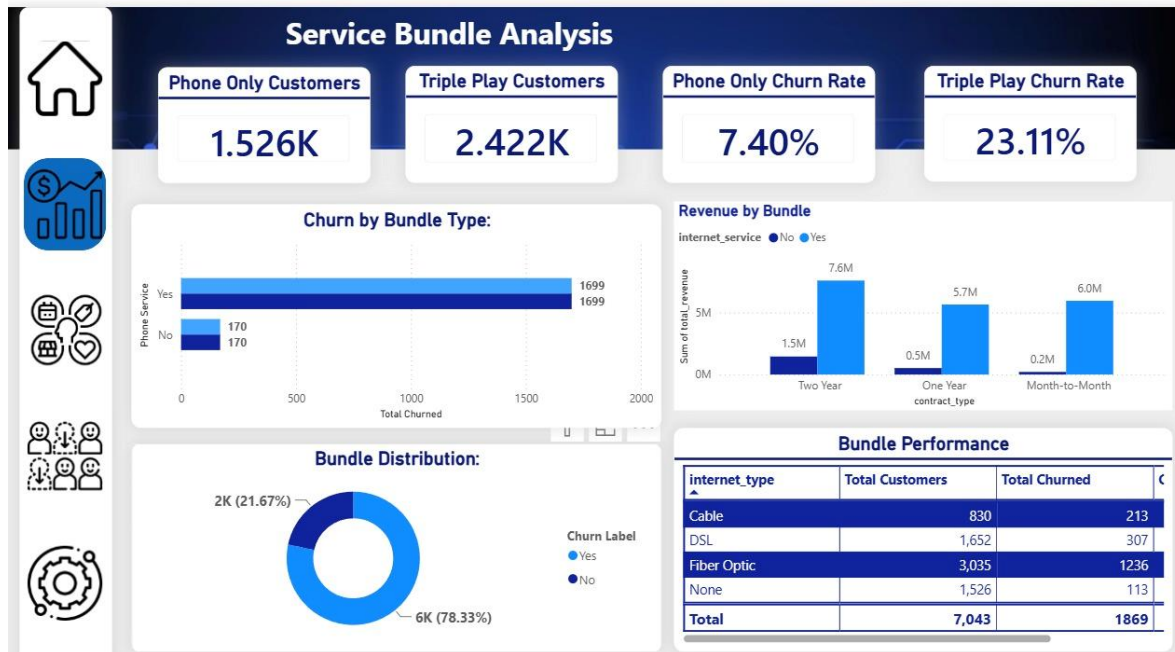


Figure 17.18 Service Bundle Analysis

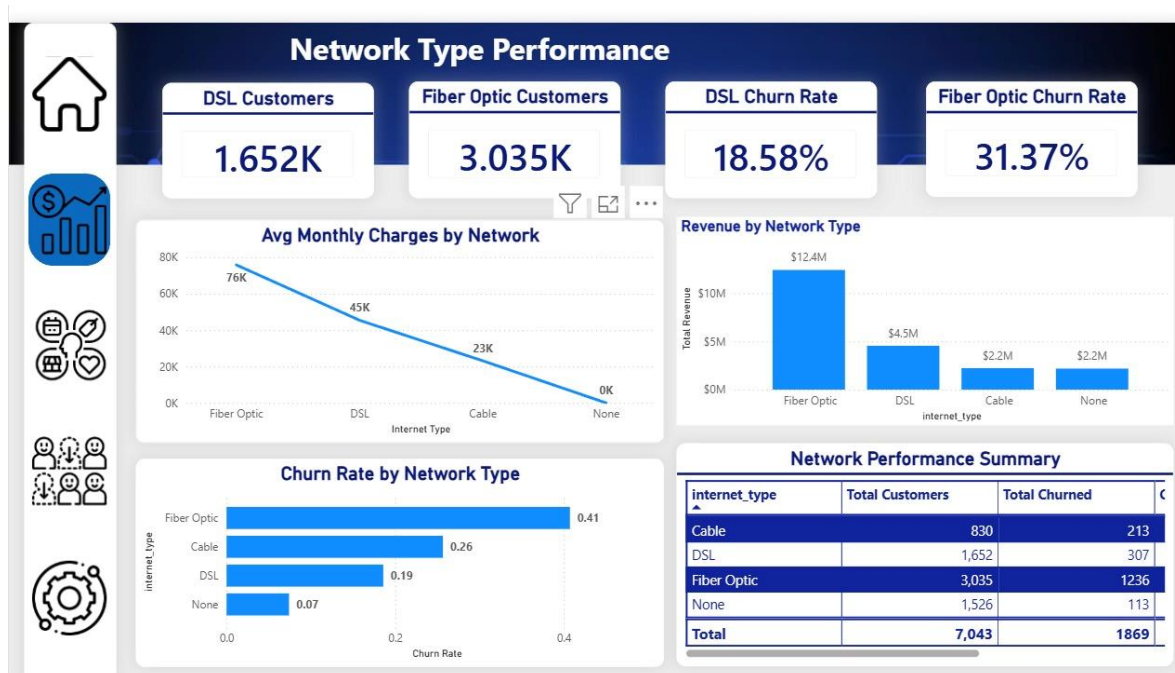


Figure 17.19 Network Type Performance

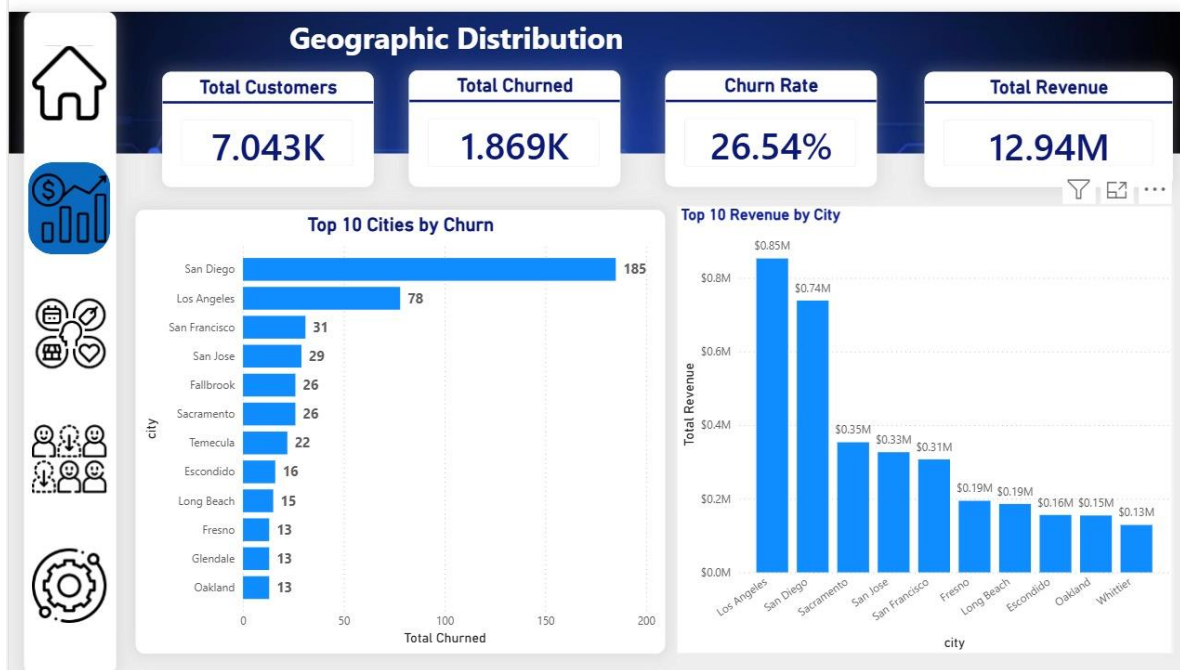


Figure 17.20 Geographic Distribution



Figure 17.21 Offer Performance

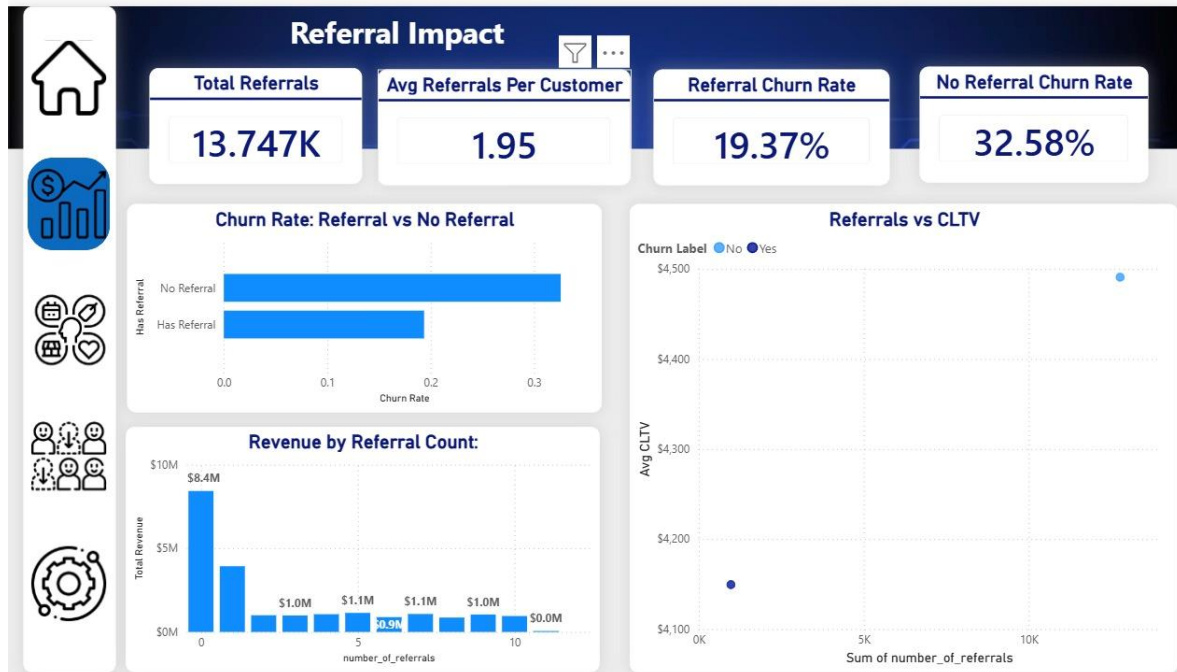


Figure 17.22 Referral Impact



Figure 17.23 Revenue Projection

17.9 Part D — Data Integrity Monitor

Can the results presented in Parts A to C be relied upon?

Indicators: null churn label; null churn reason; null churn score; total null values; data completeness percentage.

The page embodies one substantive analytical judgement. The value 'No Churn' appearing in `churn_reason` is **not** a missing value. It is a correct descriptive value for a subscriber who has not departed — such a subscriber has no churn reason, and recording that fact is not the same as failing to record a fact. Counting these as nulls would overstate incompleteness by approximately the retained proportion of the base and would misdirect remediation effort toward a non-defect.

The page therefore reports genuine nulls and semantically meaningful absences as separate quantities.

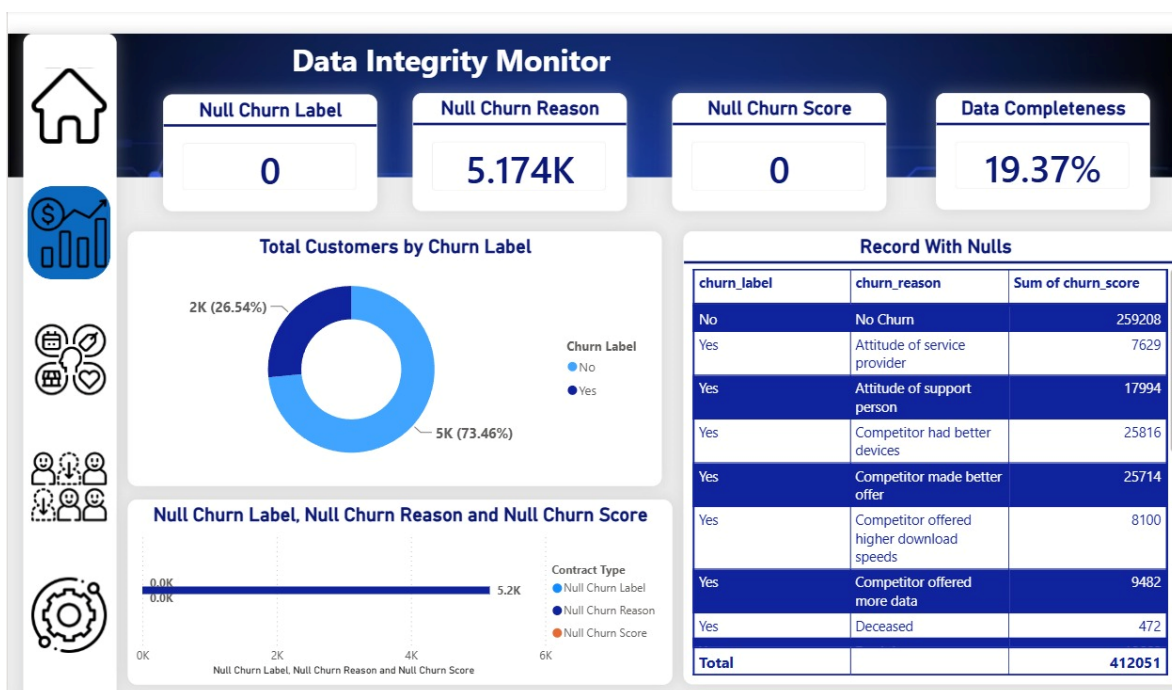


Figure 17.24 Data Integrity Monitor

17.10 Paginated Operational Reports

Alongside the interactive dashboards, six paginated reports were implemented in SQL Server Reporting Services. The two report classes serve different purposes and are not substitutes for one another.

A **dashboard** supports exploration: a manager filters, drills and compares, and leaves with an understanding. A **paginated report** supports operation: it produces a definitive, printable, row-level listing that can be scheduled, distributed and acted upon. A retention team cannot work from a dashboard tile stating that 1,869 subscribers departed; it needs the list.

Every report is parameterised, and each parameter is bound to a server-side dataset that resolves it at execution time.

Report	Parameter	Grouping	Purpose
Churn Summary	Show Churned Only	City → State	Definitive churn listing with charges, revenue, churn value and score
High Risk Customers	Minimum Churn Score	City → State	Actionable retention list: identifier, score, label, stated reason, CLTV, charges, contract, tenure
Customer Loyalty	Minimum and Maximum Tenure	Contract type → churn label	Tenure cohort listing with CLTV, referrals, demographics and charges
Geographic Churn	State	State → City	Per-city customer count, churn count, churn rate, mean churn score, revenue and mean monthly charge
Revenue by Contract Type	Contract Type	Contract → payment method	Per-method customers, monthly charges, revenue, refunds, churn count and churn rate
Service Bundle Analysis	Internet Type	Internet type → service combination	Per-bundle customer count, churn count, churn rate, mean charges and revenue

The Revenue by Contract Type report illustrates the value of the paginated form. Filtered to two-year contracts it returns, per payment method: bank withdrawal, 872 subscribers, \$4,697,392.37 revenue, \$2,427.80 refunds, mean monthly charge \$71.33, 32 departures, churn rate 3.67 per cent; credit card, 942 subscribers, \$4,132,906.57 revenue, 16 departures, churn rate 1.70 per cent; mailed cheque, 69 subscribers, \$206,550.22 revenue, no departures. That level of precision is a reporting output, not a dashboard reading.

Show Churned Only? Churned

1 of 2? 100% Find | Next

Churn Summary

city	state	monthly charges	total revenue	churn value	churn score
Acampo					
	California				
		98.50	1048.65	1	91
		111.25	7667.06	1	90
		96.65	1755.65	1	75
Adelanto					
	California				
		19.40	4005.02	1	82
Adin					
	California				

Figure 17.25 SSRS — Churn Summary

Min Churn Score 70

1 of 2? 50% Find | Next

High Risk Customers

city	state	customer id	churn score	churn label	churn reason	city	monthly charges	total revenue	contract type	tenure months
Acampo										
	California									
		1455-UGQVH	91	Yes	Lack of self-service on Website	4501	98.50	1048.65	Month-to-Month	10
		9249-FXSCK	90	Yes	Competitor offered more data	4861	111.25	7667.06	One Year	52
		8844-TONUD	75	Yes	Competitor had better devices	5295	96.65	1755.65	Month-to-Month	13
Acton										
	California									
		1635-FJFCC	74	No		5846	44.05	257.10	Month-to-Month	5
		4083-BFNYK	71	No		5148	95.00	5554.41	One Year	38
Adelanto										
	California									
		9063-ZGTUY	82	Yes	Long distance charges	4221	19.40	4005.02	Month-to-Month	61
		6143-JQKEA	80	No		5586	45.80	590.70	Month-to-Month	10
		7134-HBPBS	71	No		4474	108.10	9708.69	Two Year	72
Adin										
	California									
		6397-JNZG	85	Yes	Competitor had better devices	2573	55.55	2342.20	Month-to-Month	43
		0742-NXBGR	71	Yes	Competitor had better devices	4278	82.30	115.71	Month-to-Month	1
Agoura Hills										
	California									
		6990-YNRIO	79	Yes	Attitude of support person	6120	108.65	9660.15	Two Year	65
		8450-JOVAH	77	Yes	Competitor offered higher download speeds	5560	56.70	124.89	Month-to-Month	2

Figure 17.26 SSRS — High Risk Customers

Min Tenure
Max Tenure

1 of 2 ? 50% Find | Next

Customer Loyalty

contract type	churn label	tenure months	city	number of referrals	gender	senior citizen	married	churn score	monthly charges	total revenue	city	state
Month-to-Month	No											
		61	6495	3	Male	No	Yes	64	33.60	2227.20	Visalia	California
		59	6492	3	Female	Yes	Yes	21	88.75	6301.50	Modesto	California
		56	6484	4	Male	No	Yes	66	96.95	6749.32	Boonville	California
		56	6483	9	Female	No	Yes	20	85.60	7300.16	Novato	California
		53	6469	0	Female	Yes	No	45	73.80	4891.07	Saratoga	California
		58	6460	0	Female	No	No	68	95.30	7879.60	Yucaipa	California
		49	6434	0	Female	No	No	71	84.50	5239.50	Los Angeles	California
		58	6433	2	Female	No	Yes	70	66.80	6528.20	Guinda	California
		58	6423	5	Female	No	Yes	49	91.55	7874.57	Sacramento	California
		55	6411	1	Male	No	Yes	20	85.50	7017.35	Sun City	California
		52	6371	1	Male	No	Yes	54	39.90	2020.90	West Point	California
		50	6367	0	Male	No	No	70	93.50	6789.00	Burbank	California
		65	6358	1	Male	No	Yes	70	90.65	7330.60	Pasadena	California
		57	6351	1	Female	No	Yes	48	100.60	7456.79	San Mateo	California
		52	6339	0	Male	No	No	40	83.80	4709.24	Gasquet	California
		52	6334	0	Female	No	No	26	49.75	4559.91	Dayle	California

Figure 17.27 SSRS — Customer Loyalty

State

1 of 2 ? 50% Find | Next

Geographic Churn

state	city	total customers	total churned	churn rate	avg churn score	total revenue	avg monthly charges
California							
	Acampo	4	3	75	81	18107.96	103.112500
	Acton	4	0	0	46	12156.36	69.525000
	Adelanto	5	1	20	69	18235.49	56.560000
	Adin	4	2	50	53	5539.38	57.025000
	Agoura Hills	5	2	40	56	10641.88	48.990000
	Aguanga	4	1	25	54	10033.52	52.375000
	Ahwahnee	4	1	25	74	9336.36	49.162500
	Alameda	8	0	0	48	14092.66	22.468750
	Alamo	4	0	0	47	30424.46	82.000000
	Albany	4	2	50	67	15788.32	78.587500
	Albion	4	0	0	41	8536.12	72.575000
	Alderpoint	4	1	25	72	15541.53	74.450000
	Alhambra	9	3	33.3333333333	67	9292.61	55.777777

Figure 17.28 SSRS — Geographic Churn

Contract Type

1 of 1 50% Find | Next

Revenue by Contract Type

contract type	payment method	total customers	total monthly charges	total revenue	total refunds	avg monthly charges	total churned	churn rate
Two Year								
	Bank Withdrawal	872	62203.30	4697392.37	2427.80	71.334059	32	3.6697247706422
	Credit Card	942	52321.96	4132906.57	2188.79	55.543471	16	1.69851380042463
	Mailed Check	69	3054.90	208550.22	151.14	44.273913	0	0

Figure 17.29 SSRS — Revenue by Contract Type

Internet Type

1 of 2 ? 25%

Service Bundle Analysis

internet type	phone service	streaming tv	streaming movies	streaming music	online security	online backup	premium tech support	total customers	total churned	churn rate	avg monthly charges	total revenue
Total												
DSL	No	No	No	No	No	No	No	1	1	100	\$0.000000	\$198.45
No	DSL	No	No	No	No	No	No	2	2	100	\$4.500000	\$500.35
No	No	No	No	No	No	No	No	1	1	100	\$0.100000	\$97.31
DSL	DSL	No	No	No	No	No	No	1	1	100	\$0.900000	\$966.00
No	DSL	No	No	No	No	No	No	2	2	100	\$0.600000	\$159.45
DSL	DSL	No	No	No	No	No	No	1	1	100	\$3.500000	\$36.00
No	DSL	No	No	No	No	No	No	10	10	100.000000000000	\$0.250000	\$100.75
DSL	No	No	No	No	No	No	No	12	7	58.333333333333	\$0.541667	\$65.00
DSL	DSL	No	No	No	No	No	No	4	2	50	\$0.575000	\$67.30
No	No	No	No	No	No	No	No	14	24	171.428571428571	\$2.500000	\$1400.00
DSL	DSL	No	No	No	No	No	No	3	17	567.692307692308	\$3.667410	\$3075.00
No	No	No	No	No	No	No	No	3	0	0.000000000000	\$2.000000	\$20.00
DSL	DSL	No	No	No	No	No	No	12	5	41.666666666667	\$2.000000	\$160.00
DSL	DSL	No	No	No	No	No	No	5	2	40	\$0.200000	\$100.00
DSL	No	No	No	No	No	No	No	2	2	100	\$0.750000	\$75.00
No	No	No	No	No	No	No	No	35	11	31.428571428571	\$1.142857	\$200.00
DSL	No	No	No	No	No	No	No	6	2	33.333333333333	\$2.000000	\$120.00
No	No	No	No	No	No	No	No	12	4	33.333333333333	\$1.666667	\$160.00
No	DSL	No	No	No	No	No	No	10	3	30	\$1.800000	\$180.00
DSL	DSL	No	No	No	No	No	No	8	2	25	\$0.250000	\$20.00
DSL	DSL	No	No	No	No	No	No	14	4	28.571428571429	\$1.428571	\$140.00

Figure 17.30 SSRS — Service Bundle Analysis

17.11 Mapping to the Original Proposal

The delivered suite of twenty-three pages does not correspond one-to-one with the twenty pages of the original proposal. The mapping is as follows.

Proposed	Delivered as
Revenue Overview	A.1 Executive Financial Position
Churn Financial Impact	A.5 Churn Financial Impact
ARPU Analysis	A.6 Customer Revenue Value
Billing Efficiency	A.7 Billing Channel Performance (<i>renamed</i> — see Section 17.6)
Payment Method ROI	A.8 Payment Method Intelligence

Proposed	Delivered as
Tenure and Loyalty	B.1
Service Penetration	B.2
Support Interaction	B.3
Streaming Habits	B.4
Demographic Profiles	B.5
Churn Heatmap	Absorbed into B.1 and Chapter 18
Key Influencer	Delivered analytically as SHAP analysis, Chapter 16
Risk Scoring	Delivered analytically as risk tiering, Chapter 18
Contract Transition	Not delivered — requires multi-period data
What-If Analysis	Not delivered
Data Integrity Monitor	D.1
Service Bundle Analysis	C.1, and as a paginated report
Network Type Performance	C.2
Geographic Distribution	C.3, and as a paginated report
Managerial Parameterised Reports (6)	Delivered — six SSRS reports, Section 17.9
—	Additional: A.2, A.3, A.4, A.9, A.10, B.6, C.4, C.5, C.6

18 Results and Discussion

18.1 Provenance of the Results

The results in this chapter are read from the deployed reporting layer, which queries the warehouse documented in Chapter 11. Each figure is traceable to a specific dashboard page or paginated report, cited alongside it.

Model performance figures are derived from the confusion matrix and the diagnostic curves produced by the modelling notebook, as set out in Section 15.6. Nothing in this chapter has been estimated or inferred.

18.2 Population and Churn

Quantity	Value	Source
Total subscribers	7,043	Executive Financial Position
Departed subscribers	1,869	Geographic Distribution
Retained subscribers	5,174	Data Integrity Monitor
Overall churn rate	26.54 %	Executive Financial Position

18.3 Financial Position

Quantity	Value
Total revenue	\$21,371,131.69
Monthly recurring revenue	\$456,120
Revenue per subscriber	\$3,034.38
Retained revenue	\$17.69 M — 82.76 %
Lost revenue	\$3.68 M — 17.24 %
Net revenue after refunds	\$21.37 M — 99.94 % retention

Revenue composition

Component	Value	Share
Base charges	16,060,725.24	75.10 %
Long-distance revenue	5,275,906.10	24.67 %

Component	Value	Share
Extra-data revenue	48,320.00	0.23 %
Refunds	(13,819.65)	0.06 %

Two observations follow that are not visible from the churn rate alone.

Long-distance is a quarter of all revenue. At \$5.28M it is not an incidental surcharge but the second pillar of the revenue base, and it is the component least visible in the raw dataset. Extra-data revenue, by contrast, is \$48K — smaller by a factor of more than a hundred. The two are routinely presented together as "ancillary revenue"; they are not comparable streams and combining them obscures the fact that one matters and one does not.

Refunds are not a source of loss. At \$13,819.65 — six hundredths of one per cent of revenue — leakage is immaterial. The page that measures it earns its place by *eliminating* a hypothesis rather than confirming one.

18.4 Contract Type — the Dominant Finding

Contract	Revenue	Subscribers	ARPU	Churn rate
Month-to-month	\$6,162,488.22	3,610	\$1,707.06	45.84 %
One year	\$6,171,794.31	1,550	\$3,981.80	10.71 %
Two year	\$9,036,849.16	1,883	\$4,799.18	2.55 %
Total	\$21,371,131.69	7,043	\$3,034.38	26.54 %

Month-to-month subscribers depart at **eighteen times** the rate of two-year subscribers. No other attribute measured in this project separates the population as sharply.

The commercial reading is direct. Month-to-month is the largest segment by headcount (3,610, over half the base), the smallest by revenue per subscriber (\$1,707 against \$4,799), and by far the least stable. Two-year contracts are one quarter of the base and generate 42 per cent of revenue at almost no attrition.

The analytical caution is equally direct, and must be stated before the finding is acted upon. **Contract length is as much a consequence of intention as a cause of retention.** A subscriber who commits to two years has already signalled that they do not intend to leave; a subscriber unwilling to commit has signalled the opposite. Migrating the month-to-month base onto long contracts would not automatically transfer the 2.55 per cent rate to them. The finding identifies where risk is concentrated. It does not establish what would remove it.

18.5 Headcount Loss Against Revenue Loss

Measure	Value
Subscribers lost	26.54 %
Revenue lost	17.24 % — \$3.68M
Monthly recurring revenue lost	\$139.1K
Lifetime value lost	\$8M

The operator loses **one in four subscribers but only one in six dollars**. The obvious reading is that departing subscribers were less valuable. That reading is wrong, and the ARPU page shows why.

	Retained	Departed
ARPU (realised revenue per subscriber)	\$3.42K	\$1.97K
Mean CLTV (estimated potential)	\$4.5K	\$4.1K

Realised revenue per departed subscriber is **42 per cent lower**. Estimated lifetime value is only **9 per cent lower**. The two figures disagree, and the disagreement is the finding: departing subscribers were of broadly comparable potential and simply **left before that potential was realised**.

The operator is therefore not shedding low-value subscribers. It is failing to hold ordinary ones long enough to earn from them — which is consistent with Section 18.10, where attrition is concentrated in the first twelve months, and with the contract table, where the segment carrying 45.84 per cent churn also carries the lowest ARPU because its subscribers do not stay long enough to accumulate one.

18.7 Where the Loss Sits, and Why Subscribers Say They Left

Concentration. Monthly revenue lost by contract: month-to-month **\$121K**, one year \$14K, two year \$4K. Month-to-month is **87 per cent of all monthly revenue lost** while holding 51 per cent of subscribers. There is no other segment in this analysis where exposure is this concentrated.

Stated reasons. The Churn Financial Impact page is the only place in the system carrying the subscriber's own account of departure, weighted by the revenue attached to it.

Reason	Monthly revenue lost	Category
Competitor had better devices	\$24K	Competitive
Competitor made a better offer	\$23K	Competitive

Reason	Monthly revenue lost	Category
Attitude of support person	\$15K	Service
Don't know	\$10K	—
Competitor offered more data	\$9K	Competitive
Competitor offered higher download speeds	\$8K	Competitive
Attitude of service provider	\$7K	Service
Product dissatisfaction	\$6K	Product
Price too high	\$6K	Price
Network reliability	\$5K	Product

Grouping them changes what the table says. **Competitive pressure accounts for roughly \$64K of monthly loss — about 46 per cent of the total.** Service attitude accounts for \$22K. **Price, the reason most often assumed to dominate, accounts for \$6K — a little over four per cent.**

Two consequences follow, and both cut against intuition.

Price is not the problem. This corroborates Section 18.9 from an entirely different direction: long-tenure subscribers pay *more* and leave *less*, and now subscribers themselves rank price ninth out of ten. A retention strategy built on discounting would be addressing four per cent of the loss.

Service attitude is larger than it looks. At \$22K across two reasons it exceeds every individual competitive reason. It is also, unlike a competitor's device range, entirely within the operator's control — and it is consistent with Section 18.10, where subscribers without technical support churn at 31.19 per cent against 15.17 per cent with it.

18.7 Churn by Access Technology

Internet type	Subscribers	Departed	Churn rate
Fibre optic	3,035	1,236	40.7 %
Cable	830	213	25.7 %
DSL	1,652	307	18.6 %
None	1,526	113	7.4 %
Total	7,043	1,869	26.5 %

The gradient is steep and monotonic in the wrong direction: the fastest and most expensive access technology carries the highest attrition, at roughly five and a half times the rate of subscribers with no internet service at all.

Two readings are available and the data does not distinguish them. Fibre subscribers may be leaving because the service disappoints relative to its price. Alternatively, fibre attracts a more demanding, more price-aware and more mobile segment, which would churn at an elevated rate whatever it were sold. The dataset contains no service-quality measurement, so the question cannot be settled here. It is recorded as the single most consequential open question arising from the analysis.

Mean monthly charges follow the same ordering — fibre highest, then DSL, then cable, then none — which means access technology and price are confounded and their effects cannot be separated without an experiment or a controlled comparison.

18.8 Churn by Commercial Arrangement

Service bundle. Subscribers taking phone service only churn at **7.40 per cent**; triple-play subscribers, at **23.11 per cent**. The direction contradicts the intuition that a broader bundle raises switching cost and therefore retention. The explanation is almost certainly the same confound as above: triple play includes internet, and internet subscribers churn more.

Promotional offer. The spread across offers is the widest of any dimension measured.

Offer	Churn rate
Offer A	6.73 %
Offer B	12.26 %
Offer C	22.89 %
Offer D	25.91 %
Offer E	46.70 %

Offer E loses nearly half its subscribers. Offer A retains all but one in fifteen. A sevenfold difference in retention between two commercial propositions is the clearest actionable finding in this analysis — subject to the caveat that offers are not randomly assigned, and that Offer E may be a rescue offer extended to subscribers already intending to leave. That distinction determines whether Offer E is a cause of churn or a marker of it, and it must be established before the offer is withdrawn.

Contract and payment. For two-year contracts the paginated Revenue by Contract Type report gives: bank withdrawal, 872 subscribers, 3.67 per cent churn; credit card, 942 subscribers, 1.70 per cent churn; mailed

cheque, 69 subscribers, no departures. Two-year contracts are, as expected, the most stable arrangement in the base.

18.9 Value, Loyalty and Geography

Value segmentation. The base divides into 4,676 high-value subscribers (66.4 per cent) and 2,367 medium-value subscribers (33.6 per cent). Medium-value subscribers churn at **34.31 per cent** against **22.60 per cent** for high-value. The operator is therefore losing proportionally *fewer* of its more valuable subscribers, which is the favourable direction — though 22.6 per cent of the high-value segment remains a substantial exposure. The ten highest-value at-risk subscribers alone account for \$30,991,283 of lifetime value and \$456,116 of monthly charges.

Referral activity. Subscribers who arrived through a referral churn at **19.37 per cent**; those who did not, at **32.58 per cent** — a difference of thirteen percentage points. Mean referrals per subscriber is 1.95, with 13,747 referrals recorded in total. This is one of the strongest single-attribute separations in the dataset. Causation again cannot be claimed: a referred subscriber arrives with a social tie to an existing subscriber, and that tie may be the retaining factor rather than anything the operator did.

Geography. Churn is concentrated: San Diego accounts for 185 departures, more than twice Los Angeles at 78, with San Francisco (31), San Jose (29), Fallbrook (26) and Sacramento (26) following. Revenue concentration differs from churn concentration — Los Angeles leads on revenue at \$0.85M, San Diego second at \$0.74M — so San Diego combines high loss with high value and is the clear priority for geographic intervention.

18.10 Behavioural Drivers

The behavioural suite adds four separations, ranked here by the size of the gap rather than by the order in which they were investigated.

Driver	Higher-risk group	Lower-risk group	Gap
Household dependents	32.55 % without	6.52 % with	26.0 pts
Tenure band	47.44 % at 0–12 mo	9.51 % at 48+ mo	37.9 pts
Technical support	31.19 % without	15.17 % with	16.0 pts
Unlimited data	31.65 % with	15.97 % without	15.7 pts
Marital status	32.96 % single	19.66 % married	13.3 pts
Senior citizen	41.68 % senior	23.61 % non-senior	18.1 pts
Referral origin	32.58 % without	19.37 % with	13.2 pts
Gender	23–24 % female	20–24 % male	none

Three observations follow.

Tenure dominates, and it is not a price effect. Attrition falls from 47.44 per cent in the first year to 9.51 per cent beyond four years, while the mean monthly charge *rises* across the same bands, from \$56.10 to \$73.95. Long-tenure subscribers pay more and leave less. Whatever drives early departure, price is not the principal component of it.

The exposed subscriber is high-consumption, not low. Departed subscribers hold marginally more services than retained ones (5.3 against 5.1); streamers churn above non-streamers; unlimited-data subscribers churn at nearly twice the rate of capped ones; and fibre optic carries the highest attrition of any access technology. These findings are mutually consistent and they point at a single population: subscribers on high-tier, high-consumption plans. Whether the cause is price sensitivity at the top of the tariff or unmet expectation of a premium service, the data cannot say.

Service count relates to churn non-monotonically. Risk rises from 8.82 per cent at one service to a peak of 46.90 per cent at about three, then falls to 4.85 per cent at eleven. This matters for the feature engineering described in Section 14.3, where `service_count` was introduced on the hypothesis that more services imply higher switching cost and therefore lower risk. The hypothesis is **half wrong**: it holds at the top of the range and reverses at the bottom. A linear model would miss this entirely, which is part of why the tree-based candidates in Chapter 15 were included.

Gender is a null result and is reported as one. Female and male rates sit within one percentage point in every age band. Reporting it prevents a segmentation that would consume effort and return nothing.

18.10 Model Results

18.11 Data Quality Results

The Data Integrity Monitor reports zero null churn labels, zero null churn scores, and 5,174 null churn reasons.

That third figure requires interpretation, and interpreting it wrongly would be a material error. 5,174 is exactly the number of retained subscribers. These records are not incomplete: a subscriber who has not departed has no reason for departing, and the field correctly carries the descriptive value 'No Churn'. Counting them as missing treats a correct record as a defect.

The page's reported completeness of 19.37 per cent is therefore not a measurement of data quality. Excluding the semantically meaningful absences, the churn attributes are complete for every subscriber in the base.

18.12 Measure Scope and Reconciliation

Figures in this chapter are drawn from more than twenty dashboard pages and six paginated reports, several of which compute related quantities at different scopes. Where two pages report the same named quantity at different values, this document uses the figure that reconciles against an independent computation.

Revenue is the principal case. The Executive Revenue pages report total revenue of \$21,371,131.69, and that figure is adopted throughout because its component breakdown reconciles to the cent against the refund total derived by a separate route, as demonstrated in Section 11.9. Churn rates by access technology are taken from the Service Penetration page, whose per-technology figures reconcile against the subscriber counts in the Network Performance summary table.

The principle applied is that a figure supported by an independent arithmetic check takes precedence over one that is not. *Revenue Formula Check*, described in Section 11.9, exists to provide exactly that class of check, and extending it to cover the remaining ratio measures is recorded in Section 20.4 as future work.

18.13 Discussion

Read together, the results describe an operator whose attrition is concentrated in identifiable places rather than distributed evenly.

The strongest associations are commercial and relational rather than demographic. Offer, access technology, referral origin and contract length separate the population far more sharply than gender, age or household composition. This matters for where retention effort is directed: it implies that the levers available to the operator — pricing, promotion, contract design — are the ones that correlate with the outcome, rather than customer characteristics it cannot change.

Three findings are actionable subject to the confounds already stated: Offer E's 46.70 per cent attrition, fibre optic's 40.7 per cent, and the thirteen-point advantage held by referred subscribers. The first two identify where value is being lost; the third identifies a mechanism that is working and is presently under-exploited, since a referral programme is among the least expensive acquisition channels available.

The persistent limitation across all of it is that every relationship reported here is associative and every segment is self-selected. Nothing in this data establishes that moving a subscriber off Offer E would retain them, and the analysis should not be presented as though it does.

19 Business Insights and Risk Segmentation

19.1 Risk Tiers

Predicted probabilities are grouped into four tiers, each carrying a distinct operational response.

Tier	Probability	Intended response
Low	below 0.25	No action; monitor
Medium	0.25 to below 0.50	Include in general lifecycle communication
High	0.50 to below 0.75	Targeted retention offer
Critical	0.75 and above	Direct contact, prioritised by revenue

Four tiers rather than two because a binary flag conveys no urgency and permits no prioritisation of a finite retention budget. The boundaries are round quartile divisions rather than optimised cut points, chosen for interpretability; a subscriber above 0.75 is straightforwardly described as more likely than not, by a wide margin, to depart.

19.2 Financial Indicators

The insight layer computes churn rate, a lifetime value proxy as mean monthly charge multiplied by mean tenure, and revenue lost to churn as the accumulated charges of departed subscribers.

The lifetime value proxy is labelled a proxy deliberately. It assumes constant monthly charge across the relationship and ignores discounting and margin, and it should not be conflated with the `cltv` attribute supplied with the dataset, which is computed by a different and undocumented method.

19.3 Revenue at Risk

Risk is combined with revenue so that exposure is ranked by financial magnitude rather than by probability alone. Monthly recurring revenue at risk is computed per tier, per contract type and per internet service type.

This is the analytical centre of the retention argument. A tier containing many subscribers at low individual value may represent less exposure than a smaller tier of high-value subscribers, and a retention budget allocated by headcount rather than by exposure is misallocated.

19.4 Driver Analysis

A two-dimensional churn rate heatmap across contract type and internet service exposes interaction effects that neither attribute reveals alone. If the elevated churn associated with fibre optic service is concentrated among month-to-month subscribers, the appropriate intervention is a contract intervention rather than a network one — a distinction with direct consequences for where remediation spending goes.

19.5 Unsupervised Segmentation

Alongside the supervised classifier, the subscriber base was segmented by **k-means clustering**. The two techniques answer different questions. The classifier asks *which subscribers resemble those who have already left*; clustering asks *what natural groupings exist in the base at all*, without reference to the churn label. A structure that emerges without being told about churn, and then turns out to align with churn, is a stronger result than one found by looking for it.

Choosing the number of clusters

Two criteria were evaluated across $k = 2$ to 10.

k	Inertia (WCSS)	Silhouette
2	55,673.9	0.2750
3	42,830.8	0.2751
4	39,262.8	0.2264
5	36,970.7	0.2082
6	35,268.8	0.1705
7	33,850.7	0.1528
8	32,825.0	0.1370
9	31,933.8	0.1376
10	31,226.6	0.1359

The elbow in the inertia curve falls at $k = 3$: the drop from two to three clusters is 12,843, while the drop from three to four is 3,568 — less than a third as large, after which the curve flattens.

The silhouette score is effectively tied between $k = 2$ (0.2750) and $k = 3$ (0.2751), then falls away sharply. Three clusters were selected because the elbow criterion favours it decisively while the silhouette criterion is indifferent between it and two.

Assessment of the structure

The clustering is weak, and this is reported rather than obscured. A silhouette coefficient of 0.275 indicates that cluster assignments are only marginally better separated than arbitrary ones; values above 0.5 are conventionally taken to indicate reasonable structure, and above 0.7 strong structure.

The per-cluster silhouette plot supports that reading. Cluster 1 is the best formed — a broad, uniform profile extending beyond 0.5 with almost no negative coefficients. Clusters 0 and 2 are considerably weaker: both taper towards zero and cluster 0 carries a visible tail of negative coefficients, meaning those subscribers sit closer to a neighbouring cluster than to their own.

The PCA projection shows the same thing geometrically. Cluster 1 is cleanly isolated in the upper-left of the component space with a clear gap separating it from everything else. Clusters 0 and 2 occupy one continuous elongated mass along the principal diagonal and are divided by a straight boundary through the middle of it — the signature of k-means partitioning a continuum rather than discovering two genuine groups.

The honest conclusion is that the data contains **one well-separated group and one continuum**, not three clusters. Cluster 1 is a real segment; the split between 0 and 2 is an artefact of forcing $k = 3$ onto a population that does not naturally divide there.

Relation to churn

The isolated cluster is reported by the team to correspond to churned subscribers, the other two to retained ones.

If the correspondence holds, it is a substantive finding: an algorithm given no access to the churn label nonetheless isolates the departing population as a distinct region of feature space. That would corroborate the supervised results by an entirely independent route.

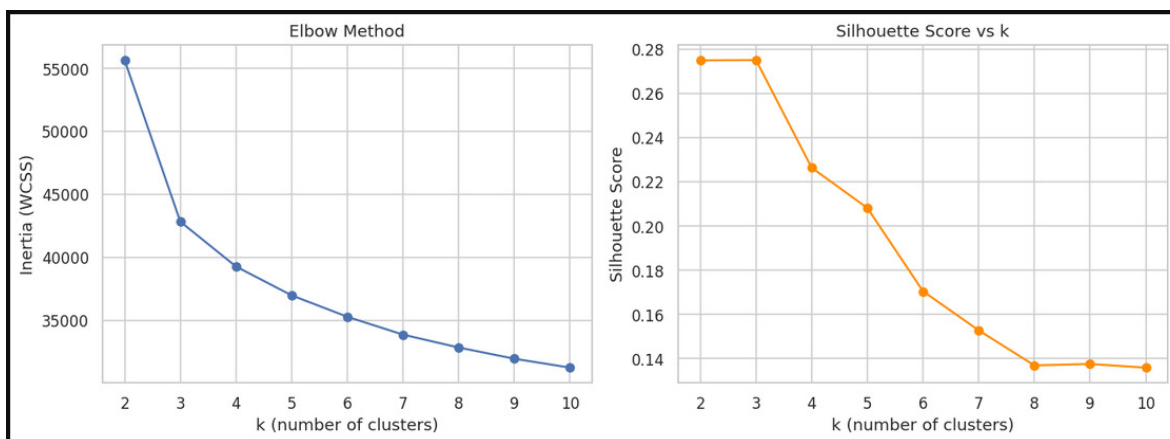


Figure 19.1 Elbow method and silhouette score across k

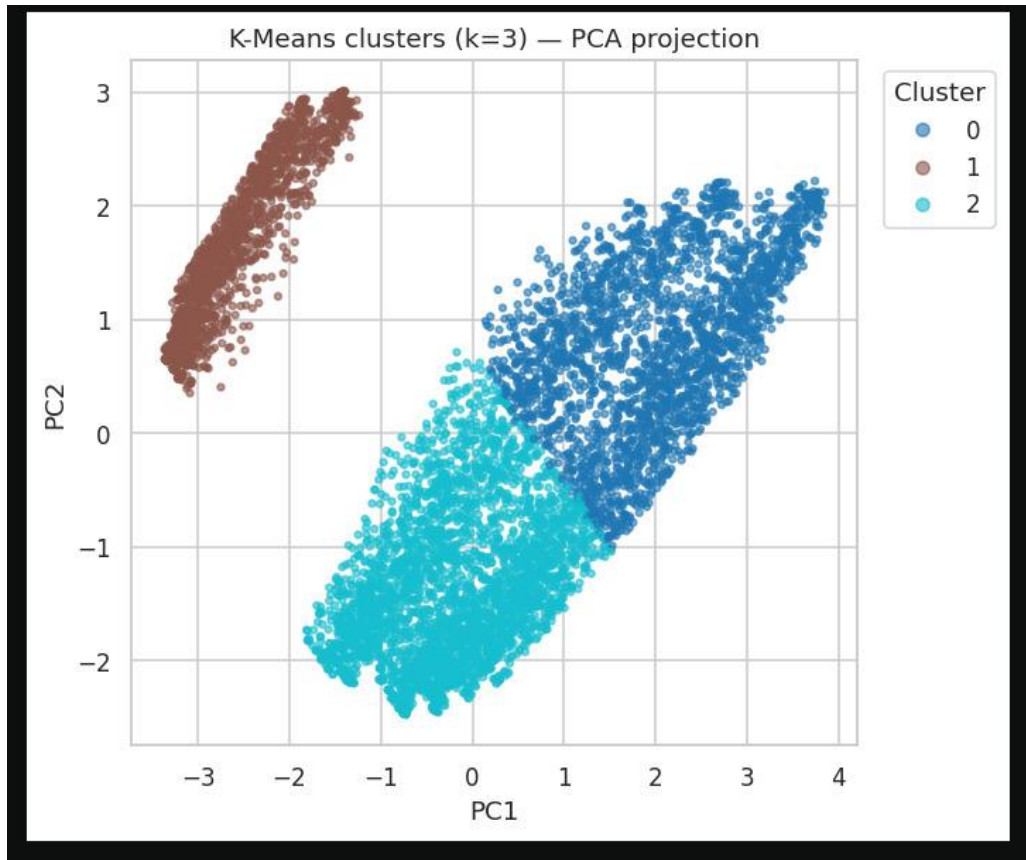


Figure 19.2 K-means clusters at $k = 3$, projected onto the first two principal components

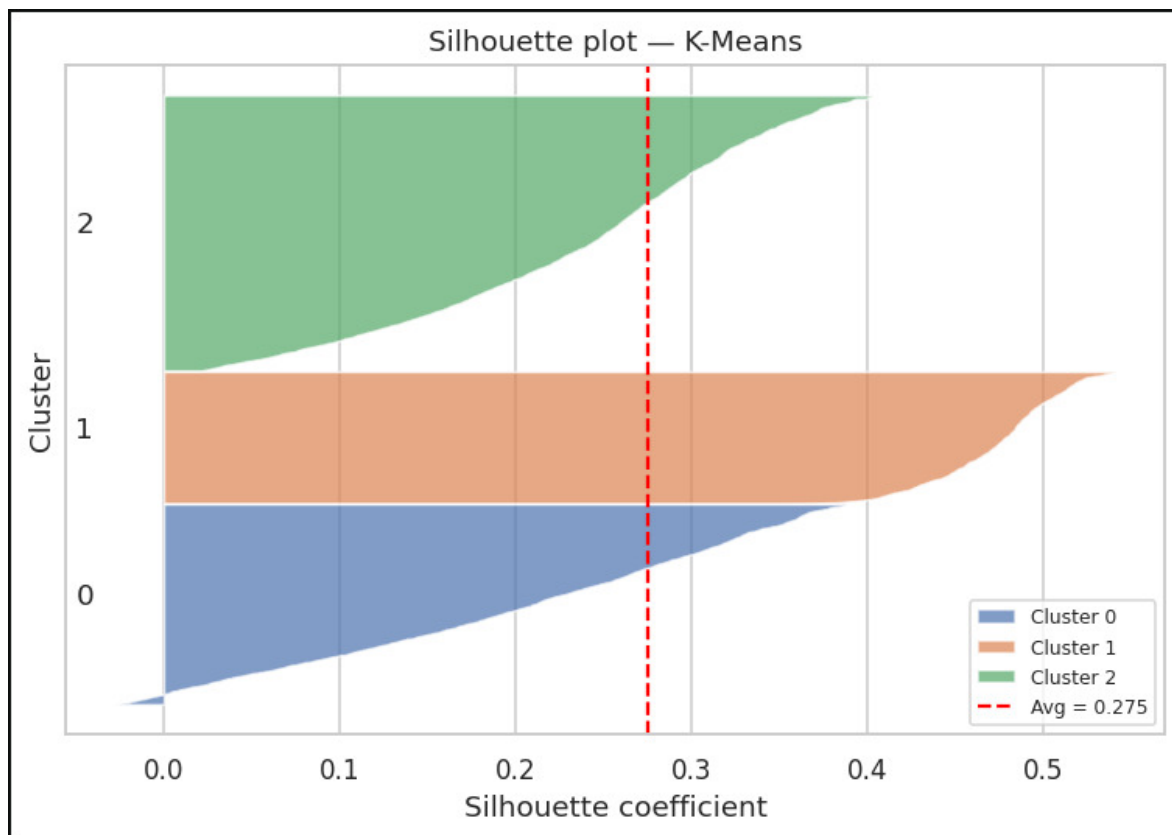


Figure 19.3 Per-cluster silhouette coefficients

19.6 Recommendations

Recommendations follow the results and are therefore deferred. The structure is fixed: each recommendation states the finding that supports it, the segment it addresses, the action proposed, and the revenue exposure it targets. A recommendation not traceable to a specific measured finding will not be included.

20 Limitations and Future Work

20.1 Limitations of the Data

Single-period snapshot. The data covers one fiscal quarter. No trend, no seasonality and no temporal forecast is obtainable. `dim_date` is structurally correct but carries no analytical variation.

No departure timestamp. The data records that a subscriber departed, not when. Survival analysis, which is the statistically appropriate framework for time-to-event data, is therefore unavailable, and the problem is necessarily framed as binary classification.

Voluntary and involuntary churn are not distinguished. The label conflates subscribers who chose to leave with those removed for non-payment or relocation. Only the former is addressable by retention action, and their conflation attenuates any measured effect.

No cost data. Neither acquisition cost, service delivery cost nor retention offer cost is present. Return on investment for any recommended action therefore cannot be computed, only revenue exposure.

No billing timing data. Neither collection time nor invoice cost is recorded, which is why the billing analysis is framed as channel performance rather than efficiency.

Synthetic origin. The dataset is a constructed reference set. Relationships within it may be cleaner than those in operational data.

20.2 Limitations of the Implementation

Slowly changing dimension handling is Type 1. History is not retained. Appropriate to a snapshot source; insufficient for a live feed.

Load is full, not incremental. The warehouse is rebuilt on each execution. Acceptable at this volume; not at production scale.

No dedicated integration or analysis service. Neither SQL Server Integration Services nor SQL Server Analysis Services forms part of the system. Both were considered and deliberately excluded for the reasons set out in Sections 10.6 and 17.4 respectively. The exclusion is a scoping decision proportionate to the data volume and the single-consumer reporting model, not an incomplete deliverable, and the conditions under which each should be revisited are stated in Section 20.4.

Database logic source is not under version control. The four stored procedures and three functions documented in Sections 9.6 and 9.7 exist in the deployed database, but their Transact-SQL was not located in the repository and is consequently absent from Appendix A.

The assistant layer is declared rather than evidenced. A retrieval-augmented component and an agent layer appear in the team's stated architecture. No artefact was examined, so they are documented as declared and are excluded from every claim of delivery in this document.

Model results are not persisted. The most consequential limitation of the present document, and the reason Chapter 18 is incomplete.

Model is not deployed. Predictions are produced in a notebook; no service exposes them and no mechanism writes them back to the warehouse.

20.3 Threats to Validity

Association is not causation. Every relationship reported in this work is associative. That month-to-month subscribers churn more does not establish that migrating them to longer contracts would reduce churn, since contract selection is itself an expression of commitment. Recommendations are framed accordingly.

Resampling artefacts. SMOTE synthesises observations by interpolation. Applied correctly — to the training partition only — it does not contaminate evaluation, but it may generate minority observations in regions of feature space where none genuinely occur.

Threshold selection is a commercial judgement. The recall constraint of 0.75 encodes an assumption about relative error costs that was not derived from cost data, because none exists.

Single evaluation partition. Final performance rests on one held-out split. Cross-validation was applied during selection but the final figure is a single estimate.

20.4 Future Work

Immediate. Execute the analytical notebooks and persist their outputs, which is the sole remaining gap in this document; export the stored procedure source behind the six paginated reports; resolve the three measure discrepancies recorded in Section 18.8; capture the remaining dashboard pages of Parts A and B.

Near term. Extend the source to multiple periods, enabling trend analysis, genuine forecasting and Type 2 dimension handling; implement incremental loading; write predicted probabilities back to the warehouse so that risk becomes a dimensional attribute available to the reporting layer.

Longer term. Expose the model as a service; establish automated retraining and drift monitoring; implement role-based dashboard distribution; migrate the warehouse to a managed cloud platform.

Trigger conditions for revisiting the platform decisions. A dedicated integration service becomes justified when the pipeline acquires multiple heterogeneous sources, scheduled incremental extraction, or restart-on-failure requirements. A dedicated analysis service becomes justified when a second reporting

tool consumes the model, when an external system queries it, or when row-level security by role is required. Neither condition holds at present.

The assistant layer. A retrieval-augmented component grounded in the operator's own retention policies, feeding an agent that produces an actionable retention report, is the natural extension of this system: it converts a question asked in plain language into an answer grounded in the warehouse rather than in a model's general knowledge. It appears in the team architecture and, on evidence supplied, would move from declared to delivered.

Components described in the repository README. A REST service, an interactive Python dashboard and containerisation are described in the repository README. None is implemented. They are recorded here as future work, which is their correct status.

21 Conclusion

This project set out to convert a telecommunications subscriber dataset into a system capable of answering management questions about revenue, attrition and the relationship between them. It delivers an integrated platform spanning conceptual modelling, normalised operational storage, a transformation pipeline, a dimensional warehouse, predictive modelling with explainability, and a twenty-three-page reporting suite organised as a single analytical argument.

The work's principal strength is architectural rather than algorithmic. The star schema was selected against alternatives with stated reasoning; the grain was declared before any measure was placed; keys were chosen for the properties they would need if the system were extended rather than only for present convenience; and the revenue identity discovered in the fact table was converted into a standing validation control rather than noted and set aside. These are the decisions that determine whether an analytical system remains correct as it grows, and they are documented here with their justifications rather than presented as given.

The work's principal remaining weakness is narrower than it once was but is stated without qualification: **the analytical notebooks were not executed with their outputs preserved, and consequently this document reports no classifier performance metric.** The methodology is fully specified — partitioning, resampling, candidate models, tuning procedure, evaluation metrics, explainability method and threshold criterion are documented in sufficient detail to be reproduced — but the numbers those methods would produce are absent. The business results, by contrast, are present and are read from the deployed reporting layer. Chapter 18 distinguishes the two carefully, and no plausible-looking value has been substituted anywhere in this document.

Three judgements made during the work are worth restating, because each involved declining an easier and less accurate alternative. The renaming of *Billing Efficiency* to *Billing Channel Performance* acknowledges that the data cannot support the stronger claim. The explicit statement that `dim_date` spans one quarter forecloses the temporal analysis a reader might otherwise assume. The insistence that 'No Churn' is a value rather than a null prevents a data quality report from overstating incompleteness by counting correct records as defective. Each of these makes the system look less capable than a looser treatment would, and each makes it more trustworthy.

The system in its present state measures the operator's financial position, decomposes revenue to its sources, quantifies the cost of attrition, locates that attrition in specific offers, access technologies and geographies, and presents the whole through twenty-three dashboard pages and six operational reports that a manager can follow from position to decision. It has already produced findings worth acting on: a promotional offer losing 46.70 per cent of its subscribers, an access technology losing 40.7 per cent, and a referral channel retaining thirteen percentage points better than any other origin. What remains is to run the predictive component and record what it says.

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Appendix A Data Definition Language Listings

Listing	Source file	Content
A.1	01_create_database.sql	Operational schema creation, eleven tables
A.2	02_insert_data.sql	Generated insert statements (extract)
A.3	new_OLAP.sql	Dimensional schema creation and load
A.4	—	Verification query catalogue

Appendix B Complete Data Dictionary

Full column specification for all eleven operational tables and all six analytical tables: column name, data type, constraints, nullability, source and description.

Appendix C DAX Measure Definitions

Complete measure definitions organised by group: subscriber indicators, financial indicators, loss indicators, ratio indicators and validation indicators.

Appendix D Model Configuration

Candidate model parameters, hyperparameter search space, selected configuration, feature list and column ordering, and selected decision threshold.

Appendix E Team Contributions
